MGM University

Chhatrapati Sambhajinagar

Name of Faculty – Engineering & Technology

Name of College – School of Engineering & Technology

Name of Department – Computer Science and Engineering

Name of Programme – B.Tech Integrated Computer Science and Engineering Data Science

CURRICULUM BOOKLET

(According to NEP 2020 with effect from Academic Year 25-26)

(Applicable for Academic Batch: 2022-28, 2023-29 & 2024-30)

MGM University

Chhatrapati Sambhajinagar

Published by –

Academic Section,
Registrar Office,
MGM University.

Contents

Sr. No.	Particulars	Page no.
1.	University's vision	5
2.	University's mission	5
3.	University song	6
4.	SOET at a glance	7
5.	SOET's vision	8
6.	SOET's mission	8
7.	Programs offered at SOET	9
8.	Department of Computer Science and Engineering (Data Science)	10
9.	Admission eligibility for Maharashtra candidate	11
10.	Admission eligibility for All India candidate	11
11.	Course structure for I semester	12
12.	Course structure for II semester	12
13	Course structure for III semester	13
14	Course structure for IV semester	13
13.	Course structure for V semester	14
14.	Course structure for VI semester	15
15.	Course structure for VII semester	16
16.	Course structure for VIII semester	17
17.	Course structure for IX semester	18
18.	Course structure for X semester	19
19.	Course structure for XI semester	20
20.	Course structure for XII semester	21

21.	Syllabus of Semester I	23
22.	Syllabus of Semester II	43
23.	Syllabus of Semester III	58
24.	Syllabus of Semester IV	81
25.	Syllabus of Semester V	96
26.	Syllabus of Semester VI	118
27.	Syllabus of Semester VII	142
28.	Syllabus of Semester VIII	160
29.	Syllabus of Semester IX	178
30.	Syllabus of Semester X	203

MGM University

Vision

- To ensure sustainable human development which encourages self-reliant and self-content society.
- To promote activities related to community services, social welfare and also Indian heritage and culture.
- To inculcate the culture of non-violence and truthfulness through vipassanna meditation and Gandhian Philosophy.
- To develop the culture of simple living and high thinking

Mission

- To impart state of art education and technical expertise to students and give necessary training to teachers to create self-reliant society for future.
- To encourage students to participate in Indian and International activities in sports, literature, etc. so that future generation becomes base for free and liberal society
- To educate students in areas like Management, Finance, Human relations to inculcate philosophy of simple living and high thinking value of simple economic society.
- To inculcate culture of non-violence and truthfulness through Vipassana.

  To sustain activities of Indian culture (viz. classical dance, music and fine arts) through establishing institutes like Mahagami, Naturopathy, etc.

विद्यापीठगीत

अत्तदिपभवभवप्रदिपभव,

स्वरूपरूपभवहो

ज्ञानसब्बविज्ञानसब्बभव ,

सब्बदिपभवहो

अत्ताहिअत्तनोनाथो ,

अत्ताहिअत्तनोगति

अत्तमार्गपरअप्रमादसेहैतुझेचलना

सब्बकाकल्याणहो ,

वोकार्यकुशलकरना

सब्बकाउत्तममंगल , पथप्रदर्शकहो

अत्तदिपभवभवप्रदिपभव ,

स्वरूपरूपभवहो

ज्ञानसब्बविज्ञानसब्बभव ,

सब्बदिपभवहो

बुद्धमंशरनंगच्छामि :

धम्मंशरनंगच्छामि :

संघंशरनंगच्छामि :

School of Engineering and Technology (SOET) at a Glance

The School of Engineering and Technology (SoET), a self-aided, non-minority, constituent institution of MGM University is the first institution to provide integrated B.Tech (Integrated) and B.Tech Programs in the Marathwada region of Maharashtra. A sister institution of the illustrious Jawaharlal Nehru Engineering College (JNEC), Aurangabad, it was established as MGM Polytechnic College in 2010.

However, recognizing the rising need of quality engineering education after schooling in the region and beyond, it was transformed into the School of Engineering and Technology (SoET) in 2022. Inheriting a 42 years rich legacy of imparting world-class professional education from MGM Trust, SoET provides ultramodern research and innovation facilities to its students. Its integrated programs eliminate the need to appear for 12" standard and various engineering entrance tests, thus saving valuable time and energy of the students. It allows the students to utilize the same time to gain more practical knowledge of the field and a strong edge over the competition.

The institute offers a wide variety of Undergraduate (Six Years Integrated Programs after SSC), Undergraduate (Four Years Programs after HSC) programs to fulfill the dreams of the students. All these are advanced engineering programs founded on industry-oriented curriculum.

The Institution employs many modern teaching-learning methods like Project-based Learning, Research-based Learning and Collaborative Learning to develop comprehensive understanding and real-life skills in the students. The School has also signed several memoranda of understanding with renowned institutions, associations and industries to provide internship and project work opportunities to the students. It also joined hands with NPTEL, SWAYAM, ATAL and other MOOC platforms to facilitate blended learning facilities for the students. Institute is unique in its structure, methods and goals, the institution is strongly rooted in the philosophy of training and research that enhances the relationship between knowledge and its application and seeks to promote the creation ofan ideal society through holistic education

Vision

Enrich the students in technical education to work globally with human values.

Mission

- To inculcate technical skills by continuous assessment practices.
- To strengthen interaction between industry and institute.
- To involve the students in social activities with environmental awareness.

Programs offered at SOET

B.Tech (Integrated) UG Program 6 Years (after 10th)	**B.Tech UG Program 4 Years (after 12th)**
Civil Engineering	Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)
Computer Science and Engineering (Data Science)	Computer Science and Engineering (Cyber Security)
Computer Science and Engineering (IoT, Cyber Security Including Block Chain Technology)	Electrical Engineering
Robotics and AI	Electronics Engineering (VLSI Design and Technology)
--	Robotics and Automation

Department of Computer Science and Engineering

The Bachelor of Technology (B. Tech.) in Computer Science and Engineering degree program started in the year 1983 and is approved by the All India Council for Technical Education (AICTE), New Delhi and offers Choice Based Credit System education. In addition to core courses, students can opt for discipline specific elective subjects, open elective subjects from different institutes of the University. In addition, this program is uniquely designed to increase the employability and to prepare students to work in a Multi-disciplinary work environment. The program offers Major degrees in ComputerScience and Engineering and is open to students opting for minor specializations as per their interests. Pedagogies concentrating on student's active participation are extensively used in the teaching-learning process.

Vision

To provide update technology in computer engineering to fulfill needs of industry and society.

Mission

- To impart technical knowledge and skills by involving students in educational activities.
- To enable students to be entrepreneurs enhancing interaction between institute and industry.
- To nurture moral and professional values to build up society and nation.

Name of Program – B.Tech (Integrated) Computer Science and Engineering (DataScience)

Duration – Six Years

Eligibility –

Admissions to First Year:-

(a) The Candidate should be an Indian National.

(b) Candidate should have passed 10th Std. examination or its equivalent examination with 50% and 45% (for reserve categories and Persons with Disability candidates belonging to Maharashtra State only) aggregate marks and the candidate should obtain non-zero positive score in MGMU-CET.

Lateral Entry Admissions (Direct Second Year):-

(a) The Candidate should be an Indian National.

(b) Passed 10+2 examination with Physics / Mathematics / Chemistry / Computer Science / Electronics / Information Technology / Biology / Informatics Practices Biotechnology/Technical Vocational Subject/Agriculture/ Engineering Graphics / Business Studies / Entrepreneurship (Any of the three)

OR

10th + (2 years ITI) with appropriate Trade in that order shall be eligible for admission to Second Year B.Tech (Integrated) course(s) of appropriate programme.

(c) Any other eligibility condition/qualification declared, from time to time, by the appropriate authority.

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology (SOET)
Name of the Programme: B.Tech (Integrated) Computer Science and Engineering (DataScience)
Programme Type (UG/PG): UG
Duration: Six Years

Program Name: B.Tech Integrated Computer Science and Engineering (Data Science)

(Semester:First)

Coursecode	Course Title	Type	TeachingScheme			EvaluationScheme(Marks)						MinimumPassing(Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCC0101B	Mathematics-I	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCC0102B	Physics-I	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCC0103B	Chemistry-I	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4	
22IUCC0104H	English-I	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCC0105T	Fundamentals of Computer System	Theory	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Audit	
22IUCC0106T	Engineering Graphics	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3	
22IUCC0102L	Physics-I Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0103L	Chemistry-I Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0104L	Language Lab-I	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0105L	Fundamentals of Computer System Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0106L	Engineering Graphics Lab	Practical	-	-	4	-	-	100	-	-	100	-	-	40	-	-	40	2	
Total			**17**	**3**	**12**	**100**	**100**	**300**	**300**	**0**	**800**	**0**	**0**	**120**	**120**	**0**	**320**	**24**	

(Semester:Second)

Coursecode	CourseTitle	Type	TeachingScheme			EvaluationScheme(Marks)						MinimumPassing(Marks)						Credit	Remarks
						Internal			External		Total	Internal			External		Total		
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR			
22IUCC0201B	Mathematics-II	Theory	3	1	-	20	20	-	60	-	100			-	24	-	40	4	
22IUCC0202B	Physics-II	Theory	3	1	-	20	20	-	60	-	100			-	24	-	40	4	
22IUCC0203B	Chemistry-II	Theory	3	1	-	20	20	-	60	-	100			-	24	-	40	4	
22IUCC0204H	Universal Human Value	Theory	2	-	-	-	-	-	50	-	50			-	20	-	20	2	
22IUCC0205T	Computer Workshop	Theory	2	-	-	-	-	-	-	-	-			-	-	-	-	Audit	
22IUCC0202L	Physics-II Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0203L	Chemistry-IILab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0204L	Universal Human Value Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0205L	Computer Workshop Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1	
22IUCC0206L	Workshop Practices Lab	Practical	-	-	4	-	-	100	-	-	100	-	-	40	-	-	40	2	
Total			**13**	**3**	**12**	**60**	**60**	**300**	**230**	**0**	**650**	**0**	**0**	**120**	**92**	**0**	**260**	**20**	

Program Name:B.Tech Integrated Computer Science and Engineering (DataScience)																		
(Semester:Third)																		
			Teaching			Evaluation Scheme (Marks)						Minimum Passing (Marks)						
Course Code	CourseTitle	Type	Scheme			Internal			External		Total	Internal			External		Total	Credit
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR		
22IUCD0301T	Engineering Mathematics	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4
22IUCD0302T	Introduction to Biology	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0303T	Computer Programming	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0304T	Web Page Designing with HTML	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0305T	Basic Electronics	Theory	2	-	-	20	20	-	60	-	100	-	-	-	24	-	40	2
22IUCD0306T	Effective Communication and soft skills	Theory	2	-	-	20	-	-	-	-	20	8	-	-	-	-	8	Audit
ENV-C	Environmental Studies	Theory	2	-	-	20	20	-	60	-	100	-	-	-	24	-	40	Audit
22IUCD0303L	Computer Programming Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2
22IUCD0304L	Web Page Designing with HTML Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2
22IUCD0305L	Basic Electronics Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1
22IUCD0306L	Effective Communication and softskills Lab	Practical	-	-	2	-	-	30	-	-	30	-	-	12	-	-	12	Audit
Total			18	1	12	140	120	180	360	100	900	8	0	72	144	40	360	20
(Semester:Fourth)																		
			Teaching			Evaluation Scheme(Marks)						Minimum Passing (Marks)						
Course code	Course Title	Type	Scheme			Internal			External		Total	Internal			External		Total	Credit
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR		
22IUCD0401T	Discrete Mathematics	Theory	3	1	-	20	20	-	60	-	100	-	-	-	24	-	40	4
22IUCD0402T	Object Oriented Programming in C++	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0403T	Computer Graphics	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0404T	Digital Electronics	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0402L	Object Oriented Programming in C++ Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2

22IUCD0403L	Computer Graphics Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1
22IUCD0404L	Digital Electronics Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1
Total			12	1	8	80	80	140	240	60	600	0	0	56	96	24	240	17

Program Name:B.Tech Integrated Computer Science and Engineering (DataScience)																		
(Semester:Fifth)																		
			Teaching			Evaluation Scheme (Marks)						Minimum Passing (Marks)						
Course Code	CourseTitle	Type	Scheme			Internal			External		Total	Internal			External		Total	Credit
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR		
22IUCD0501T	Statistics and Probability	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0502T	Java Programming	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0503T	Software Engineering	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0504T	Programming in Python	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0505T	Data Structure and Algorithms	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
GS-01	Gandhian Studies	Theory	2	0	0	20	-	-	30	-	50	-	-		12		20	Audit
22IUCD0502L	Java Programming Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2
22IUCD0503L	Software Engineering Lab	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1
22IUCD0504L	Programming in Python Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1
22IUCD0505L	Data Structure and Algorithms Lab	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1
Total			17	0	10	120	100	170	330	80	800	0	0	68	132	32	320	20
(Semester:Sixth)																		
			Teaching			Evaluation Scheme(Marks)						Minimum Passing (Marks)						
Course code	Course Title	Type	Scheme			Internal			External		Total	Internal			External		Total	Credit
			L	T	P	CA	MSE	TW	ESE	PR		CA	MSE	TW	ESE	PR		
22IUCD0601T	Data Communication and Networking	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0602T	Advanced Java Programming	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0603T	Software Testing	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0604T	Data Science With Python	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
22IUCD0605T	Database Management System	Theory	3	-	-	20	20	-	60	-	100	-	-	-	24	-	40	3
CSS	Communication Skill	Theory	2	-	-	10	10		30	-	50				12		20	2
CSS-L	Communication Skill	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1

22IUCD0601L	Data Communication and Networking Lab	Practical	-	-	2	-	-	50	-	-	50		-	20	-	-	20	1
22IUCD0602L	Advanced Java Programming Lab	Practical	-	-	4	-	-	60	-	40	100	-	-	24	-	16	40	2
22IUCD0603L	SoftwareTesting	Practical	-	-	2	-	-	50	-	-	50	-	-	20	-	-	20	1
22IUCD0604L	Data Science With Python	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1
22IUCD0605L	Database Management System	Practical	-	-	2	-	-	30	-	20	50	-	-	12	-	8	20	1
Total			17	0	14	110	110	270	330	80	900	0	0	108	132	32	360	24

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B. Tech (Integrated) Computer Science and Engineering(DataScience)
Programme Type (UG/PG): UG
Duration: 06 Years

ProgramName:B.Tech Integrated Computer Science and Engineering(DataScience)												
Semester VII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No of Credits**	**Teaching (Contact hrs/ week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CSI24PCL401	Artificial Intelligence	Lecture	3	3	-	60	40	100	-	16	40
PCC	CSI24PCL402	Operating System	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCL403	Data Science With R	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCP401	Artificial Intelligence Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP402	Operating System Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP403	Data Science with R Lab	Practical	1	-	2	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
OE		Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
VEC	MGM56VEL102	Constitution of India	Lecture	2	2	-	30	20	50	-	8	20
EEMC	CSI24PCL404	Business Management & Financial Accounting	Lecture	2	2	-	60	40	100	-	16	40
FP	CSI24FPJ405	Field Project	Practical	2		4	50	-	50	20	-	20
		TOTAL		**22**	**17**	**10**	**530**	**320**	**850**	**-**	**-**	**-**

Nature of Course: L-Lecture, P-Practical

Course Category: BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B. Tech (Integrated) Computer Science and Engineering(DataScience)

Programme Type (UG/PG): UG

Duration: 06 Years

Semester - VIII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CSI24PCL451	Machine Learning	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCL452	Design and Analysis of Algorithm	Lecture	3	3	-	60	40	100	-	16	40
PCC	CSI24PCL453	Data Analytics	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCP451	Machine Learning Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP452	Design and Analysis of Algorithm Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP453	Data Analytics Lab	Practical	1	-	2	30	20	50	-	8	20
VSEC	CSI24VSP455	UI-UX Desing Lab	Practical	2	-	4	60	40	100	-	16	40
EEMC	CSI24HSL456	Entrepreneurship Development	Lecture	2	2	-	60	40	100	-	16	40
AEC		Refer MGMU AEC Basket	Lecture	2	2	-	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
OE	OE-III	Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
VEC	MGM24VEL201	Energy Conservation & Management System	Lecture	2	2	-	30	20	50	-	8	20
		TOTAL		**22**	**17**	**10**	**540**	**360**	**900**	**-**	**-**	**-**

Nature of Course: L-Lecture, P-Practical

Course Category: BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Name of Faculty: Engineering and Technology

Name of the College/Institute/Department/School: School of Engineering and Technology

Name of the Programme: B. Tech (Integrated) Computer Science and Engineering(DataScience)

Programme Type (UG/PG): UG

Duration: 06 Years

Semester - IX												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CSI24PCL501	Data Story Telling and Visualization	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCL502	Big Data	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCL503	Advance Database Management	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCL504	Data Privacy and Security	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCP501	Data Story Telling and Visualization Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP502	Big Data Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP503	Advance Database Management Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCP504	Data Privacy and Security Lab	Practical	1	-	2	30	20	50	-	8	20
PEC	CSI24PCL505	Basic of CC	Lecture	2	2	-	60	40	100	-	16	40
	CSI24PCL506	DevOps-I										
PEC	CSI24PCP505	Basic of CC Lab	Practical	1	-	2	30	20	50	-	8	20
	CSI24PCP506	DevOps-I Lab										
MDM		Refer MGMU MDM Basket	Lecture	4	4	-	60	40	100	-	16	40
OE	OE-IV	Refer MGMU Open Elective Basket	Lecture	2	2	-	30	20	50		8	20
		TOTAL		**21**	**16**	**10**	**540**	**360**	**900**	-	-	-

Nature of Course: L-Lecture, P-Practical,**Course Category:** BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B. Tech (Integrated) Computer Science and Engineering (DataScience)
Programme Type (UG/PG): UG
Duration: 06 Years

Semester - X												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CSI24PCL551	Business Intelligence	Lecture	1	1	-	30	20	50	-	8	20
PCC	CSI24PCL552	Natural Language Processing	Lecture	3	3	-	60	40	100	-	16	40
PCC	CSI24PCL553	Deep Learning	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCP551	Business Intelligence Lab	Practical	2	-	4	60	40	100	-	20	40
PCC	CSI24PCP552	Natural Language Processing Lab	Practical	1	-	2	30	20	50	-	8	20
PCC	CSI24PCL553	Deep Learning	Practical	1	-	2	30	20	50	-	8	20
PEC	CSI24PCL554	CC using AWS	Lecture	2	2	-	60	40	100	-	16	40
	CSI24PCL555	DevOps-II										
PEC	CSI24PCL556	Image Processing	Lecture	3	3	-	60	40	100	-	16	40
	CSI24PCL557	Quantum Computing										
PEC	CSI24PCP554	CC using AWS Lab	Practical	1	-	2	30	20	50	-	8	20
	CSI24PCP555	DevOps-II Lab										
PEC	CSI24PCP556	Image Processing Lab	Practical	1	-	2	30	20	50	-	8	20
	CSI24PCP557	Quantum Computing Lab										
VSEC	CSI24VSP559	Mobile Application Development Lab	Practical	2	-	4	60	40	100	-	16	40
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
		TOTAL		**21**	**13**	**16**	**570**	**380**	**950**	**-**	**-**	**-**

Nature of Course: L-Lecture, P-Practical,**Course Category:** BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B. Tech (Integrated) Computer Science and Engineering(DataScience)
Programme Type (UG/PG): UG
Duration: 06 Years

Semester - XI												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CSI24PCL601	Distributed Computing System	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCL602	Artificial Neural Network	Lecture	2	2	-	60	40	100	-	16	40
PCC	CSI24PCP602	Artificial Neural Network Lab	Practical	1	-	2	30	20	50	-	8	20
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
PEC	CSI24PCL603	CC using Azure IBM /	Lecture	2	2	-	60	40	100	-	16	40
	CSI24PCL604	DevOps- on AWS										
PEC	CSI24PCP603	CC using Azure IBM Lab /	Practical	1	-	2	30	20	50	-	8	20
	CSI24PCP604	DevOps- on AWS Lab										
PEC	CSI24PEL605	AR VR Technology	Lecture	2	2	-	60	40	100	-	16	40
	CSI24PEL606	Quantum Algorithm										
PEC	CSI24PEP605	AR VR Technology Lab	Practical	1	-	2	30	20	50	-	8	20
	CSI24PEL606	Quantum Algorithm Lab										
RM	CSI24RML606	Research Methodology	Lecture	4	4	-	60	40	100	-	16	40
RP	CSI24RPJ607	Project	Practical	4	-	8	60	40	100	-	16	40
		TOTAL		**21**	**14**	**14**	**510**	**340**	**850**	**-**	**-**	**-**

Nature of Course: L-Lecture, P-Practical,**Course Category:** BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

Name of Faculty: Engineering and Technology
Name of the College/Institute/Department/School: School of Engineering and Technology
Name of the Programme: B. Tech (Integrated) Computer Science and Engineering (DataScience)
Programme Type (UG/PG): UG
Duration: 06 Years

Semester - XII												
Course Category	**Course Code**	**Course Title**	**Nature of Course**	**No. of Credits**	**Teaching (Contact Hrs / Week)**		**Evaluation Scheme (Marks)**			**Minimum Passing (Marks)**		
					L	**P**	**Internal**	**External**	**Total**	**Internal**	**External**	**Total**
PCC	CSI24PCL651	MOOCS-I	Lecture	4	4	-	60	40	100	-	16	40
PEC	CSI24PCL652	MOOCS-II	Lecture	2	2	-	60	40	100	-	16	40
OJT	CSI24JTI653	Internship	Practical	12	-	24	90	60	150	-	24	60
MDM		Refer MGMU MDM Basket	Lecture	2	2	-	60	40	100	-	16	40
		TOTAL		**20**	**8**	**24**	**270**	**180**	**450**	-	-	-

Nature of Course: L-Lecture, P-Practical,**Course Category:** BSC – Basic Science Course, ESC – Engineering Science Course, AEC – Ability Enhancement Course, PCC – Program Core Course, VSEC - Vocational Skill and Skill Enhancement Course, CCA-Co-Curricular Activity

SEM-I

Course code: **22IUCC0101B**	**Course Name:** Mathematics-I
Credits: 4 **Teaching scheme:** L-3 T-1	**Evaluation scheme:** CA–40, ESE–60
Duration of End Sem. Exam: 3 hr.	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the basic concepts, principles, rules of algebra to solve the engineering problem in real world.	
CO2: Apply the basic concept of trigonometry to solve the engineering problem.	
CO3: Solve the engineering problem under the given geometrical condition.	
CO4: Utilize the basic concept of statistics to solve engineering problem.	
CO5: Understand the basic concept of system of complex numbers and illustrate simple problems.	
CO6: Understand the basic concept of system of complex numbers and illustrate simple problems.	

Contents:

Unit	Content	Teaching hours
1	**Algebra** Logarithm: Concept and Definition of Logarithm, Rules of Logarithms (Simple examples only),Permutation and Combination: Introduction, Fundamental Principle of counting, Permutations, Combinations. (Simple examples only), Binomial Theorem: Introduction, Binomial theorem for positive integral indices, General and middle terms. (Simple examples only) Partial Fraction: Introduction, Definition and types of fraction, Resolve into partial based on nature of factors and related problems	8
2	**Determinant and Matrices** Determinant :Definition and simple examples of determinant of order 2 & 3, Properties of Determinant, Minor and Cofactors, Crammer's Rule ,Matrices: Concept and definition of matrices Types of matrices, Algebra of matrices, Transpose, orthogonal, Adjoint of matrices, Inverse of matrix, Equivalent matrices, Matrix inversion method	8
3	**Trigonometry** Trigonometric ratios of compound, allied, multiple and sub multiple angles, Factorization and De- Factorization formulae, Inverse trigonometric ratios and related problems, Principle values and relation between trigonometric and inverse trigonometric ratios. Properties of triangle: sin, cosine, projection and tangent rule.(Without proof)	10
4	**Co-ordinate Geometry** Distance Formula, Section Formula, Different centers of triangle (without proof), Shifting of origin, Slope of line, Forms of Straight lines, Point of intersection of two lines, Angle between two lines (without proof) and Condition of parallel and perpendicular lines, Perpendicular Distance of a point from the line, Perpendicular Distance between two parallel lines. Conic Section: Circle, Parabola, Ellipse, Hyperbola.	10
5	**Data Analysis** Measure of central tendency: Mean (AM, GM, HM,) Mode, Median. , Measure of dispersion: Range, MD, SD, and Test of consistency	6
6	**Complex Numbers** Definition of complex number, Algebra of Complex number, Argand diagram, Forms of Complex number, Properties of modulus and arguments, De Moivre's Theorem, Cube root of unity.	6

Reference Books:

1.Hall and Knight, Higher Algebra Arihant Publications India limited,
2. Loney, SL (1912), Plane Trigonometry, Cambridge: University Press, JFM 25.0892.12
3. Loney, SL (1895), The elements of coordinate geometry, London: Macmillan and Co.
4. S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi.
5. David Freedman, Robert Pisani and Roger Purves, Statistics, 4th Edition.
6. S.C. Gupta, Fundamental of Statistics, Himalaya.
7. H.C. Saxena, Mathematical Statistics, Sultan Chand & Sons, New Delhi.

Online Resources:

1. www.scilab.org/-SCI Lab
2. www.mathworks.com/products/matlab/ -MATLAB
3. www.allmathcad.com/- MathCAD
4. www.wolfram.com/mathematica/ - Mathematica

Course code: 22IUCC0102B	**Course name: Physics-I**	**Course category:** PCC
Credits: 4 **Teaching scheme:** L-3 T-1		**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: dentify different systems of units and convert units from one system to another as well as conversant with Practical units.		
CO2:Understand equations of motion and their applications.		
CO3:Understand equations of motion and their applications.		
CO4:Acquire basic knowledge on semiconductor & application of P-N junction diode		
CO5:Use the properties of photoelectric effect, optical fiber &nano particle for various engineering application		
CO6: Understand concept of LIGHT		

Contents:

Unit	Content	Teaching hours
1	**Units & Measurements** Introduction of Unit, fundamental and derived quantities and their units, different system of Units (CGS, MKS, FPS and SI), Definition of accuracy and error, Estimation of Errors, absolute error, relative error and percentage error. Numerical problems. Explanation of Dimension of physical quantity Dimensional Equations of physical quantities and their uses with examples.	**8**
2	**Mechanics** Introduction to scalar and vector Quantities, representation of vector, addition, subtraction and multiplication of vectors, parallelogram law of vector Newton's Laws of motion: First law, Second law, &Third law, Statements & examples, Principle of conservation of linear momentum, statement and simple examples. Work, power and energy, its concept, units and dimension, Potential and Kinetic energy, its mathematical relations, Principle of conservation of energy, Recoil of gun.	**8**

3	**Circular Motion** Circular motion, and angular velocity, relation between angular velocity and linear velocity, Rotational motion, Simple Harmonic Motion, derivation of its equationsdefinition of amplitude, time period, frequency, phase etc., mathematical relations and units, Simple Pendulum	**6**
4	**Heat And Thermodynamics** Concept of heat and temperature, Unit of heat, Joule and calorie, thermometer, its interval and fixed point, different scales of temperatures and their conversion formulae. Transmission of heat, three modes of heat transfer, conduction, convection and Radiation, good and bad conductor of heat, coefficient of thermal conductivity, its S.I. unit and dimension. 1st law and 2nd law of thermodynamics, Joule's law & its derivation.	**8**
5	**Modern Physics** Photo electricity, Photon (quantum), Plank's hypothesis, energy of photon, properties of photons. Photo electric effect: Circuit diagram, process of photoelectric emission, definitions:- threshold frequency, threshold wavelength, Stopping potential, characteristics of photoelectric effect, Application of photoelectric cell & LDR.	**6**

Text Books:
Reference Books: 1. Engineering Physics- R K Gour& S L Gupta, DhanpatRai Publication- New Delh 2.Engineering Physics- M N Avadhanulu& P G Kshirsagar, S Chand & Company Ltd.- New Delhi 3.Fundamental of Physics- Jearl Walker, Wiley India Edition 4.Engineering Physics- V Rajendran, Tata McGraw-Hill Publishing Company Ltd- New Delhi 5.Applied Physics- Manpreet Singh, Dr. Major Singh, Mrs.Hitashi Gupta, S K Kataria& Sons- New Delhi
Online Resources 1. http://nptel.ac.in/course.php?disciplineId=115 2.http://nptel.ac.in/course.php?disciplineId=104 3.http://hperphysics.phy-astr.gsu.edu/hbase/hph.html 4.www.physicsclassroom.com 5.www.physics.org 6.www.fearofphysics.com

Course code: 22IUCC0103B	**Course name:**Chemistry-I
Credits: 4 **Teaching scheme:** L-3, T-1	**Evaluation scheme:** CA–40, ESE–60
Duration of End Sem Exam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday Experiences.	
CO2: Illustrate the Basic Concept of Structure of Atom	
CO3: Describe nature of bonding in molecular compounds.	
CO4: To know the common analysis techniques used in metallurgical industries.	
CO5: Students know how to approach the problem of choosing an alloy for a particular application.	
CO6: Use Relevant Electrolyte in Electrolysis for different Applications	

Contents:

Unit	Content	Teaching hours
1	**Basic principles of organic chemistry** Introduction, Structural representation of organic molecules-Condensed formula, structural formula, Bond line or Zigzag formula with examples. Classification of organic compound. Acyclic/Open chain compound. Cyclic compound , Homocyclic,Heterocycliccompound,Alicyclic,Aromatic(Benzenoid,Nonbenzenoid),Heterocyclic (Aromatic . Nonaromatic),(Definition with example.) ,Functional group in organic compounds. Halides,Cyanides,Isocyanides, Nitro compound,Alcohol,Phenol,Amines.Ether,Aldehydes,Ketones,Carboxylic acid, Ester, Amide, Sulphonic acid, Acyclic chloride,Acidunhydride,Alkyne,Alkene,Alkane.(Structure and Functional group with examples).	8
2	**Atomic Structure and Chemical bonding** Introduction ,Definition of atom, Bohr's atomic model, definition of atomic number, atomic mass number and their differences, , Isotopes and Isobars: Definitions, examples and distinction , Orbits and orbitals: s, p, d, f orbitals, shapes and description of s-orbital and p-orbital. Distribution of electrons in orbitals: , Definition of electronic configuration, Aufbau's principle, Hund's rule, orbital Electronic configurations (s, p, d, f) of elements having atomic number 1 to 30	6
3	**Chemical bonding** Introduction, Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 , Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 , Definition of covalence, single, double and triple covalent bonds, formation of Covalent compounds H2O,CO2,N2	6
4	**Metals and Ores**	8

	Introduction , Occurrence of metals in free and combined state, definitions- mineral, ore, Gangue, flux and slag, metallurgy. , Ores of Iron , Steps involved in the extraction of metals: Crushing, Concentration, Reduction, Refining , Mechanical properties of metals- Hardness, ductility, malleability, tensile strength, toughness, mach inability, weld- ability, forging, soldering, brazing, cast ability.	
5	**Alloys and its applications** 5.1 Introduction , Definition of alloy, purposes of making alloys with examples. Preparation methods- Fusion, Compression , Classification of Alloys- Ferrous and nonferrous alloys with examples. , Examples of alloys-Composition, properties and application of duralumin, Woods metal. Brass, Solders. ,Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. ,. pH and pOH: Definition of pH, pOH, pH Scale	6
6	**Electrochemistry** Introduction , Basic concepts of electrolysis: Electrolyte, types of electrolyte- strong and weak electrolyte, their difference. , affecting degree of ionization., Applications of electrolysis- Electroplating of silver, electro refining of blister copper, , Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. , pH and pOH: Definition of pH, pOH, pH Scale	6

Text Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
3. Atkins, Physical Chemistry, OUP Oxford.
4. Engineering Chemistry, Wiley India Pvt, Ltd, First edition

Reference Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi.
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New D
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technical Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media

Online Resources:

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)

Course Code: 22IUCC0104H	**Course Name:**English-I
Credits:3 **Teaching Scheme:** L-3	**Evaluation Scheme:** CA–40, ESE–60
Duration of EndSemExam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Student will be able to apply basic concept of grammar in day to day communication	
CO2: Student will be able to Summaries and comprehend the given passages	
CO3: Student will be able to compose dialogue and paragraph for different situations	
CO4: Student will be able to use appropriate words as per the context	
CO5: Student will be able to deliver prepared speeches to express ideas, emotions and thoughts.	

Contents –

Unit	Content	Teaching hours
1	**Basic Grammar** 1.1 Articles, 1.2 Preposition, 1.3 Conjunction, 1.4 Tenses, 1.5 Direct indirect speech, 1.6 Change the voice.	10
2	**Comprehension Skills** 2.1 Definition of Reading 2.2 Definition of Reading Comprehension 2.3 Kinds of Reading 2.4 Extensive Reading 2.5 Intensive Reading 2.6 Effective Strategies for Reading Comprehension	10
3	**Paragraph and dialogue writing** 3.1 Paragraph Writing: Elaborate and expand the ideas with cohesion, coherence and use of correct punctuation marks 3.2 Types of Paragraph: Narrative, Descriptive, Technical, Comparison and Contrast 3.3 Dialogue Writing: Based on various situations	10
4	**Vocabulary building** 4.1 Words Often Confused 4.2 Collocation 4.3 Prefix and Suffix 4.4 Synonyms and Antonyms	6
5	**Speeches** Speech Writing based on situations	6

	5.1 Welcome Speech, 5.2 Farewell Speech, 5.3Vote of Thanks and Introducing a Guest	
6	**Introduction to the sounds of English** 6.1 English Phonetics 6.2 Phonetics symbols 6.3 Consonant and vowel sounds 6.4 Accent, rhythm and intonation. 6.5 The functions of intonation	6

Reference Books:

1. English Grammar and Composition Book (Wren & Martin)
2. Oxford Modern English Grammar
3. Fundamentals of English Grammar (N.C.Sinha)

Online Resources:

1. https://www.britishcouncil.in/english/learn-online
2. http://learnenglish.britishcouncil.org/en/content
3. http://www.talkenglish.com
4. Languagelabsystem.com
5. www.wordsworthelt.com

Course Code: 22IUCC0105T		**Course Name:** Fundamental of Computer System
Credits: 2	Teaching **scheme:** L-2	**Evaluation Scheme:** Audit
Duration of End sem Exam: --		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Use Computer System and its Peripherals**.**		
CO2: Prepare Business Document using word processing tool.		
CO3: Interpret data and represent it graphically using spreadsheet.		
CO4: Prepare Professional Presentations.		
CO5: Identify the layout of wired and wireless LAN Environment		
CO6: Prepare and interpret database using csv file.		

Contents:

Unit	Content	Teaching hours
1	**Introduction to Computer System** Basic of computer system: Overview of Hardware and Software, block diagram of Computer System, Input/output unit CPU, Arithmetic logic Unit (ALU), System of software Memory Unit. 1.2 Generation of O.S.: Overview of Different Types and Generation of Operating system 1.3 Working with OS: Create and manage file and folders, Copy a file, renaming and deleting of files and folders, Searching files and folders, application installation, creating shortcut of application on the desktop. 1.4 Printer and Scanner: Overview of printer and scanner, Features of printer and Scanner, Difference between Printer and Scanner.	4
2	**Word Processing** Word processing: Overview of Word processor Basics of Font type, size, color, Effects like Bold, italic, underline, Subscript, superscript, Case changing options Previewing a document, Saving a document, Closing a document and exiting application. 2.2 Editing a document: Format and align text, Formatting Paragraphs, Line and paragraph spacing, using FIND and REPLACE, Setting line spacing, add bullet and numbers in lists, add borders and shading, document views, Page settings and margins, Spelling and Grammatical checks. 2.3 Inserting an image to word document: Insert and delete a page break, Insert page numbers, Insert the date and time, Insert special characters (symbols), Insert a picture from a file, Resize and reposition a picture 2.4 Working with table: Insert a table, Convert a table to text, Navigate and select text in a table, Resize table cells, Align text in a table, Format a table, Insert and delete columns and rows, Borders and shading, Repeat table headings on subsequent pages, Merge and split cells.	4
3	**Spreadsheet** Working with spread sheet: Overview of workbook and worksheet, Create Worksheet Save, Copy Worksheet, Delete Worksheet, Close and open Workbook 3.2 Editing worksheet: Insert and select data, adjust row height and column width, delete, move data, insert rows and columns, Copy and Paste, Find and Replace, Spell Check Zoom In-Out, Special Symbols, Insert Comments, Add Text Box, Undo Changes, Freeze Panes, hiding/unhanding rows and columns. 3.3 Formatting cells and sheet: Setting Cell Type, Setting Fonts, Text options, Rotate Cells,	4

	Setting Colors, Text Alignments, Merge and Wrap, apply Borders and Shades, Sheet Options, Adjust Margins, Page Orientation, Header and Footer, Insert Page Breaks, Set Background. 3.4 Working with formula: Setting Cell Type, Setting Fonts, Text options, Rotate Cells, Setting Colors, Text Alignments, Merge and Wrap, apply Borders and Shades, Sheet Options, Adjust Margins, Page Orientation, Header and Footer, Insert Page Breaks, Set Background 3.5 Advanced operation: Conditional Formatting, Data Filtering, Data Sorting, Using Ranges, Data Validation, Adding Graphics, Printing Worksheets, print area, margins, header, footer and other page setup options 3.6 Working with chart: Introduction to charts, overview of different types of charts, Bar, Pie, Line charts, creating and editing charts. Using chart options: chart title, axis title, legend, data labels, Axes, grid lines, moving chart in a separate sheet.	
4	**Presentation tool** Creating presentation : Outline of an effective presentation, Starting a Presentation Files, Creating a Basic Presentation, Working with textboxes, Apply Character Formats, Format Paragraphs View a Presentation, Saving work, creating new Slides, Applying a theme, Changing Colors, fonts and effects, apply custom Color and font theme, changing the background, Arrange Slide sequence. 4.2 Working with table: Insert a Table in a Slide, Format Tables, and Import Tables from Other Office Applications. 4.3 Animation: Adding and Modifying animation to a Presentation - Insert Images into a Presentation, insert audio clips, video/animation, Add Shapes, Add Visual Styles to Text in a Presentation, Edit Graphical Objects on a Slide, Format Graphical Objects on a Slide, Group Graphical Objects on a Slide, Apply an Animation Effect to a Graphical Object, Add Transitions, Add Speaker Notes, Print a Presentation. 4.4 Working with Charts: Insert Charts in a Slide, Modify a Chart, Import Charts from Other Office Applications.	4
5	**Basic of Networking** Introduction of networking Devices: Application of Network interface card (NIC), HUB, Switches, Routers, Modem 5.2 Concept of Network :- LAN,MAN,WAN Wireless network and devices; Wi-Fi, Access point, Repeaters, Bluetooth 5.3 Characteristics of Cable: CAT5,CAT6,fibre optic cable, use of crossover and straight cable,RJ-45 connectors, SC.ST,FC,LC type fiber Connectors 5.4 Types of Servers	4
6	**Databases** Create an appropriate database structure : Import data from existing files (including .csv, .txt) using specified field names to create tables. Create relationships between tables Create and use a data entry form . Types of database Characteristics, uses, advantages and disadvantages of a flat file and a relational database .Primary and foreign keys Characteristics of primary key and foreign keys. Form design Characteristics of good form design.Perform calculations Use arithmetic operations or numeric functions to perform calculations including calculated fields, calculated controls Use formulae and functions to perform calculations at run time including addition, subtraction, multiplication, division, sum, average, maximum, minimum, count .Sort data Use a single criterion, or multiple criteria to sort data into ascending or descending order.Search and select data Use a single criterion, or multiple criteria to select subsets of data using a query perform searches using a variety of operators including AND, OR, NOT, LIKE, >, =, <=, <> perform searches using wildcards.Display data Produce reports to display data including displaying all the required data and labels.	6

Text Books:
1. B Ram –Computer Fundamental architecture and Organization, New age international Publications
2. Dr RajendraKawale ,"Computer Fundamental", Devraj Publications Dist. Solapur

Course Code: 22IUCC0106T	**Course Name:** Engineering Graphics
Credits:3 **Teaching Scheme:** L-3	**Evaluation Scheme:** CA–40, ESE–60
Duration of EndSemExam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Student will able to draw geometrical figures with proper dimensioning technique	
CO2: Student will able to draw the views of given object by using principles of orthographic projection	
CO3: Student will able to draw isometric views of given component or from orthographic projection	
CO4: Student will able to draw Engineering curves	
CO5: Student will able to draw free hand sketches of given engineering Elements.	
CO6: Student will able to create 2D computer drawings on CAD software	

Contents –

Unit	Content	Teaching hours
1	**Basics of Graphics:** Introduction to Engineering Graphics, Need of Engineering Drawing, Drawing Instruments, BIS code of practice for general engineering drawing, standard sizes of drawing sheets (ISO-A Series) , single stroke vertical Letters and Numbers, convention of lines and their applications, Scale- reduced, enlarged and full size, Dimensioning technique, and Geometrical construction	4
2	**Orthographic Projection:** Orthographic projection-first angle and third angle method their symbol, conversion of pictorial view into orthographic views- Orthographic projection of machine parts containing plain surfaces, slanting surfaces slots, ribs, cylindrical surfaces (First angle method is to be used)	7
3	**Isometric Projections:** Introduction to Pictorial views, Isometric scale and natural scale, Converting Orthographic Projections into Isometric Projections and Isometric views. Illustrative problems related to simple objects having plain, slanting , cylindrical surfaces and slots on slanting surfaces	7
4	**Engineering Curves** Curves used in Engineering Practice, Conic sections, Construction of conics by different methods, Rectangular- hyperbola, Cycloidal curves, Epi and hypo-cycloids, Involutes and Archimedean spiral, Loci of points on single slider crank mechanism with given specification	7
5	**Free Hand Sketches of Engineering objects** Free hand sketches of machine elements; Thread profiles, nuts bolts, studs set screws, washers, locking arrangments.	3
	Introduction to CAD	

6	Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects	4

Reference Books:

1. N. D. Bhatt and Panchal V. M., "Engineering Drawing", Charotar Publishing House, Anand
2. P. J. Shah , " A Text Book of Engineering Drawing", S.Chand, New Delhi
3. Shah M.B. &RanaB.C,"Engineering Drawing & Computer Graphics", Pearson Publications,
4. .Agrawal B.&Agrawal C.M, "Engineering Graphics", TMH Publications
5. Narayana K.L. & P. Kannaiah, "Engineering Drawing", Scitech Publications
6. .P.I Vargese, "Engineering Graphics", McgrawHill Publications
7. D.A.Hindoliya," Engineering Graphics", B.S.Publications
8. Dabhade M. L., "Engineering Graphics", Vol.-I and Vol.-II, Vision Publications, Pune .
9. K.Venugopal, "Engineering Drawing and Graphics" ,New Age International Publishers

Online Resources:

1. https://www.machinedesignonline.com
2. https://www.me.umn.pdu/courses/me2018/handouts/engg%20graphics.pdf
3. https://www.youtube.com/watch?v=fvjk7PlxAuo
4. https://www.youtube.com/watch?v=3WXPanCq9LI

Course code: 22IUCC0102L	**Course name:** Physics-I Lab
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1. Identify different systems of units and convert units from one system to another as well as conversant with Practical units.	
CO2. Understand equations of motion and their applications.	
CO3: Understand equations of motion and their applications.	
CO4. Acquire basic knowledge on semiconductor &applicatation of P-N Junction diode.	
CO5. Use the properties of photoelectric effect, optical fiber &nano particle for various engineering application	
CO6: Understand concept of LIGHT	

Contents –
Perform any 10 practicals from the list.

Practical	Content	Teaching hours
1	Measure the dimensions of given objects using Vernier caliper.	2
2	Measure the dimensions of given objects using micrometer screw gauge	2
3	Measure the dimensions of given objects using spherometer	2
4	To determine the valve of acceleration due to gravity [g] of a place with simple pendulum.	2
5	Determination of force constant using Helical Spring	2
6	Use Searle's thermal conductivity apparatus to find coefficient of thermal conductivity of a given material	2
7	Use Joule's calorimeter to determine Joule's Mechanical/ electrical equivalent of heat	2
8	To study the I-V characteristics of photoelectric effect	2
9	To determine the refractive index of the material of the glass slab by pin method.	2

Course code:22IUCC0103L	**Course name:**Chemistry-I Lab
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to	
CO1: Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday Experiences. **CO2:** Illustrate the Basic Concept of Structure of Atom **CO3:** Describe nature of bonding in molecular compounds. **CO4:** To know the common analysis techniques used in metallurgical industries. **CO5:** Students know how to approach the problem of choosing an alloy for a particular application. **CO6:** Use Relevant Electrolyte in Electrolysis for different Application	

Contents –

Unit	Content	Teaching hours
1	Do you know chemistry laboratory	2
2	Determine preliminary test and functional group of organic compound No.1	2
3	Determine preliminary test and functional group of organic compound No.2	2
4	Determine preliminary test and functional group of organic compound No.3	2
5	Determine preliminary test and functional group of organic compound No.4	2
6	Identify anion in the given ionic solution (non- metallic ions)	2
7	Identify cation in the given ionic solution (metallic ion)	2
8	Determine the percentage of iron in the given sample using redox titration	2
9	Determine the pH value of given solution using universal indicator	2
10	Determine the voltage generated from chemical reaction using Daniel cell.	2
11	Determine the electrode potential of Iron metal.	2
12	Determine the rate of corrosion on different temperatures for Aluminum	2

Course code: 22IUCC0104L	**Course name:** Language Lab-I
Credits: 1**Teaching scheme:** P-02	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Student will be able to apply basic concept of grammar in day to day communication	
CO2: Student will be able to Summaries and comprehend the given passages.	
CO3: Student will be able to Compose dialogue and paragraph for different situations.	
CO4: Student will be able to use appropriate words as per the context	
CO5: Student will be able to Deliver prepared speeches to express ideas, emotions and thoughts	

Contents:

Unit	Content	Teaching hours
1	Make sentences using correct articles	2
2	Make sentences using correct Preposition	2
3	Make sentences using correct Conjunctions	2
4	Make sentences using correct active and passive voice.	2
5	Make sentences using correct direct and in direct speech.	2
6	Make sentences using correct tenses	2
7	Make oral presentations using correct grammar.	2
8	Write short paragraphs emphasizing on syntax and sentence structure.	2
9	Write different types of dialogues for role plays.	2
10	Construct sentences using various collocations.	2
11	Write different types of speeches using new vocabulary.	2
12	Deliver extempore short speeches of 3to5 minutes.	2

Course Code: 22IUCC0105L	**Course Name:** Fundamental of Computer System Lab
Credits: 1 Teaching **Scheme:** P-2	**Evaluation Scheme:** TW–50, PR–0
Duration of End Sem Exam:	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Use Computer System and its Peripherals**.**	
CO2: Prepare Business Document using word processing tool.	
CO3: Interpret data and represent it graphically using spreadsheet.	
CO4: Prepare Professional Presentations.	
CO5: Identify the layout of wired and wireless LAN Environment	
CO6: Prepare and interpret database using csv file.	

Contents –

Unit	Content	Teaching hours
1	Identify various Input/output devices. Connections and peripherals of computer system.	2
2	Install and work with wifi printer and scanner setting.	2
3	Create, edit and save documents: apply formatting features on the text, line, paragraph. Use bullets, numbering, page formatting.	2
4	Insert, edit images and shapes, sizing, cropping color, background, group/ungroup. Insert and apply various table formatting features on it.	2
5	Create, open and edit worksheet 1.Enter data and format it adjust row height and column width. 2. Insert and delete cells, rows and columns. 3. Apply wrap text, orientation feature on cell.	2
6	Insert formulas, ”IF” conditions, functions and named ranges in worksheet. Apply data sort, Filter and data validation features.	2
7	Create charts to apply various chart options.	2
8	Create slide presentation. 1. Apply design themes to given presentation. 2.Add new slides and insert pictures / images, shapes. 3. Add tables and charts in slides. 4.Run slides presentation in different modes. 5. Print slide presentations on handouts.	2
9	Apply animation effects to text and slides Add audio and video files in presentation.	2

10	Connect/Disconnect LAN cable, external hard disk modem. Connect desktop computers and laptops with LCD/DLP projectors.	2
11	Prepare and test crossover and straight cable, CAT5,CAT6 cable, using crimping Tools, splicer. Connect two switches/Hubs/switches/r outers physically.	2
12	Identify server by its type and verify its Specification	2
13	Create new database, table and build relationship between them, saving tables	2
14	Managing data in tables and performing calculation using functions, and sorting data from tables.	2

Course Code: 22IUCC0106L	**Course Name:** Engineering Graphics Lab
Credits:2 **Teaching Scheme:** P-4	**Evaluation Scheme:** TW–100, PR–0
Duration of End SEM Exam:	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Student will able to draw geometrical figures with proper dimensioning technique	
CO2: Student will able to draw the views of given object by using principles of orthographic projection	
CO3: Student will able to draw isometric views of given component or from orthographic projection **CO4:** Student will able to draw Engineering curves	
CO5: Student will able to draw free hand sketches of given engineering Elements	
CO6: Student will able to create 2D computer drawings on CAD software	

Contents –

Unit	Content	Teaching hours
1	Draw different types of lines, dimensioning styles. Draw regular geometric construction	2
2	Draw one figure showing dimensioning techniques. Draw one problem on redraw the figure.	2
3	Draw two problem on orthographic projection using first angle method of projection having plain and slanting surface	2
4	Draw two problem on orthographic projection using first angle method of projection having cylindrical surface, rib	2
5	Draw two problems on Isometric view of simple object having plain and slanting surface using natural scale	2
6	Draw two problems on Isometric view of simple object having plain and slanting surface using natural scale	2
7	Draw two problem on Isometric projection of object having cylindrical surface by using isometric scale	2
8	Draw two problem on isometric projection of object having slot on slanting surface by using isometric scale	2
9	Draw Engineering curves generated from conics 1. Ellipse 2. Parabola 3. Hyperbola	2
10	Draw Involutes of circle and pentagon. Draw cycloid curve	2

11	Draw free hand sketches/conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washer, locking arrangement	2
12	Draw free hand sketches/conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washer, locking arrangement	2
13	Draw basic 2D entities like: rectangle, polygon, circle, Arc, circular and rectangular array by using AutoCAD	2
14	Draw basic branch specific components in 2D using AutoCAD	2
15	Draw complex branch specific components in 2D using Autocad	2

SEM-II

Course Code: 22IUCC0201B	**Course Name:** Mathematics-II
Credits: 4 **Teaching scheme:** L-3 T-1 P-0	**Evaluation Scheme:** CA–40, ESE–60
Duration of End Sem Exam: 3 hr	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the concept to formulate a mathematical model using of function, and concept of Limits and continuity which rise to derivative.	
CO2: Estimate, to graph derivatives of the function and used rules of differentiation to solve problem involving rates of change, approximation of function and optimization problem.	
CO3: Understand the concept of integration to find the arc length, area, volume and surface area using integrals.	
CO4: Solve ordinary differential equations and also apply these methods to solve engineering problems.	
CO5: Trace the Cartesian curve, parametric curve and Polar curve	

Contents –

Unit	Content	Teaching hours
1	**Function limit and Continuity** Definition of Function, Ways to represent a function, Types of functions, Mathematical modeling, and transformation of functions. Limit: Concept and definition of limit using tangent and velocity problem, Laws of limits with simple problems. Concept of continuity	8
2	**Differential Calculus** Concept and definition of Derivatives through physical and geometrical problem. And Rates of Change, The Derivative as a Function. Rules of Differentiation, Derivative of inverse functions, Implicit Differentiation, Logarithmic Differentiation, Parametric Differentiation, Second order differentiation	8
3	**Application of Derivative** Rate of Change in the Natural and Social Sciences, Exponential Growth and Decay, Linear Approximations and Differentials, Maximum and Minimum Values, Indeterminate Forms and L'Hospitals Rule. Optimizing problems**.**	8
4	**Integral Calculus** Indefinite and Definite integrals. Methods and Properties, Applications to find arc length, Area, Volume and surface area.	10
5	**Differential Equation** Introduction to Initial Value problems. Linear and Bernoulli's equations, Exact equations	8

	and reducible to exact form. Applications of first order differential equations- Orthogonal trajectories, growth and decay problems, Electrical circuits, and falling bodies.	
6	**Curve Tracing** Tracing of standard curves in Cartesian form, Parametric form and Polar form	6

Reference Material:

1. Narayan, Shanti and Mittal, P.K. Differential Calculus, S Chand publication.
2. Narayan, Shanti and Mittal, P.K. Integral Calculus, S Chand publication
3. S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi.
4. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley.
5. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York.
6. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi.
7. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II), Pune VidyarthiGrihaPrakashan, Pune.
8. H. K. Das and RajnishVerma, Higher Engineering Mathematics, S. Chand & CO. Pvt. Ltd., New Delhi.

Online Resources:

1.www.scilab.org/-SCI Lab
2.www.mathworks.com/products/matlab/ -MATLAB
3.www.allmathcad.com/-MathCAD
4.www.wolfram.com/mathematica/ - Mathematica
5.www.dplot.com/-DPlot

Course Code: 22IUCC0202B	**Course Name:**Physics-II
Credits: 4 **Teaching Scheme:** L-3 T-1	**Evaluation Scheme:** CA–40, ESE–60
Pre-requisites: Soil	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: To learn about the useful solid material in industry & understand its physical properties.	
CO2: To learn about the useful liquid material & understand its physical properties.	
CO3: Understand the concepts of sound wave	
CO4: To learn about the concepts of radioactive and ultrasonic principles.	
CO5: To learn about the concepts of electrostatic & X rays.	
CO6: To learn about the concepts of the principles of electricity , magnetism & semiconductors to solve engineering problems.	

Contents –

Unit	Content	Teaching hours
1	**Properties of Solid** 1.1 Deforming Force and Restoring Force, Elasticity, Plasticity, Rigidity 1.2 Stress and Strain and their types, Elastic limit and Hooke's law Stress -Strain diagram, Poisson's ratio, factors affecting elasticity, With numerical problem	6
2	**Properties of Liquid** 2.1 Fluid friction, pressure, pressure- depth relation, Pascal's law, Archimedes' principle 2.2 Viscosity, velocity gradient, Newton's law of viscosity. 2.3 Free fall of spherical body through viscous medium and Stokes' law, derivation of coefficient of viscosity 'ç' by Stokes' method, effect of temperature and adulteration on viscosity of liquids With numerical problem	8
3	**Wave Theory & Sound** 3.1 Wave , Longitudinal Wave, Transverse Wave, , wave length, amplitude, frequency, time-period and velocity of wave, relation between velocity, frequency and time - period of wave. 3.2 Sound waves, Newton's formula for velocity of sound, Musical sound & noise, Loudness, Pitch, Quality of sound (Timbre), Musical Scale.	8
4	**Radioactivity** 4.1 Radioactivity, α, β and γ rays and their properties. 4.2 Radioactive decay law, half-life period. Energy Mass Relation, Nuclear fission, Nuclear fusion. 4.3 Ultrasonic waves, properties of Ultrasonic waves, Applications of ultrasonic waves, Doppler Effect and its applications.	6
	Electrostatics & X Ray	

5	5.1 Capacitors and capacitance, Parallel plate capacitor, effect of dielectric on capacitance, Series Combination of capacitors, Parallel Combination of capacitors, energy stored in a capacitor. Cells, emf of cell, internal resistance of cell 5.2 X ray Definition, Production of X-rays by modern Coolidge tube, Continuous & Characteristic X ray, Properties of X ray and Engineering , Medical, & Scientific Applications of X ray.	6
6	**Electricity , Magnetism & Semiconductor** 6.1Coulomb's law, Ampere's law Faraday's law motion of charges in electric & magnetic fields to determine e/m. Electrostatic focusing system. Magneto static focusing system. 6.1 Conductors, Insulators And Semiconductors, Energy bands, intrinsic and extrinsic semiconductors p-n junction diode, I-V characteristics of p-n junction, applications of p-n junction diode.	6

Reference Material :

1. Engineering Physics- R K Gour& S L Gupta, DhanpatRai Publication- New Delh

2. Engineering Physics- M N Avadhanulu& P G Kshirsagar, S Chand & Company Ltd.- New Delhi 3. Fundamental of Physics- Jearl Walker, Wiley India Edition

4. Engineering Physics- V Rajendran, Tata McGraw-Hill Publishing Company Ltd- New Delhi

5. Applied Physics- Manpreet Singh, Dr. Major Singh, Mrs.Hitashi Gupta, S K Kataria& Sons- New Delhi

E-Resources:

1. http://nptel.ac.in/course.php?disciplineId=115
2. http://nptel.ac.in/course.php?disciplineId=104
3. http://hperphysics.phy-astr.gsu.edu/hbase/hph.html
4. www.physicsclassroom.com
5. www.physics.org
6. www.fearofphysics.com

Course Code: 22IUCC0203B		**Course Name:** Chemistry-II
Credits: 4	**Teaching scheme:** L-3 P-0 T-1	**Evaluation Scheme:** CA–40, ESE–60
Duration of End sem Exam:3hr.		
Pre-requisites:		

Course Outcomes: At the end of the course, the students will be able to –
CO1: Describe different processes for manufacturing metals and control of their corrosion
CO2: Use relevant water treatment process to solve industrial problems
CO3: Justify the selection of lubricants for different types of machines
CO4: Describe different types of fuels and their properties and uses.
CO5: Justify the selection of lubricants for different types of machines
CO6: Used Relevant engineering materials in Industry

Contents:

Unit	Content	Teaching hours
1	**Corrosion 1.1** Definition, causes of Corrosion Different types of Corrosion and factors affecting corrosion rate. **1.2 Protection of metal by**: Galvanising ,Tinning ,Electroplating ,colourizing 1.5 classification of plain carbon steel, low carbon, medium carbon,high carbon steel with their properties and Application. 1.3 **Paint:** Definition characteristics, constituent of paint and their function.	10
2	**Metallurgy** 2.1 Introduction 2.2 Definition of metallurgy Properties and application of commercial forms of iron- pig iron ,cast iron and wrought iron 2.3 Steel: Definition of steel ,Preparation of steel from pig iron using open hearth process, basic oxygen process. 2.4 classification of plain carbon steel, low carbon, medium carbon, high carbon steel with their properties and application.	10
3	**Water treatment:** 3.1 Introduction 3.2 Hardness : Classification, 3.3 Hard water in boilers and prevention: Boiler corrosion, Caustic embrittlement, priming and foaming, scales and sludge's 3.4 Water softening : lime soda process (hot lime soda and cold lime soda process) Zeolite process, Ion exchange process (Cation exchange process and anion exchange) 3.5 Potable water treatment: sedimentation, coagulation, filtration, and sterilization. Waste water treatment, BOD and COD of sewage water, Reverse Osmosis, recycling of waste water	10
4	**Fuels** 4.1 Introduction 4.2 Definition of fuel, calorific and ignition temperature. Characteristic of good fuel classification of fuels 4.3 Solid fuels: Coal classification and composition, proximate analysis, Ultimate analysis. Determination of gross calorific value by using Bomb calorimeter. 4.4 Liquid fuel: Fractional distillation of crude	10

	petroleum , boiling range, composition, properties, application of Bio- diesel 4.5 Gaseous fuel: Composition, properties ,application of Biogas, LPG, CNG	
5	**Lubricants** 5.1 Introduction Definition of lubricants .Function of lubricants, Classification of lubricants. 5.2 Characteristic : Physical characteristic of lubricants : Viscosity, Viscosity index, oiliness, flash point, fire point, 5.3 chemical characteristic of lubricants- acid value, emulsification, saponification value.(only definition) 5.4 Selection of Lubricants for road rollers, steam engine, sewing machine, concrete mixture, I.C. engine, cutting tools, gears.	10
6	**Non -metallic Engineering material: Polymers (Plastics, Rubber) :** 6.1Introduction 6.2 Plastics: Definition of plastic, polymer, polymerization, types of Polymerization with examples. Types of plastic- thermo softening plastics and thermosetting plastics and their difference, properties and applications of plastics. 6.3 Rubber: Types of rubber. Natural Rubber- definition, drawbacks of natural rubber, vulcanization of rubber with chemical reaction, applications of vulcanized rubber Synthetic rubber- definition, difference between natural and synthetic rubber, examples of synthetic rubber, properties of synthetic rubber like -elasticity, tack, and abrasion resistance, their definition and related application Thermal Insulators 6.4 Thermal Insulators -Definition, characteristics of thermal insulators, classification- organic and inorganic thermal insulators, their examples, preparation, properties and applications of polystyrene and Distilled or pure water	10

Reference Materials

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi.
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technica Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media
8. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
9. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
10. Atkins, Physical Chemistry, OUP Oxford.
11. Engineering Chemistry, Wiley India Pvt, Ltd, First edition.

E-Resources:

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-andengineering/chemistry?languages=en&page=2
4. “Introduction to Corrosion” NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq
5. “Tribology &LubricationNPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFqJAltJvWAfEVU4gcUK27hEA7FD)
6. “Electrochemistry” NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)

Course Code: 22IUCC0204H	**Course Name:** Universal Human Values
Credits: 2**Teaching Scheme:** L-2 P-0	**Evaluation Scheme:** ESE–50
Duration of End Sem. Exam: 1 hr.	
Pre-requisites:	
Course Objectives:	

Course Outcomes: At the end of the course, the students will be able to –
CO-1 To create awareness on Engineering Ethics and Human Values.
CO-2 Understand the significance of value inputs in a classroom and start applying them in their life and profession
CO-3 Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body.**C**
O-4 Understand the role of a human being in ensuring harmony in society and nature.
CO-5 Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Contents –

Unit	Content	Teaching hours
1	**HUV Introduction** 1.1 Need, Basic Guidelines Content and Process for Value Education, 1.2 Continuous Happiness and Prosperity, 1.3 Self-Exploration, Continuous Happiness and Prosperity 1.4 Right Understanding, Relationship and Physical Facilities 1.5 Understanding Happiness and Prosperity Correctly.	8
2	**Harmony of Human Being** 2.1 Harmony in Myself 2.2 Understanding Human Being as a co-existence of the Sentient 'I' and the material 'Body' 2.3 Programs to ensure Sanyam and Swasthya.	8
3	**Harmony in Human-Human Relationship** 3.1 Understanding Harmony in the Family 3.2Understanding the meaning of Vishwas, 3.3Understanding the meaning of Samman, 3.4Understanding the Harmony in the Society, 3.5Visualizing a Universal Harmonious order in Society.	8
4	**Harmony in the Nature and Existence** 4.1 Understanding the Harmony in the Nature 4.2 Interconnectedness and Mutual Fulfilment among the Four Orders of Nature 4.3 Understanding Existence as Co-existence.	8
5	**Harmony on Professional Ethics** 5.1 Natural Acceptance of Human Values, 5.2 Definitiveness of Ethical Human Conduct, 5.3 Basis for Humanistic Education, 5.4 Humanistic Constitution and Humanistic Universal Order, 5.5 Competence in Professional Ethics, Case Studies of Typical Holistic Technologies.	8
6	**SELF DEVELOPMENT**	8

	6.1 Goal setting and its importance. 6.2 Characteristics of Goal setting (SMART- Specific, Measurable, Attainable, Realistic, Time bound) 6.3 Time Management - Importance, prioritization of work, time matrix, time savers, and time wasters. 6.4 Stress Management - Definition, types of stress, causes of stress, managing stress, and stress busters.	

Reference Material:

1 Corliss Lamont, Philosophy of Humanism
2.Gaur. R.R. ,Sangal. R, Bagaria. G.P, A Foundation Course in Value Education, Excel Books, 2009.
3.Gaur. R.R. ,Sangal. R ,Bagaria. G.P, Teachers Manual Excel Books, 2009.
4.I.C. Sharma . Ethical Philosophy of India Nagin& co Julundhar
5.Mortimer. J. Adler, – Whatman has made of man
6.William Lilly Introduction to Ethic Allied Publisher

Online Resources:

1. https://www.britishcouncil.in/english/learn-online
2. http://learnenglish.britishcouncil.org/en/content
3. http://www.talkenglish.com
4. Languagelabsystem.com
5. www.wordsworthelt.com

Course Code: 22IUCC0205T	**Course Name:**Computer Workshop
Credits: 2**Teaching Scheme:** L-2 P-0 T-0	**Evaluation Scheme:** Audit
Duration of End Sem Exam:	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Troubleshoot common motherboard problems.	
CO2: Select processors required for relevant systems.	
CO3: Partition/Format hard disk drives.	
CO4: Troubleshot peripherals and networks.	
CO5: Test Power supplies	
CO6: Networking and Maintenance	

Contents:

Unit	Content	Teaching hours
1	**Motherboard** Motherboard: Components Layout Connections Motherboard Types: Types and features Enhancing features of motherboard: Adding and or replacing components. Troubleshooting: problems of motherboard.	4
2	**Processor and BIOS** Processor: Common features, Types, Basic structure of CPU,Different levels of cache, system bus, clock speed, packaging. Multiple core processors: Description, Two core processor architecture and multicourse architecture. Co- Processors: Graphics, math. BIOS: Basic I/O system services, Bios interaction, date and time, Boot device priority, Boot setting configuration, password security.	4
3	**Hard Disk** Hard disk Interfaces:EIDE,Serial ATA,SCSI,USB and IEEE 1394,RAID,Solid state Drive. Disk Structures: Heads, tracks, sectors, cylinders, clusters, landingzone, MBR,Zone bit recording. Disk Performance parameters characteristics: Seeks and latency, Data transfer rate. File system: FAT16,FAT32,NTFS,Unix file system,EXT2/EXT3,RAID	4
4	**I/O and Modem** Troubleshoot I/O devices: Keyboard, Switches, Mouse, Scanners, webcam,monitors,printers,speaker and Mike,LCD projector. I/O cables: Specifications of I/O cables,Types of I/O cables,Types of I/O ports,Internal and external modem, Block diagram and specification. Network Interfaces: Definition of network interface, Types of network interface, troubleshooting of network connectivity, Antivirus.	4

5	**Power supply** Purpose and features of SMPS,Working of SMPS Fault finding in power supply Uninterrupted Power supply: Characteristics of UPS,Types of UPS,online and offline. Preventive Maintenance of power supply.	4
6	**Networking and Maintenance** Share and control resource and Internet connection through network. Implement Network Security to protect from various attacks on networking.MAC Address and its features. Firewall : Technology, implementaion	6

Reference Material:

1. James. K. L. "The Computer hardware installation, interfacing, troubleshooting and maintenance", PHI Learning ,New Delhi,2014 ISBN:978-81-203-4798-4

2. Gupta, Vikas ,"Comdex: Hardware and Networking course kit" Dreamtech Press, New Delhi ISBN:978-93-5119-265-7

3. Minasi,Mark, "The Complete Pc Upgrade and maintenance Guide" BPB Publication, Mumbai, New Delhi ISBN:978-81-265-0627

Online Resources:

1. https://www.tutorialspoint.com/computer_fundamentals/computer_motherboard.htm
2. https://www.javatpoint.com/what-is-a-multicore-processor
3. https://www.geeksforgeeks.org/co-processor-computer-architecture/
4. https://www.digitalocean.com/community/tutorials/an-introduction-to-networking terminology-interfaces-and-protocols
5. https://www.tutorialspoint.com/information_security_cyber_law/network_security.htm
6. https://www.tutorialspoint.com/what-is-a-proxy-server

Course Code: 22IUCC0202L	**Course Name:** Physics-II Lab
Credits: 1 **Teaching scheme:** L-0 P-2 T-0	**Evaluation scheme:** TW–50
Pre-requisites:	

Course Outcomes: At the end of the course, the students will be able to –
CO 1: To learn about the useful solid material in industry & understand its physical properties.
CO 2: To learn about the useful liquid material & understand its physical properties.
CO 3: Understand the concepts of sound wave
CO 4: To learn about the concepts of radioactive and ultrasonic principles.
CO 5: To learn about the concepts of electrostatic & X rays
CO 6: To learn about the concepts of the principles of electricity , magnetism & semiconductors to solve engineering problems.

Contents:

Unit	Content	Teaching hours
1	Use Searle's method to determine the Young's modulus of given wire	2
2	Determine the coefficient of viscosity of given liquid by Stoker's materials	2
3	Find the downward force, along an inclined plane, acting on a roller due to gravity and its relationship with the angle of inclination.	2
4	Determine the refractive index of glass slab	2
5	Determine the refractive index of prism using Pin Method	2
6	Use resonance tube to determine velocity of sound.	2
7	Determine the specific resistance of given wire.	2
8	Use magnetic compass to draw the magnetic lines of forces of magnet of different shapes.	2
9	Use p-n junction diode to draw forward bias and reverse bias I-V characteristics.	2

Course Code:22IUCC0203L	**Course Name:** Chemistry-II Lab
Credits: 1 Teaching scheme: L-0 P-2 T-0	**Evaluation scheme: TW–50**
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to	
CO1: Describe different processes for manufacturing metals and control of their corrosion **CO2:** Use relevant water treatment process to solve industrial problems **CO3:** Justify the selection of lubricants for different types of machines **CO4:** Describe different types of fuels and their properties and uses. **CO5:** Used Relevant engineering materials in Industry	

Contents –

Unit	Content	Teaching hours
1	Prepare the corrosive acidic medium for Aluminum	2
2	Determine the rate of corrosion on different temperatures for Aluminum	2
3	Determine the acid base titration neutralization reaction	2
4	Determine the percentage of iron present in given Hematite ore by KMnO4 solution	2
5	Determine the percentage of copper in given copper ore	2
6	Determine Chloride content in given water sample by Mohr's method.	2
7	Determine total hardness, temporary hardness and permanent hardness of watersample by EDTA method	2
8	Determine the alkalinity of given water sample	2
9	Determine the effect of temperature on viscosity for given lubricating oil using Redwood Viscometer	2
10	Determine the moisture and ash content in given coal sample by using proximate analysis	2
11	Prepare phenol formaldehyde resin used in manufacturing of Bakelite plastic	2
12	Prepare Urea formaldehyde resin and understand the structure and properties for its applications in engineering	2

Course code: 22IUCC0205L	**Course name: Computer Workshop Lab**
Course category: PCC	
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:** TW–50, PR–00	
Pre-requisites: 1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Troubleshoot common motherboard problems.	
LO2: Select processors required for relevant systems.	
LO3: Partition/Format hard disk drives.	
LO4: Troubleshot peripherals and networks.	
LO5: Test Power supplies	
LO6: Networking and Maintenance	

List of Experiments–

Sr.No.	Name of Experiment
1	Identify different hardware components onmotherboard.
2	Troubleshoot common problems of motherboard.
3	Configure BIOS setting.
4	Partition and manage hard disk: Format harddrives with different file systems
5	Install operating system –Windows family
6	Install operating system –Unix family
7	Troubleshoot hard disk problems
8	Set Keyboard, Mouse, Monitor, Speaker, Webcam, Microphone and LCD Projector ect.
9	Install SMPS,measure voltage levels in mainconnectors of SMPS connecting various subsystems.
10	Troubleshoot computer system by diagnosing the problem.
11	Use diagnostic software for fault finding andviruses
12	Undertake preventive maintenance of PC using vacuum cleaner and simple tools.
13	Configure internet connection to the PC usingwireless technology and troubleshoot various connection related problems
14	Install Proxy Server and configure it.
15	Set up basic protection using public keys and MAC address filters.
16	Practice on firewall technologies to secure the network perimeter

Reference Books:

1. B Ram –Computer Fundamental architecture and Organization, New age international Publications
2. DrRajendraKawale ,"Computer Fundamental", Devraj Publications Dist. Solapur

Online Resources: 1. https://www.tutorialspoint.com

Course Code: 22IUCC0204L	**Course Name:** Universal Human Values Lab
Credits: 1 Teaching scheme: P-2	**Evaluation scheme: TW–50**
Course Objectives:	

Course Outcomes: At the end of the course, the students will be able to –
CO-1 To create awareness on Engineering Ethics and Human Values.
CO-2 Understand the significance of value inputs in a classroom
and start applying them in their life and profession
CO-3 Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body.
CO-4 Understand the role of a human being in ensuring harmony in society and nature.
CO-5 Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Contents –

Unit	Content	Teaching hours
1	Introduce yourself in detail	2
2	Explain your ideas of success/progress/development.	2
3	Write down 5 needs of your self and 5 needs of your body	2
4	Explain self- esteem with two examples	2
5	Chalk out programs to ensure that you are responsible to your body- for the nurturing, protection and right utilization of the body.	2
6	Form small groups in the class and in that group initiate dialogue and ask the eight questions related to trust	2
7	Write a note in the form of story, poem, skit, essay, narration, dialogue to educate a child. Evaluate it in a group.	2
8	Choose any one subject being taught today. Evaluate it and suggest suitable modifications to make it appropriate and holistic.	2
9	Explain harmony between body and mind in your own words.	2
10	Suggest some ideas to be a happy human being.	2
11	Collect information on any two great humanitarians.	2
12	Explain importance of goal setting .	2

Course Code:22IUCC0206L	**Course Name:**Workshop practice Lab
Credits: 2 Teaching scheme: P-4	**Evaluation scheme: TW–100**
Pre-requisites: Basic knowledge of computer, Computer Aided Drawing I lab	
Course Outcomes: At the end of the course, the students will be able to –	
CO 1: Select tool and machinery according to job.	
CO 2: Use hand tools in different shops for performing different operation.	
CO 3: Opera equipment and machinery in different shop.	
CO 4: Prepare composite or utility job according to drawing.	
CO5: Maintain workshop tools and equipment's	

Contents –

Unit	Content	Teaching hours
1	Prepare fitting job (male and female type) as per given drawing or job with following operation marking operation punching operation drilling operation cutting operation filling operation fitting operation (male and female assembly) checking correctness of fit of mating parts	8
2	Prepare bill of material along with estimate cost according given drawing of job such as repairing of class room furniture / book shelves / metallic doors / motor sari guard / battery stand with locking device considering the following applicable operation. a) marking operation as per drawing . b)cutting operation as per drawing . c) cleaning operation as per drawing d) edge preparation operation as per drawing e) filling operation as per drawing f) bending operation as per drawing g) welding operation as per drawing	8
3	Prepare steel frame / structure of utility job (like stool, benches, tables, drawing desk, window grill, ventilator, door frame or similar job) involving arc welding joints as per given drawing or job.	8
4	Prepare simple turning job with facing/taper turning/knurling/grooving/threading operation as per drawing	8
5	Prepare sheet metal utility job using following operation - Part I a. Cutting and Bending b. Edging c. End Curling d. Lancing e. Soldering f. Riveting	8

SEM-III

Course code: 22IUCD0301T	**Course name: Engineering Mathematics Course category: PCC**
Credits: 4	**Teaching scheme:** L-3 T-1 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic knowledge of Mathamatics	
Course Outcomes: At the end of the course, the students will be able to -	
CO.1: Find the partial derivatives of functions and apply the concepts of partial differentiation to find maxima and minima of functions containing two variables	
CO.2: Evaluate the double and triple integrals.	
CO.3: Calculate area, volume, surface area using double and Triple integral.	
CO.4: Solve problems on directional derivatives, gradient, Divergence, Curl of vector field.	
CO.5: Solve the problems on fundamental theorems of vector calculus viz. Green's, Stokes and Divergence theorems.	
CO.6: Find the solution of linear differential equations having their applications in mechanical and electrical systems.	

Contents –

Unit	Content	Teaching hours
1	**Partial Differentiation**: Function of several variables, Partial derivatives, Chain rule, Total Derivative, Change of variables, Euler's theorem on Homogeneous function. Level curves. Applications of Partial differentiation: Jacobian, Tangent planes, Errors and Approximations, Differentials. Extreme value problems. Lagrange's multipliers.	**10**
2	**Multiple Integrals:** Double integration in Cartesian &polar coordinates, Evaluation of double integrals by changing the order of integration and changing to polar form, triple integration.	**08**
3	**Applications of Multiple integrals:**To find Area by double integration, Surface Area, Volume of Solid of Revolution, Volume by triple integration.	**08**
4	**Vector differential Calculus**:Vector fields, Gradient and directional derivatives. Independence of path and potential functions. Divergence of vector field, Curl of vector field, vector identities, Applications.	**10**
5	**Vector Integral Calculus:** Line and Surface integrals, Green's theorem, Divergence theorem, Stokes theorem. Applications.	**08**
6	**Differential equations of higher orders: Linear**: Differential equations of second and higher order with constant coefficients, The method of Variation of parameters. Linear differential equations with variable coefficients – Cauchy's Linear Equation and Legendre's Linear Equation, Initial and boundary value problems. Applications of second order differential equations- vibrating of spring, electrical circuits, and Bending of Beam problems.	**08**

Text Books:

1. James Stewart, Calculus Early Transcendental, 7th edition, Cengage.

2. George B. Thomas, Ross L. Finney, Calculus and Analytical Geometry, 9th edition, Pearson.

3. Shanti Narayan, Differential Calculus, S. Chand & Co.

Reference Books:

1. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley.
2. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, NewYork.
3. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.
4. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II), Pune VidyarthiGrihaPrakashan,Pune.
5. H. K. Das and RajnishVerma, Higher Engineering Mathematics, S. Chand & CO. Pvt. Ltd., New Delhi.
6. RudraPratap, Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers, Oxford University Press, 2010.

Online Resources: 1. @ NPTEL lecture on Calculus of several variable.
2. @NPTEL lectures on Multivariable calculus variable.
3. @ Khanacademy lectures on Multivariable calculus variable.

Course code: 22IUCD0302T	**Course name: Introduction to Biology**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Evaluate the Basic Biological Concepts viarel event Industrial Applications and Case Studies	
CO2: Evaluate the Principles of Design and Development for Exploring Novel Bioengineering Projects.	
CO3: Corroborate the Concepts of Biomimetic for Specific Requirements.	
CO4: Think Critically towards Exploring Innovative Biobased Solutions for Socially Relevant Problems.	

Contents –

Unit	Content	Teaching hours
1	**BiomoleculesandtheirApplications** Carbohydrates(CellulosebasedWaterFilters,PHAandPLAasBioplastics) NucleicAcids(DNAVaccineforRabiesandRNAVaccinesforCovidl9,Forensics,DNAFingerprinting) Proteins(ProteinsasFood, WheyProteinand MeatAnalogs, Plantbased Proteins) Lipids(Biodiesel,CleaningAgents/Detergents) Enzymes(Glucose-oxidaseinBiosensors,LignolyticEnzymeinBio-Bleaching)	**10**
2	**HumanOrganSystemsandBioDesigns-I** BrainasaCPUSystem(Architecture,CNSandPeripheralNervousSystem,SignalTransmission,EEG,RoboticArmsforProsthetics,EngineeringSolutionsforParkinson's Disease) EyeasaCameraSystem(ArchitectureofRodandConeCells,OpticalCorrections,Cataract, LensMaterials,BionicEye) HeartasaPumpSystem(Architecture,ElectricalSignalling-ECGMonitoringandHeartrelatedIssues,ReasonsforBlockagesofBloodVessels,DesignofStents,PaceMakers,Defibrillators)	**10**
3	**HumanOrganSystemssndBioDesigns-II** LungsasPurificationSystem (Architecture,GasExchange Mechanisms, Spirometry,AbnormalLungPhysiology- COPD,Ventilators,Heart-LungMachine) KidneyasaFiltrationSystem(Architecture,Mechanism ofFiltration,CKD,Dialysis Systems) Muscular and Skeletal Systems as Scaffolds (Architecture, Mechanisms, Bioengineering solutions for MuscularDystrophyandOsteoporosis)	**10**

4	**TrendsInBioengineering** BioprintingTechniquesandMaterials 3DprintingofEar,Bone,SkinandFoods ElectricalTongueandElectricalNoseinFoodScience DNAOrigamiandBiocomputing Bioimagingand ArtificialIntelligenceforDiseaseDiagnosis Self-HealingBioconcrete(BasedonBacillusSpores,CalciumLactateNutrientsandBiornineralizationProcesses)	**10**

Text Books:

1."BiologyforEngineers"byThyagarajanS., SelvamuruganN.,RajeshM.P.,Nazeer R.A.,ThilagarajW.,BarathiS.,andJaganthanM.K.ofTataMcGraw-Hill, New Delhi,2012.
2. "BiologyforEngineers"bySohiniSinghandTanuAllenof VayuEducationofIndia, New Delhi,2014.

Reference Books:

1."HumanPhysiology"byStuartFox, Krista Rompolski of McGraw-HilleBook, 16th Edition, 2022.
2."BiomedicalInstrumentation"byLeslie CromwellofPrentice Hall2011.
3. BiologyforEngineers"byArthurT. Johnson, CRC Pressof Taylor and Francis, 2011.
4."Biomimetics: Nature-Based Innovation"byYosephBar-CohenofCRC Press1stEdition, 2012.
5."BioInspiredArtificialIntelligence"byTheories,MethodsandTechnologies,D.Floreanoand C.MattiussiofMIT Press, 2008.
6."3DBioprinting: Fundamentals, Principles and Applications"byIbrahimOzbolatofAcademicPress, 2016.
7."Electronic Nosesand TonguesinFood Science"byMariaRodriguezMendeofAcademicPress, 2016.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0303T	**Course name: Computer Programming**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Computer Knowledge	
Course Objective: Develop logical thinking and problem-solving skills to analyse problems and design appropriate algorithms.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Develop flowchart and algorithm to solve problem logically.	
CO2: Write simple C programs using arithmetic expressions	
CO3: Develop C programs using control structures.	
CO4: Develop C programs using arrays and structures.	
CO5: Develop C programs using Pointers.	

Contents –

Unit	Content	Teaching hours
1	**Fundamentals of C Language :** Introduction to C Language, Importance of C Language, Use of C Language, Application of C Language, .Features of C Language	**4**
2	**Overview of C language :** Introduction to Algorithm, Introduction to flowchart, Basic Structure of C Programming, Tokens in C, Keywords, Identifiers, Format Specifies	**8**
3	**C Language Fundamentals :** Character set, Identifiers, Keywords, Operators, Constants, Variables, Data Types, Statements, Decision Making of C language, If Statement, If else Statement, Nested if Statement, if else if ladder ,.Switch case,, Looping, while loop, do while loop, For Loop	**10**
4	**Arrays and String :** Introduction to Array, Array types, Introduction to string, String functions in C.	**8**
5	**Functions & Pointers:** Function: Concept of Function, User defined Function, System Defined Function, and Types of parameter passing in function. **Pointers:** Need of Pointers, Types of Pointers, Pointer Expression, Arrays of Pointers, Pointers and Function.	**10**
6	**Structure and Unions :** Structure in C, Nested structure in C, Array of structure in C, Introduction To Unions, Difference between Structure and Unions.	**5**

Text Books:

1.C in Depth – Deepali Srivastava & S. K. Srivastava
2.Programming in C — Ashok Kamthane & Amit Kamthane

Reference Books:

1. Title of Book: Programming in ANSI C Author: Balagurusamy. E.
2. Title of Book: Let us C Author: Kanetkar, Yashwant

Online Resources:

1. https://www.tutorialspoint.com
2. https://javatpoint.com
3.https://www.w3schools.com

Course code: 22IUCD0304T **Course name: Web Page Designing with HTML** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites:
Course Objectives:
1**.** To understand basic concepts of website design and web page content.
2. To create lists and apply hyperlinks on text and images.
3. To use HTML block-level and text-level formatting tags effectively.
4. To organize web content using tables and frames.
5. To apply CSS for styling and presentation of web pages.
6. To publish websites on Internet or Intranet.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Use of website and content of website designing in web page.
CO2: Use block level and text level formatting tags to present content in web page.
CO3: Arrange list and apply hyperlink on images.
CO4: Organize and arrange content using table and frames.
CO5: Apply presentation schemes on content of web page using CSS.
CO6: Publish website on Internet or Intranet.

Contents –

Unit	Content	Teaching hours
1	**Introduction to Web Design** Introduction of Internet, WWW, Website, Working of Websites, Webpages, Front End, Back End, Client and Server Scripting Languages, Responsive Web Designing, Types of Websites (Static and Dynamic Websites). Editors: Downloading free Editors like Notepad++, Sublime Text Editor, Making use of Editors, File creation and editing, saving. Why to create Web site: use of Website, History & Feature of HTML What is HTML : HTML Documents , Basic structure of an HTML document ,Creating an HTML document	**4**

2	**Introduction to HTML Tag** What is HTML Tag : Use of Tags, Types of Tags, Cascading of tags Block Level tags and Horizontal Rules: Headings. Paragraphs. Line Breaks. Divisions. Text Level Tags And Special Characters: Centered Text, Block Quotes. Preformatted text, types of Address, HR tag Bold Italic. Teletype ,Underline, Strikethrough, Superscript Subscript DIV tag, Displaying special characters	**6**
3	**Lists, URL & Images** Lists: Ordered Lists Unordered Lists. Definition Lists, Nested Lists URL: URL And Anchor Tag:URL Types of URL. Absolute URL, Relative URLS, pros and cons of relative and absolute URLs, Anchor Tag Linking various documents for internal and external links Images, Colors And Backgrounds: Inserting Images, formatting image for sizing, alignment Border and using other attributes with IMG tag Inserting image as page background Creating solid color page background.	**6**
4	**Tables and Frames** Table Table tag with attributes TABLE, TR. TH, TD tags. Border, cell spacing, cell padding, width, align, bgcolor attributes. Frames: Types of Frames with their attributes, Creating frames: FRAMESET tag - rows, cols attributes.	**4**
5	**Cascading Style Sheets** CSS Introduction to CSS, Types of CSS, CSS Selectors: Universal Selector, ID selector, Tag Selector, Class Selector, Sub Selector, Attribute Selector, Group Selector CSS Properties: Back Ground properties, Block Properties, Box properties, List properties, Border Properties, Positioning Properties Data base connectivity, Form designing with database	**6**
6	**Hosting of Website** Concept of Internet and Intranet Publishing website on Intranet. Installing and configuring web server, uploading files on intranet site, access intranet-based website; Publishing website site on Internet, hiring Web space, uploading files using FTP, Virtual Hosting, access internet-based website	**4**
Text Books:		

Reference Books:

1. HTML and XHTML- the complete reference – Powell, Thomas – Tata McGraw Hill, New Delhi,2014, ISBN: 9780070701946
2. HTML and CSS – The Complete Reference -- Tata McGraw Hill, New Delhi

Online Resources:

1. https://www.tutorialspoint.com
2. https://javatpoint.com
3. https://www.w3schools.com

Course code: 22IUCD0305T	**Course name: Basic Electronics**	**Course category:** PCC
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Students should have prior knowledge of: Basic understanding of electrical quantities (voltage, current, resistance), basic circuit concepts, and elementary mathematics.		
Course Objectives:		
1. Introduce the fundamentals of electronic components, semiconductor devices, and signals**.**		
2. Enable understanding of the working principles of diodes, transistors, and amplifiers.		
3. Develop the ability to analyze basic electronic circuits and their applications.		
4. Familiarize students with sensors and transducers used in electronic systems.		
5. Build a foundation for advanced courses in digital electronics, communication and embedded systems.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Identify Electronics componants in Electronics Group.		
CO2: Apply Electronics componants safety		
CO3: Use Relevant Electrical or Electronic Protective Device Safety.		
CO4: Use Various Sensors and Transducsers Curriculum		

Contents –

Unit	Content	Teaching hours
1	**Electronics Componants & Siganls** A brief history of Electronics, Defination of Semiconductor , Types of Semiconductor Active and Passive Componants Resistor , Inductor , Capacitor Symbol , Colour Code , and Specification Voltage & Current sources, Ideal and Practical Voltage Source and Current Source Signal , Types of Signal (Analog and Digital)	**6**
2	**Diode Circuits** P N Junction Diode : Symbol , Construction, Working and Applications Zener Diode : Symbol, Working ,appliaction , Zener Diode as A Voltage Regulator Rectifier Circuit : Half Wave Rectifier , Full Wave Rectifier, Bridge Wave Rectifier and Related performance parameter Diffeence between HWR & FWR Introduction to Filter : Working of Inductor (L) & Capacitor (C) Filter , Application of Various Filter.	**6**
3	**Bipolar Junction Transistor and Amplifier** BJT : Symbol Construction and Working , N-P-N & P-N-P Types, Transistor as A Switch and Amplifier Operating Region of Transistor , Input and Output Charactristics of Transistor. Amplifier**:** Voltage Amplifier, Current Amplifier and Power Amplifier its Application. Introduction to OP-AMP (IC741)	**6**

4	**Sensors And Transducers** Defination ,Working and Applications of Sensors. Selection Criterian For Sensor & Transducers . Defination ,Working and Applications of Transducers Types Of Transducers (Inductive, Capacitive, Restive and Piezo Electric) , Active and Passive Transducers. Advantages and Disadvantages of Sensor and Transducers.	**8**

Text Books: 1. "Principle of Electronics" by V. K. Mehta. 2. "Applied Electronics" by R. S. Sedha of S Chand Publication. 3. "Digital Electronics" by J.S. Katre of Tech-Max Publication, Pun**e.**
Reference Books: 1. "Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0306T **Course name: Effective Communication and Soft skills**
Course category: PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–20, ESE–00
Pre-requisites: Basic knowledge of English
Course Outcomes: At the end of the course, the students will be able to -
CO1: Student will be able to utilizeeffectivecommunication skills.
CO2: Student will be able to select and apply the appropriate methods of communication in various situations.
CO3: Student will be able to develop presentation skills and body language.
CO4: Student will be able touse appropriate words as per the context
CO5: Student will be able todraft correct format of emails and business letters

Contents –

Unit	Content	Teaching hours
1	**Introduction to communication** Definition of communication Process of communication Types of communication (Formal,Informal,Verbal,Nonverbal,Vertical, Horizontal, Diagonal) Principles of communication- seven C's of communication Barriers to communication-Physical Barrier, Mechanical, Psychological, Language.	**06**
2	**Non verbal communication** Introduction to nonverbal communication Aspects of body language (Kinesics) (**Facial expression, eye contact, vocalic, paralanguage,gesture,posture, dress & appearance, haptics**) Non-verbal codes- Proxemics, Chronemics, Artefacts Body language and confidence Use of body language at workplace.	**06**
3	Formal Written Communication Office Drafting- Notice, memo & e-mail. Job application with resume. Business correspondence-(Enquiry letter, order letter ,complaint letter, adjustment letter.) Report writing- Accident report, fall in production, investigation report.	**06**

4	Presentation skills What is Presentation Importanceand types of Presentation. Elements of effective presentation. Tips for effective Presentation	**06**
5	**Listening skills:** Importance of listening skill, Listening Process. Active and Passive listening Barriers to listening Strategies of effective listening.	**04**
6	**Soft skills:** Public Speaking, Interview skills. Group discussion Team work and interpersonal skills	**04**

Text Books:
Reference Books: 1. Text book of Communication Skills. (Sanjay Kumar, PushpaLata) Oxford University Press. 2.Communication Skills (URMILA RAI) Himalaya Publications 3. English for success (E.Sureshkumar, P. Sreehari, J.Savithri) Foundation Books. 4.Effective communication skills (M.AshrafRizvi)
Online Resources: 1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills 2. https://www.skillsyouneed.com/ips/communication-skills.html 3.https://corporatefinanceinstitute.com/resources/management/communication/ 4. https://www.coursera.org/articles/communication-skills

Course code: ENV-C **Course category:** PCC	**Course name: Environmental Studies**
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic knowledge of Environmental	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the need of Environment Study	
CO2: Analyze the use of resources	
CO3: Understand the concept of an ecosystem	
CO4: Understand the concept ofBiodiversity and its conservation	
CO5: Analysis the concept of Pollution	
CO6: Analysis is the Social Issues and the Environment	

Contents –

Unit	Content	Teaching hours
1	**The multidisciplinary nature of environmental studies** Definition, scope and importance Need for public awareness	2
2	**Natural Resources:** Natural resources and associated problems. (a) Forest resources: Use and over-exploitation, deforestation, case studies. Timber extraction, mining, dams and their effects on forests and tribal people. (b) Water resources: Use and over-utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems. (c) Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. (d) Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. (e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources, case studies. (f) Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification. · Role of an individual in conservation of natural resources. · Equitable use of resources for sustainable lifestyles.	8

3	**Ecosystems** Concept of an ecosystem · Structure and function of an ecosystem · Producers, consumers and decomposers · Energy flow in the ecosystem · Ecological succession · Food chains, food webs and ecological pyramids · Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, ocean estuaries)	**6**
4	**Biodiversity and its conservation** Introduction – Definition: genetic, species and ecosystem diversity · Biogeographical classification of India · Value of biodiversity: consumptive use, productive use, social, ethical aesthetic and option values · Biodiversity at global, national and local levels · India as a mega-diversity nation · Hot-spots of biodiversity · Threats to biodiversity: habitat loss, poaching of wildlife, man wildlife conflicts · Endangered and endemic species of India · Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity	
5	**Environmental Pollution** Definition • Causes, effects and control measures of: a. Air pollution b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear pollution • Solid waste management: Causes, effects and control measures of urban and industrial wastes. • Role of an individual in prevention of pollution • Pollution case studies • Disaster management: floods, earthquake, cyclone and landslides	**8**

6	**Social Issues and the Environment** From unsustainable to sustainable development • Urban problems and related to energy •Water conservation, rain water harvesting, watershed management • Resettlement and rehabilitation of people; its problems and concerns. Case studies. • Environmental ethics: Issues and possible solutions • Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case studies. • Wasteland reclamation • Consumerism and waste products • Environmental Protection Act • Air (Prevention and Control of Pollution) Act • Water (Prevention and control of Pollution) Act • Wildlife Protection Act • Forest Conservation Act • Issues involved in enforcement of environmental legislation • Public awareness	

Text Books:
Reference Books: 1.Basic Environmental Sciences- Michael Allaby 2. Environmental Sciences- Y.K. Singh 3. Environmental Studies- Rajgopalan.
Online Resources: www.cpcb.nic.in

Course code: 22IUCD0303L	**Course name: Computer Programming Lab**
Course category: PCC	
Credits: 02 **Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites: Basic knowledge of Computer, Number System	
Course Outcome: Students will be able to design, write, compile, and debug basic C programs using fundamental programming constructs	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Develop Flowchart and Algorithm to solve problems logically.	
LO2: Apply concept of operators and expression	
LO3: Develop C Programs using control Structures.	
LO4: Demonstrate concept of functions	
LO5: Implement concept of pointer	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	To understand the concept of Algorithm and Flowchart.	2
2	Develop C Program using operators.	2
3	Develop program using increment/decrement operator.	2
4	Develop C Program to print ASCII value of character.	2
5	Develop C program to find whether the entered number is positive negative or zero using Else if	2
6	Write a program to find whether the given year is leap year or not.	2
7	C Program to Find the greatest of 3 numbers.	2
8	Develop C program to print Half Pyramid of Number.	2
9	Using switch statement print day of week by taking number from 1 to 7.	2
10	Write a program to find sum of digits of given number.	2
11	Develop a C Program using While loop	2
12	Prepare a program using concept of structure.	2
13	Develop a program using the concepts of pointers.	2
14	Write a program to Print student grade by accepting student's percentage.	2
15	Develop a program to sort 10 numbers	2
16	Program to find largest element in array	2
17	C Program to calculate average of all elements present in array	2
18	C Program to find length of String	2

Text Books: 1.C in Depth – Deepali Srivastava & S. K. Srivastava 2.Programming in C — Ashok Kamthane & Amit Kamthane
Reference Books: 1. Title of Book: Programming in ANSI C Author: Balagurusamy. E. 2. Title of Book: Let us C Author: Kanetkar, Yashwant
Online Resources: 1.https://www.tutorialspoint.com 2. https://javatpoint.com 3. https://www.w3schools.com

Course code: 22IUCD0304L	**Course name: Web Page Designing with HTML Lab**
Course category: PCC	
Credits: 02 **Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites:1 Basic of HTML 2. Different HTML Tags 3.Web Page Designing	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Use of website and content of website designing in web page.	
LO2: Use block level and text level formatting tags to present content in web page.	
LO3: Arrange list and apply hyperlink on images.	
LO4: Organize and arrange content using table and frames.	
LO5: Apply presentation schemes on content of web page using CSS.	
LO6: Publish website on Internet or Intranet	

List of Experiments–

Sr.No.	Name of Experiment	Practial Hours
1	Install any one editor and display hello message on screen.	2
2	Create a web page using Block Level tags and HR tags.	2
3	Create a web page using Text Level tags and display special character in web page.	2
4	Create a web page for implementation of different types of list.	2
5	Create a web page for inserting image in web page and change attribute of image.	2
6	Create a web page to link: a) Different web page of same web site b) Different web page of different web site c) Images	2
7	Create a web page for implementation of table tags	2
8	Create a web page for implementation of frame tags	2
9	Create a web page by implementation of different selector	2
10	Create a web page by using different CSS properties	2
11	Create a web page to create different forms: a) Login form b) Student Entry From c) Forms with linking of different tags	2
12	Create above web page and connect with database	2
13	Create a web page for multimedia a) Audio b) Viedo	2

	c) Source	
14	Install a web server for publishing website	2
15	Publish a web site on internet by using free hosting site.	2
16	Mini Project	2

Reference Books:

1. HTML and XHTML- the complete reference – Powell, Thomas – Tata McGraw Hill, New Delhi, 2014, ISBN: 9780070701946

2.HTML and CSS – The Complete Reference - Tata McGraw Hill, New Delhi

Online Resources:

1.https://www.tutorialspoint.com

2.https://javatpoint.com

3.https://www.w3schools.com

Course code: 22IUCD0305L	**Course name: Basic Electronics Lab**
Course category: PCC	
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Students should have prior knowledge of: Basic understanding of electrical quantities (voltage, current, resistance), basic circuit concepts, and elementary mathematics.	
Course Objectives:	
1. Introduce the fundamentals of electronic components, semiconductor devices, and signals**.**	
2. Enable understanding of the working principles of diodes, transistors, and amplifiers.	
3. Develop the ability to analyze basic electronic circuits and their applications.	
4. Familiarize students with sensors and transducers used in electronic systems.	
5. Build a foundation for advanced courses in digital electronics, communication and embedded systems.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Test Various Active and Passive Componants.	
LO2: Build Simple and Functional Circuits on Breadboard.	
LO3: Build Transistor and Amplifier Circuit.	
LO4: Use Various Sensors And Transducsers.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Introduction to Various Equipment in the Laboratory	2
2	Measure Voltage And Frequency With Cathod Ray Oscilloscope (CRO)	2
3	Use of Analog and Digital Multimeter to measure Voltage and Current.	2
4	Identify Various Active and Passive componants in given circuit.	2
5	Use of Multimeter to measure value of Various componants.	2
6	Use of LCR-Q Meter to measure value of Inductor (L) and Capacotor .	2
7	Test P N Junction Diode using DMM (Digital Multi Meter)	2
8	Test Performane of N-P-N Transistor.	2
9	Identify Three Terminals of Transistor using Digital Multi Meter.	2
10	Build Half Wave Rectifier on Bread Board.	2

Text Books:

1.”Basic Electronics “by Dr. V K Mehta Books.
2."Digital Electronics and Logic Design" by Dr. Adsul A.P. and Jagdale R.H. of Nirali Publication.
3."Digital Electronics" by J.S. Katre of Tech-Max Publication, Pune.

Reference Books:

1. "Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.

2. www. Electrical & Electronics for u 3. www.Electronics-Tutorials 4. www.wikipedia.com
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0306L **Course name: Effective Communication and Soft skills Lab**
Course category: PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–00
Pre-requisites: Basic knowledge of English
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Use of website and content of website designing in web page.
LO2: Use block level and text level formatting tags to present content in web page.
LO3: Arrange list and apply hyperlink on images.
LO4: Organize and arrange content using table and frames.
LO5: Apply presentation schemes on content of web page using CSS.
LO6: Publish website on Internet or Intranet

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Explain the importance of the business communication.	2
2	Explain the importance of body language at workplace.	2
3	Case study on nonverbal and verbal communication	2
4	Prepare guidelines for presentation.	2
5	Design power point presentation on any topic.	2
6	Prepare interview questionnaire.	2
7	Organize debate on given topic.	2
8	Deliver speech on the given topic. (Welcome speech and farewell speech)	2
9	Prepare job application with resume.	2
10	Draft a complaint letter on the given topic.	2

Reference Books:

1. Text book of Communication Skills. (Sanjay Kumar, PushpaLata) Oxford University Press.
2. Communication Skills (URMILA RAI) Himalaya Publications
3. English for success (E.Sureshkumar, P. Sreehari, J.Savithri) Foundation Books.

Online Resources:

1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills
2. https://www.skillsyouneed.com/ips/communication-skills.html
3.https://corporatefinanceinstitute.com/resources/management/communication/
4. https://www.coursera.org/articles/communication-skills

SEM-IV

Course code: 22IUCD0401T	**Course name: Discrete Mathematics**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3 T-1	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Mathematics	
Course Objectives:	
1. To develop a strong foundation in logical reasoning and mathematical thinking	
2. To develop a clear understanding of relations functions, recurrence relations their properties and types.	
3.To develop an understanding of number system	
4. To develop the ability to apply fundamental counting techniquesto solve combinatorial and mathematical problems effectively.	
5. To develop the ability to understand and apply fundamental concepts of graphs and trees.	
6. To develop understanding of algebraic system to solve mathematical problems.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: apply principles of logic, set theory, and proof techniques, to solve problems systematically.	
CO2 : analyze mathematical structures by applying concepts of relations, functions, and recurrence relations	
CO3 : apply number theory concepts to solve mathematical and computational problems**.**	
CO4 : solve combinatorial and probability-based problems by applying counting techniques	
CO5 : analyze and apply graph and tree conceptsto solve problems and implement algorithms	
CO6 : apply the concepts of algebraic systems to solve mathematical	

Contents –

Unit	Content	Teaching hours
1	**Set Theory, Logic and Proofs** Set Theory, Logic and Proofs: Propositions, Conditional Propositions, Logical Connectivity, Propositional calculus, Universal and Existential Quantifiers, First order logic, Proofs: Proof Techniques, MathematicalInduction. Set, Combination of sets, Finite and Infinite sets, Un- countably infinite sets, Principle of inclusion and exclusion, strong Induction	**10**
2	**Relations, Functions, Recurrence Relations** Definitions, Properties of Binary Relations, Equivalence Relations and partitions, Partial ordering relations and lattices, Chains and Anti chains. Theorem on chain, Warshall's Algorithm & transitive closure, Recurrence relations. Functions: Definition, Domain, Range, Image, etc. Types of functions: Surjection, Injection, Bijection, Inverse, Identity, Composition of Functions.	**10**

3	**Number Theory** Basics of Modulo Arithmetic, Basic Prime Number Theory, GCD, LCM, Divisibility, Euclid's algorithm, Factorization, Chinese Remainder Theorem Fields: Naturals, Integers, Rationales, Real's, Complex Numbers Properties of operations: associative, commutative, distributive, identity, inverse	**10**
4	**Counting** Basic Counting Techniques (sum, product, subtraction, division, exponent), Pigeonhole and Generalized Pigeonhole Principle with many examples, Permutations and Combinations and numerical problems ,Binomial Coefficients Pascal's, Identity and Triangle, Generating Permutations and Combinations.	**10**
5	**Graphs & Trees** Basic terminology, multi graphs and weighted graphs, paths and circuits, Shortest path Problems, Euler and Hamiltonian paths and circuits, factors of a graph, planar graph and Kuratowski's graph and theorem, independent sets, graph coloring. Trees, rooted trees, path length in rooted trees, binary search trees, spanning trees and cut set, theorems on spanning trees, cut sets ,circuits, minimal spanning trees, Kruskal's and Prim's algorithms for minimal spanning tree.	**12**
6	**Algebraic Systems** Algebraic Systems,Groups, Semi Groups, Monoids, Subgroups, PermutationGroups, Codes and Group codes, Isomorphism and Automorphisms, Homomorphism and Normal Subgroups, Ring, Field.	**8**

Text Books:

1. Schaum's Outlines Series, Seymour Lipschutz, MarcLipson,
2. C. L. Liu and D P Mohapatra, "Elements of Discrete Mathematics A Computer Oriented Approach," 3rd Edition by, Tata McGraw –Hill.

Reference Books:

1. Kenneth H. Rosen, "Discrete Mathematics and Its application", Seventh Edition by Tata McGraw –Hill.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0402T Course Name: Object Oriented Programming using C++
Course category: PCC
Credits: 3 **Teaching scheme:** L-3 P-0 T-0 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basics of programming concept, Basics of DS, Fundamental of Procedure Oriented, Basics of DMS
Course Objectives:
1. To develop a foundational understanding of object-oriented programming concepts using C++
2. To develop the understanding for designing and implementation of classes and objects
3. To develop the ability to understand reusability concept using inheritance in OOP
4. To develop the ability to use pointers and implement polymorphism for designing efficient object-oriented programs.
5. To develop the ability to perform file operations in C++
6. To develop the ability to handle runtime errors using exception handling techniques
Course Outcomes: At the end of the course, the students will be able to -
CO1: understand and apply object-oriented programming concepts using C++ fundamentals
CO2: design, implement, and manipulate classes and objects
CO3: analyse, implement inheritance to design reusable and hierarchical object-oriented programs.
CO4: apply pointers and implement polymorphism to design efficient object-oriented programs.
CO5: use C++ stream classes to open, read, write, and manage files efficiently in object-oriented programs.
CO6: apply exception handling concept to detect, manage, and recover from runtime errors effectively.

Contents –

Unit	Content	Teaching hours
1	**Principles of Object oriented Programming :** Procedure Oriented Programming verses Object Oriented Programming (OOP), basic Concept of OOP, Application of OOP,C verses C++, Structure of C++ Program, Tokens, Keywords, variables, Constants, Built in data Types, User defined data Types, Typecasting, Operators, Expressions, Control Structures: Decision making statements and Loops, scope resolution operator, memory management operator, Array, Strings and Structures in C++,	**6**
2	**Classes and Object :** Class , Object, access specifies, Defining member Functions, Creating Objects, memory allocation for Objects, Static data members & function , friend function, Array of Objects, Object as function arguments, Concepts of Constructors. Types of constructors, Destructors	**8**

3	**Inheritance :** Introduction to Inheritance, defining derived class, visibility modes and effects, Types of Inheritance: Single ,multilevel, multiple, hierarchical, hybrid. Virtual base class, abstract class. Constructors in derived class. Function overriding.	**8**
4	**Polymorphism & Pointers :** Introduction of Polymorphism, Types of Polymorphism, Compile time Polymorphism Function overloading. Operator overloading, overloading of unary and binary operators, Rules for operator overloading. Run time polymorphism, Virtual functions, rules for virtual functions, pure virtual function. Concepts of Pointer: Pointer Declaration, Printer operator, address operator, Pointer arithmetic, Pointer to Object: Pointer to Object. this pointer, Pointer to derived class	**8**
5	**Working with Files :** C++ stream classes, Classes for file stream operations, Opening files, closing files reading from and writing to files, Detection of end of file, file modes	**8**
6	**Exception Handling and Templates :** Exception Handling: Fundamentals, other error handling techniques, simple exception handling- Divide by Zero, Multiple catching, re-throwing an exception, exception specifications, user defined exceptions, processing unexpected exceptions, constructor, destructor and exception handling, exception and inheritance, Templates: The Power of Templates, Function template, overloading Function templates, and class template, class template and Non type parameters, template and friends Generic Functions, The type name and export keywords.	**7**

Text Books: 1. Deitel, "C++ How to Program", 4th Edition, Pearson Education, ISBN:81-297-0276-
2. Robert Lafore, "Object-Oriented Programming in C++", fourth edition, Sams Publishing, ISBN: 0672323087 (ISBN 13: 9780672323089

Reference Books: 1. Herbert Schildt, "C++-The complete reference", Eighth Edition, McGraw Hill Professional, 2011, ISBN:978-00-72226805
2. Matt Weisfeld, "The Object-Oriented Thought Process", Third Edition Pearson ISBN-13:075- 2063330166
3. E.Balagurusamy, "Object-Oriented Programming with C++", 7th edition, Graw-Hill Publication, ISBN 10: 9352607996 ISBN 13: 9789352607990
4. Cox Brad, Andrew J. Novobilski, "Object-Oriented Programming: An Evolutionary Approach", Second Edition, Addison-Wesley, ISBN: 13:978-020-1548341

Online Resources:
1.https://nptel.ac.in/courses/106/105/106105151/
2. https://swayam.gov.in/nd1 noc20 cs07/preview
3. https://www.classcentral.com/course/swayam-programming-in-c-6704.

Course code: 22IUCD0403T **Course name: Computer Graphics** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Programming Concepts, Computer Fundamentals
Course Objectives: Introduce the fundamental concepts of computer graphics, including graphics systems, display devices, and graphics software.
Understand mathematical foundations required for computer graphics such as 2D and 3D coordinate systems, vectors, and matrix transformations.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Manipulate visual and geometric information of images.
CO2: Implement standard algorithms to draw various graphics objects using C program
CO3: Develop programs for 2D and 3D Transformation
CO4: Implement various clipping algorithm
CO5: Develop programs to create curves.

Contents –

Unit	Content	Teaching hours
1	**Basics of computer Graphics** Images and Objects, pixel and resolution, Text mode, Graphics mode, Basic graphic pipeline. Bitmap and vector based graphics, Application of computer graphics. Display devices: Raster scan displays, random scan display, flat panel display, LED, LCD display, plasma, touch screen. Output Primitives Graphics functions and standards. Latest trends in computer Graphics.	**8**
2	**Raster Scan graphics** Basic concept in line drawing Circle generating Algorithm Polygons-types of Polygon Scan conversions, Frame Buffer Character generation methods.	**6**
3	**Overview of transformations** Two Dimensional Transformations: Translation, Scaling, Rotation. Reflection, Shearing Matrix representations and homogeneous coordinates: Translation Scaling, Rotation, Reflection, shearing. Composite Transformations-Rotation about arbitary point. Three Dimensional transformation: Translation, Scaling, Rotation Types of Projections: Perspective and Parallel Projection	**8**

4	**Windowing and clipping** Windowing and clipping concepts: Window to viewport transformation Line clipping: Cohen Sutherland clipping algorithm. Cyrus Beck. Liang Barsky, Midpoint subdivision Polygon clipping: Sutherland –Hodgman Text clipping	**8**
5	**Introduction to curves** Curve generation: Arc generation using DDA algorithm, Interpolation Types of curves: Hilbert's Curve, Koch curve, B-Spline, Bezier curves.	**8**
6	**Animation and virtual reality** Basic Principles of Animation and Types of Animation Introduction to the flash interface Layers & Views Animation -Principles , Frame by frame animation, tweening, masks Introduction to virtual reality	**7**

Text Books: Computer Graphics: Donald Hearn, Baker M. Pauline Computer Graphics: Mourya Rajesh K.
Reference Books: Computer Graphics: Dr.Chopra Rajiv
Online Resources: **1.** https://www.tutorialspoint.com 2. https://javatpoint.com 3. https://www.w3schools.com

Course code: 22IUCD0404T **Course name: Digital Electronics**
Course category: PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Students are expected to have prior knowledge of: Basic understanding of electrical quantities, basic electronic devices and common electronic components with their circuit symbols.
Course Objectives: The objectives of this course are to
1. Understand the fundamental concepts of digital systems and number systems used in modern electronic circuits**.**
2. Analyze and design combinational and sequential logic circuits using logic gates and Boolean algebra.
3. Apply the principles of logic simplification techniques such as Boolean laws and Karnaugh maps.
4. Understand the working of flip-flops, counters, and registers used in digital systems.
5. Gain knowledge of data converters, multiplexers, demultiplexers, encoders, and decoders.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand Basics of Digital System.
CO2: Use Number System and Codes for Problem Solving of Digital Electronic System.
CO3: Use for Logic Gates and Families to Construct Logic Circuits.
CO4: Build Simple and Functional Combinational Logic Circuits.
CO5: Build Simple and Functional Sequential Logic Circuits.
CO6: Test Data Convertors and PLD's in Digital Electronics Systems.

Contents –

Unit	Content	Teaching hours
1	**Introduction to Digital Techniques** Signal - Overview of Analog and Digital Signal System - Defination, Characteristics, Classification of Various Systems Digital Systems - Positive and Negative Logic, Representation, Characteristics, Advantages, Disadvantages and Applications of Digital Systems, Digital Computers	**6**
2	**Number System and Code** Number System - Classification - Binary, Octal, Decimal, Hexadecimal Number System, Conversion of Number Systems, 1's Complement and 2's Complement Binary Arithmetic - Addition, Subtraction, Multiplication, Division Codes - BCD Code, BCD Addition, BCD Subtraction using 9's and 10's complement, Gray Code, Excess-3, ASCII Code, Hamming Code	**6**

3	**Logic Gates and Families** Basic Gates, Universal Gates, Derived Gates - Logical Symbol, Logical Expression and Truth Table of Gates- Deriving all gates using Universal Gates, Multiple Input Gates (3 - Inputs**)** **Boolean Algebra: Fundamentals of Boolean Laws** **Duality Theorem, De Morgan's Theorem** Logic Families - Characteristics of Logic Families, Noise Margin, Power Dissipation, Figure of Merit, Fan In, Fan Out, Speed of Operation, Operating Temperatures, Power Supply Requirement& Comparison between Different Logic Families, Overview	**6**
4	**Combinational Logic Circuits** Standard Boolean Representation - Sum of Product (SOP) and Product of Sum (POS), Min-Term and Max-Term, Conversion between SOP and POS Forms, Realization using NAND / NOR Gates K-Map Reduction Technique for the Boolean Expression - Minimization of Boolean Functions up to 4 Variables (SOP and POS Form) Design of Arithmetic Circuits and Code Converter using K-Map - Half and Full Adder, Half and Full Subtractor, Gray to Binary and Binary to Gray (up to 4 bits) Arithmetic Circuits - (IC 7483) Adder and Subtractor, BCD Adder Encoder / Decoder: Basics of Encoder, Decoder, Comparison, (IC 7447) BCD to 7 Segment Decoder / Driver Multiplexer and Demultiplexer - Working , Truth Table and Applications of Multiplexers and Demultiplexures, MUX Tree, IC 74151 as MUX; DEMUX Tree, DEMUX as Decoder, IC 74155 as DEMUX Buffer - Tristate Logic, Unidirectional and Bidirectional Buffer (IC741,S244,74LS245)	**8**

5	**Sequential Logic Circuits** Basic Memory Cell - RS Latch using NAND and NOR Triggering Methods - Edge Trigger and Level Trigger SR Flip Flops - SR Flip Flop, Clocked SR Flip Flop with Preset and Clear, Drawbacks of SR Flip Flop JR Flip Flops - Clocked JK Flip Flop with Preset and Clear, Race Around Condition in JK Flip Flop, Master Slave JK Flip Flop, D and T Type Flip Flop Excitation Table of Flip Flops, Block Schematic and Function Table of IC-7474, 7475 Shift Register - Logic Diagram of 4 Bit Shift Registers, Serial Input Serial Output, Serial Input Parallel Output, Parallel Input Serial Output, Parallel Input Parallel Output, 4 Bit Universal Shift Register Counters - Asynchronous Counter: 4 Bit Ripple Counter, 4 Bit Up / Down Counter, Modulus of Counter, Synchronous Counter: Design of 4 Bit Synchronous Up / Down Counter, Decade Counter: Block Schematic of IC 7490 Decade Counter, IC 7490 as MOD-N Counter, Ring Counter, Twisted Ring Counter	**8**
6	**Data Convertors and PLD's** Data Converter - DAC: Types, Weighted Resistor Circuit and R-2R Ladder Circuit, DAC IC 0808 Specifications, ADC: Block Diagram, Types and Working of Dual Slope ADC, SAR ADC, ADC IC 0808 / 0809, Specification Memory - RAM and ROM Basic Building Blocks, Read and Write Operation ,Types of Semiconductor Memories PLD: Basic Building Blocks and Types of PLDs, PLA, PAL, GAL CPLD: Basic Building Blocks, Functionality	**6**

Text Books:

1. "Digital Electronics and Logic Design" by Dr. Sanjay Sharma of KATSON Books.
2. "Digital Electronics and Logic Design" by Dr. Adsul A.P. and Jagdale R.H. of Nirali Publication.
3. "Digital Electronics" by J.S. Katre of Tech-Max Publication, Pune.

Reference Books:

1."Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.
2. "Digital Circuits and Design" by Salivahan S. and Arivazhagan S. of Vikas Publication House, Delhi.
3. "Digital Principles" by Malvino A.P. and Leach D.P. of Tata McGraw Hill Publication, New Delhi.
4.”Digital Electronics, Principles and Integrated Circuits" by Anil K. Maini of Wiely India Edition.
5. "Digital Fundamentals" by Thomas Floyd of Pearson Publication

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0402 Course name: Object Oriented Programming using C++ Lab
Course category: PCC
Credits: 02 **Teaching scheme:** P-4 **Evaluation scheme:** TW–60, PR–40
Pre-requisites: **Programming Logic, C**
Course Outcome: Students will be able to implement object-oriented concepts in C++ for real world entity.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Develop to solve problem using Procedure Oriented Approach.
LO2: Implement concept of classes and Objects for real world entity
LO3: Demonstrate concept of inheritance by sketching programs
LO4: Implement concept of polymorphism
LO5: Demonstrate file handling concept.

List of Experiments–

Sr.No.	Name of Experiment	Practial Hourse
1	Develop minimum Two Programs using constants, variables, arithmetic expression, operators, exhibiting data type conversion.	2
2	Develop a program to implement decision making statements(if-else, switch)	2
3	Develop program to implement to demonstrate control structures(for, while, do-while)	2
4	Develop program to implement 1-Dimensional array	2
5	Develop program to perform matrix operations using multi-Dimensional array	2
6	Develop programs that implements a class and use it with Objects	2
7	Develop programs that implements a class and create array of Objects.	2
8	Write a Program to implement friend Function.	2
9	Write a Program to implement inline Function	2
10	Write a Program to implement all types of constructors (constructor overloading) with destructor	2
11	Write a Program for implementing single inheritance.	2
12	Write a Program for implementing Multi-level inheritance.	2
13	Write a Program for implementing multiple inheritances.	2
14	Develop minimum 1 Program to Demonstrate pointer to Object.	2
15	Develop minimum 1 Program to Demonstrate pointer to Derived Class.	2
16	Write a Program to Demonstrate operator overloading for Unary operator	2
17	Write a Program to Demonstrate operator overloading for Binary operator.	2
18	Write a Program to Demonstrate function Overloading.	2
19	Write a Program to read and write data to and from file	2

Text Books: 1. Deitel, "C++ How to Program", 4th Edition, Pearson Education, ISBN:81-297-0276-2 2. Robert Lafore, "Object-Oriented Programming in C++", fourth edition, Sams Publishing, ISBN:0672323087 (ISBN 13: 9780672323089
Reference Books: 1. Herbert Schildt, "C++-The complete reference", Eighth Edition, McGraw Hill Professional, 2011, ISBN:978-00-72226805 2. Matt Weisfeld, "The Object-Oriented Thought Process", Third Edition Pearson ISBN-13:075- 2063330166 3. E.Balagurusamy, "Object-Oriented Programming with C++", 7th edition, Graw-Hill Publication, ISBN 10: 9352607996 ISBN 13: 9789352607990 4. Cox Brad, Andrew J. Novobilski, "Object-Oriented Programming: An Evolutionary Approach", Second Edition, Addison-Wesley, ISBN:13:978-020-1548341
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0403L		**Course name: Computer Graphics Lab**
Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Manipulate visual and geometric information of images.		
LO2: Implement standard algorithms to draw various graphics objects using C program		
LO3: Develop programs for 2D and 3D Transformation		
LO4: Implement various clipping algorithm		
LO5: Develop programs to create curves		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Write a program to draw following graphics object using build in c function: i)pixel ii) Lines iii)circle iv)Rectangle v) Ellipse	2
2	Implement DDA algorithm to draw line.	2
3	Implement Bresennham's algorithm to draw line.	2
4	Writhe a program to fill polygon using Flood fill method	2
5	Writhe a program to fill polygon using Boundary fill method	2
6	Write a program in C to perform two dimensional transformation i)Translation ii)Scaling	2
7	Write a program in C to perform two dimensional transformation(Rotation)	2
8	Write a program to clip line using Cohen-Sutherland algorithm.	2
9	Write a program to draw Hilbert's Curve	2
10	Write a program to draw Koch curve and Bezier curves.	2

Text Books:
1. Computer Graphics: Donald Hearn, Baker M. Pauline
2. Computer Graphics: Mourya Rajesh K.

Reference Books:
1.Computer Graphics: Dr.Chopra Rajiv

Online Resources:
1. NPTEL / SWAYAM lectures.

Course code: 22IUCD0404L **Course name: Digital Electronics Lab** **Course category:** PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites: Students are expected to have prior knowledge of: Basic understanding of electrical quantities, basic electronic devices and common electronic components with their circuit symbols.
Course Objectives: The objectives of this course are to
1. Understand the fundamental concepts of digital systems and number systems used in modern electronic circuits.
2. Analyze and design combinational and sequential logic circuits using logic gates and Boolean algebra.
3. Apply the principles of logic simplification techniques such as Boolean laws and Karnaugh maps**.**
4. Understand the working of flip-flops, counters, and registers used in digital systems.
5. Gain knowledge of data converters, multiplexers, demultiplexers, encoders, and decoders.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Understand Basics of Digital System.
LO2: Use Number System and Codes for Problem Solving of Digital Electronic System.
LO3: Use for Logic Gates and Families to Construct Logic Circuits.
LO4: Build Simple and Functional Combinational Logic Circuits.
LO5: Build Simple and Functional Sequential Logic Circuits.
LO6: Test Data Convertors and PLD's in Digital Electronics Systems.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Introduction to Digital Techniques and Its Applications.	2
2	Test the Functionality of Basic Logic Gates.	2
3	Test the Functionality of Universal and Derived Logic Gates.	2
4	Verify Duality Theorem and De Morgan’s Theorem.	2
5	Design and Verification of the Truth Tables of Half and Full Adder Circuits.	2
6	Design and Verification of the Truth Tables of Half and Full Subtractor Circuits	2
7	Test the Function of Multiplexer IC 74150/74151 and Demultiplexer IC 74154/74155 or any other equivalent.	2
8	Realize and Verify SR Flip Flop.	2
9	Realize and Verify Master Slave JK Flip-Flop.	2
10	Implement 4 Bit Ripple Counter.	2
11	Implement 4 Bit Universal Shift Register.	2
12	Implement 4 Bit R-2R D / A Converter.	2

Text Books:
1. "Digital Electronics and Logic Design" by Dr. Sanjay Sharma of KATSON Books.
2. "Digital Electronics and Logic Design" by Dr. Adsul A.P. and Jagdale R.H. of Nirali Publication.
3. "Digital Electronics" by J.S. Katre of Tech-Max Publication, Pune.

Reference Books:
1. "Modern Digital Electronics" by Jain R.P. of Tata McGraw Hill Publication, New Delhi.
2. "Digital Circuits and Design" by Salivahan S. and Arivazhagan S. of Vikas Publication House, Delhi.

Online Resources:
1. www.cse.yorku.ca/~mack/1011/01.NumberSystems.ppt
2.www.people.sju.edu/~ggrevera/arch/slides/binary-arithmetic.ppt
3.www.mathsisfun.com/binary-number-system.html

SEM-V

Course code: 22IUCD0501T	**Course name: Statistics and Probability**	
Course Course category: PCC		
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Algebra and Calculus, Basic Probabiplty Theory, Linear Algebra.		
Course Objectives:		
1. To understand the fundamental concepts of statistics and descriptive measures such as central tendency and dispersion.		
2. To apply probability theory and random variable concepts to model uncertain real-world situations		
3. To analyze data using standard probability distributions, correlation, and regression techniques.		
4. To perform hypothesis testing using parametric and non-parametric statistical methods for decision making.		
5. To introduce fuzzy set theory and fuzzy logic concepts for handling imprecise and uncertain information		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Apply measures of central tendency, dispersion, moments, skewness and kurtosis for statistical data analysis.		
CO2: Use probability laws and random variable concepts to solve problems involving uncertainty.		
CO3: Analyze engineering situations using appropriate standard probability distributions and apply the Central Limit Theorem for modeling and inference.		
CO4: Analyze data relationships using correlation and regression techniques.		
CO5: Perform parametric and non-parametric hypothesis testing for statistical decision making		
CO6: Apply fuzzy set theory and fuzzy logic to handle imprecision in engineering problems.		

Contents –

Unit	Content	Teaching hours
1	**Introduction to statistics** Definition, Importance and usefulness of statistics: Concept of Central Tendency, Relation between Mean, Median and Mode, Measure of Dispersion: Absolute and relative measures of dispersion, Range, Variance, Standard Deviation, Mean Deviation, Coefficient of Variation. Moments, Skewness and Kurtosis	**8**
2	**Elementary Probability** **Introduction to Probability:** Random experiment, Sample space, Events, Axiomatic Probability, Algebra of events, Addition Theorem of Probability. Conditional Probability, Multiplication theorem of Probability, Independent events, Baye's Theorem **Random variable:** Definition, Discrete random variable, Continuous Random Variable, PMF, PDF, CDF, Two-dimensional random variable, Joint probability distribution	**8**
3	**Probability Distributions** Basic concept of Bernoulli, Binomial, Poisson and Normal distributions and its mean, variance, its properties and applications. Fitting of above distributions, **Central limit** theorem.	**6**

4	**Correlation and Regression Analysis** Concept and Types of Correlation, Simple Correlation, Karl Pear so Rank correlation. Introductionofregressionanalysis, simple And multiple regression, fitting of regression line using least square method, coefficient of determination.	**7**
5	**Hypothesis Testing** **Testing of Hypothesis I**: Hypothesis, TypeIandTypeIIerrors Tests of significance - Z- test, Student's t-test Single Mean, Difference of means, paired t-test, Analysis of Variance (ANOVA),Chi-Squaretest: Test of Goodness of Fit. **Nonparametric Tests**: Nonparametric tests: Two-sample Mann— Whitney U Test, the Sign test, the signed rank test, the rank sum test, the Kolmogorov-Smirnov test, and the run test.	**8**
6	**Fuzzy Mathematics** Preview of crisp sets, Basic types of fuzzy sets, Concepts of fuzzy sets, Properties of alpha cuts, representations of fuzzy sets, Extension principle for fuzzy sets, Operations on fuzzy sets, Fuzzy numbers-Arithmetic operations on fuzzy numbers-Fuzzy equations, Fuzzy relations, Fuzzy measures-Evidence theory, Fuzzy sets and possibility theory, Overview of classical logic - Multi valued logics - Fuzzy propositions - Fuzzy quantifiers	**8**

Text Books:
1. MurrayR.Spiegel,LarryJStephens,Statistics,Fourthedition,
2. S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi.
3. J. L. Devore, Probability and Statistics For Engineering and Sciences, 8thedition, Cengage learning.
4. S.G.Purohit,S.D.Gore,S.R.Deshmukh,StatisticusingR,OxfordAlphaScienceInternational.
5. S.K.Gupta,NumericalMethods forEngineers,WileyEastern.
6. KlirandYuan,Fuzzysetsandfuzzylogic,Prentice hallofIndia

Reference Books:
1.David Freedman,RobertPisaniandRogerPurves,Statistics,4thEdition.
2.S.C.Gupta,Fundamental of Statistics,Himalaya.
3.H.C.Saxena,Mathematical Statistics,SultanChand&Sons,NewDelhi.
4.H.J.Zimmerman,FuzzySetTheoryanditsapplications,4thEdition,Springer
5.Kluwer F.MartinMcNeill,EllenThro,Fuzzy Logic A Practical Approach,AP Professional

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code:22IUCD0502T **Course name:** Java Programming **Course category:** PCC
Credits: 3 Teaching scheme: L-3 Evaluation scheme: CA–40, ESE–60
Pre-requisites: Computer Fundamentals, Programming fundamentals, Basics of problem solving and algorithms, Basics of OOP, Data Structure.
Course Objectives:
1**.** To understand the fundamentals of Java programming and its programming environment.
2. To apply object-oriented programming concepts for developing Java applications.
3. To design Java programs using decision-making, looping, and control structures
4. To implement exception handling, multithreading, and file input/output operations in Java.
5. To develop Java applications using packages, interfaces, applets, and graphics programming
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand and develop program using Object Oriented methodology in Java.
CO2: Apply and design inheritance relationships in Java classes to promote code reuse.
CO3: Apply multithreading concepts in Java programs to achieve concurrent execution.
CO4: Understand & apply Exception Handling.
CO5: Create Java programs that involve graphics and applets.
CO6: Understand and work with file streams.

Contents –

Unit	Content	Teaching hours
1	**Basic Syntactical construct sin Java** Java Features and the Java Programming Environment, Object Oriented, Compiled, Interpreted, Platform independent, Portable, Robust and Secure, Dynamic. Defining a class, creating object, accessing class members Java Tokens and Data types, Constants and Symbolic Constants, variables, dynamic initialization, data types, array and string, scope of variable, typecasting, and standard default values. Operators and Expressions, Arithmetic Operators, Relational Operators, Logical Operators, Increment and Decrement, Conditional Operators, Bit wise Operators, Instance of Operators, Dot Operators, Operator precedence and associativity , Evaluation of Expressions, Type conversions in expressions, Mathematical Functions - min(),max(), sqrt(), pow(), exp(), round(), abs().Decision making and looping: If statement, if else statement, nested if else statement, if else if ladder, the switch statement, nested switch statement, The ?:operator, The while statement, the Do while statement ,the 'for' statement, break, continue and return statement, nested loops, labelled loops, for-each version of the for loop.	**7**
2	**Derived Syntactical Construct sin Java** Constructors and methods, types of constructors, nesting of methods, argument passing the 'this' keyword, command liner arguments, varargs: variable-length arguments, garbage collection, finalize() method, the object class. Visibility Control Public, Priv. Protected, default, friendly private Protected access. Arrays and Strings: Types of arrays, creating an array. strings. string classes and string buffer. vectors, -wrapper classes. enumerated types.	**8**

3	**Inheritance , Interface and Package** Inheritance: concept of inheritance Types of Inheritance Single Inheritance. multilevel Inheritance, Hierarchical Inheritance. method and constructor overloading and overriding, dynamic method dispatch, final variables, final methods. Use of super. abstract methods and classes. static members. Interfaces: Define Interface, implementing interface ,accessing interface, variables and methods. extending interfaces, interface references, nested interfaces Package: Define package, type of package naming and creating packages, accessing package, import statement, static import ,adding class and interfaces to a package.	**8**
4	**Exception Handling and Multithreading** Errors and Exception: Types of errors ,exceptions, try and catch statement, nested try statement, throws and Finally statement, build-in exceptions. Chained exceptions, creating own exception (throw clause), subclasses. Multithreaded Programming Creating a Thread: By extending to thread class and by implementing runnable Interface. Life cycle of thread: Thread Methods: wait(), sleep(), notify(), resume(), suspend(), stop().Thread exceptions. thread priority and methods ,synchronization, inter-thread communication. deadlock.	**8**
5	**Java Applets and Graphics Programming** Introduction to applets: Applet, Applet life cycle (skeleton), Applet tag, Adding Applet to HTML file, passing parameter to applet, embedding <applet> tags in java code, adding controls to applets. Graphics Programming: Graphics classes, lines, rectangles, ellipse, circle, arcs, polygons, color and fonts, setColor(), getColor(), setForeGround(), setBackGround(), font class, variable defined by font class: name. pointsizesize, style, font methods: getFamily(). getFont(), getFontname (), getSize(). getStyle(), getAllFonts() and get available font family name()of the graphics environment tclass.	**8**
6	**Managing Input/Output Files in Java** Introduction and Concept of Streams. Stream Classes. Byte Stream Classes: Input Stream Classes, Output Stream Classes. Character Stream Classes, Using streams Using File Class: I/O Exceptions, Creation of Files, Reading/Writing characters, Reading/Writing Bytes, Handling Primitive Data types.	**6**

Text Books: 1. PatrickNaughton-HerbertScheldt,TheCompleteReferencJava2(ThirdEdition)TMH 2.Deitel&Deitel: "How to Program JAVA", PearsonEducation 3.E.Balagurusamy, Programming with Java, BPBPublication
Reference Books:
Online Resources: http://www.nptel.ac.in/courses/106105084/25 https://docs.oracle.com/javase/8/docs

Course code: 22IUCD0503T	**Course name:Software Engineering**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60	
Pre-requisites: Basic programming Concept	
Course Objectives:	
1**.** To understand and select appropriate software process models.	
2. To develop and document software requirements using SRS.	
3. To apply software modeling techniques for data and system design.	
4. To estimate software size and cost effectively.	
5. To apply project management principles in software development.	
6. To implement quality assurance and security practices in software projects.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand suitable Software Process model for software development.	
CO2: Create software requirement specifications.	
CO3: Apply Software modelling to create data designs	
CO4: Evaluate size and cost of software product.	
CO5: Apply Projectmanagement principlesinsoftwaredevelopment.	
CO6: Apply Quality Assurance and security principles in software development.	

Contents –

Unit	Content	Teaching hours
1	**Software Development Process** Software, Software Engineering as layered approach and its characteristics, Types of software. Software development framework. Software Process Framework, Process models: Perspective Process Models, Specialized Process Models. Agile Software development: Agile Process and its importance, Extreme Programming, Adaptive Software Development, Scrum, Dynamic Systems Development Method (DSDM), Crystal.Selection criteria for software process model.	8

2	**Software Requirement Engineering** Software Engineering Practices importance, Core principles. Communication Practices, Planning Practices, Modeling practices, Construction practices, software deployment (Statement and meaning of each principle for each practice). Requirement Engineering: Requirement Gathering and Analysis, Types of requirement (Functional, Product, Organizational, External Requirements), Eliciting Requirements, Developing Use-cases, Building requirement models, Requirement Negotiation, Validation. Software Requirement Specification: Need of SRS, Format, and its Characteristics.	9
3	**SoftwareModelling and Design** Translating Requirement model into design model: Data Modelling. Analysis Modelling: Elements of Analysis model. Design modelling: Fundamental Design Concepts (Abstraction, Information hiding, Structure, Modularity, Concurrency, Verification, Aesthetics). Design notations: Data Flow Diagram (DFD), Structured Flowcharts, and Decision Tables. Testing — Meaning and purpose, testing methods — Black-box and White-box, Level of testing — Unit testing. Test Documentation — Test Case Template, Test plan, Introduction to defect report, test summary report.	10
4	**Software Project Estimation** The management spectrum—4P's Metrics for' Size Estimation: Line of Code (Loc), Function Points (FP). Project Cost Estimation Approaches: Overview of Heuristic, Analytical, and Empirical Estimation. COCOMO (Constructive Cost Model), COCOMO II. Risk Management: Risk Identification, Risk Assessment ,Risk Containment, RMMM strategy.	8

5	**Software Project Management** Project Scheduling: Basic Principles Work breakdown structure Activity network and critical path Method, Scheduling techniques (CPM. PERT). Project Tracking: Timeline charts. Earned Value Analysis, Gantt Charts.	5
6	**Software Quality Assurance and Security** Software Quality Management vs. Software Quality Assurance. Phases of Software Quality Assurance: Planning, Activities, audit, and review Quality Evaluation standards: Six Sigma. ISO for software. CMMI: Levels, Process areas. Software Security ,Introduction to DevOps, Secure software engineering	5

Text Books:

1. Pressman, Roger , "Software Engineering: A Practitioner Approach", Mc Graw Hill Higher Education, New Delhi,(Seventh Edition) ISBN 978-0-07-337597-7

2.Fairly Richard,“Software Engineering Concepts”,Mc Graw Hill Education,New Delhi-2001,ISBN-13:9780074631218

Reference Books:

1.Jain,Deepak,”Software Engineering:Principles and practices”Oxford University Press,New Delhi ISBN 9780195694840

Online Resources:

1.http://www.rspa.com/spi/

2.www.tutorialspoint.com//software_engineering/

3.www.versionone.com/agile-101/

4.www.sei.cmu.edu

5.www.nptel.ac.in/courses/

Course code: 22IUCD0504T **Course name: Programming in Python** **Course Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Knowledge of programming language
Course Objectives:
1.Learn the basics of Python and how to write simple programs.
2.Use loops and conditional statements to control program flow.
3.Work with data structures like lists, tuples, sets, and dictionaries.
4.Apply functions, classes, and packages to build modular programs.
5.Explore Python libraries for data science such as NumPy, Pandas, and Matplotlib.
6.Perform file handling and exception handling to make programs reliable
Course Outcomes: At the end of the course, the students will be able to -
CO1: Display message on screen using Python Script or IDE
CO2: Developpythonprogramto demonstrate euse of operators.
CO3: Perform operations on Data Structures in Python.
CO4: Developfunctions forgivenproblem.
CO5: StudyaboutthevariousPackagesinPythonfordata science
CO6: HandleException

Contents –

Unit	Content	Teaching hours
1	**Python foreverybody** Why Program, Hardware Overview, Python as aLanguage,Features of Python, Why Python, Installation Python Jupyter notebook, Using the Python Playground ,how to write programandcompileinJupyternotebook.WritinginputandoutputStatementsinPython,outputformatting ,Writing Comments, keywords in Python, variables, indentation and comments	**6**
2	**Condition alLoops and Iteration** BasicOperators,writingexpressionsConditional Statements,inPython,LoopsandIteration,Definite Loops,range function, String Function	**8**

3	**Data Structures in Python** Lists: Defininglists,accessingvaluesinlist, deletingvaluesinlist, updating lists.b) Basic List Operations c) Built-in List functions Tuples:a) Accessing values in Tuples, deleting values in Tuples, and updating Tuples. b) Basic Tuple operations.c) Built-in Tuple Functions Sets: a) Accessing values in Set, deleting values in Set and updating Sets. b) Basic Set operations.c) Built-in Set functions Dictionaries: a) Accessing values in Dictionary, deleting values in DictionaryandupdatingDictionary.b)BasicDictionaryoperations. c)Built-inDictionaries functions	**8**
4	**Functions Classes,Modules&Packages** Functions, Types of functions, String Functions, Lamda functions, Recursivefunctions, modulesandpackages,Classes: creatingclasses,objects.DataHiding&Inheritance	**8**
5	**Packages inPython forData Science** Numpy introduction, Numerical operations on Numpy,IntroductionofMatplotlib,gettingstartedwithPandas,Dataframes basics in Pandas, key operations on dataframes, Binarysearch,FindingelementsincommoninlistsusingHash,Findinglargeste lements	**8**
6	**File I/o Handling and Exception Handling** I/O Operations: Reading keyboard input, Printing to screen File Handling: Opening file in different modes, accessing file Contents using standard library functions, Reading and writing files, Closing a file, Renaming and deleting files, Directorie sin Python, File and directory related standard functions Exception Handling: Introduction, Exception handling-lary: except:" statement, 'raise' statement, User defined exceptions	**7**

Text Books: 1.K Nageshwara "PythonProgramming", Scitech PublicationsPvt. Ltd.
2.KentD.Lee,"PythonProgrammingFundamentals",SecondEdition,SpringerPublication.
3.WesMcKinney,"PythonforDataAnalysis"O'ReillyPublication.

Reference Books:

Online Resources: 1.ThePythonLanguageReference:http://docs.python.org/2/reference/index.html
2.ThePythonStandardLibrary:http://docs.python.org/2/library/
3.https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html
4.http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab
5.http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf
6.http://seaborn.pydata.org
7.https://www.datacamp.com/community/data-science-cheatsheets
8.PEP20--The ZenofPython :https://www.python.org/dev/peps/pep-0020/
9.https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

Course code:22IUCD0505T **Course name:Data Structure and Algorithm Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites:
Programming Language such as: C, C++
Course Objectives:
1. To assess how the choice of data structures and algorithm design methods affects the performance of programs.
2. To choose the appropriate data structure and algorithm design method for a specified application.
3. To solve problems using data structures such as linear lists, stacks, queues, hash tables, binary trees, heap s, tournament trees, binary search trees, and graphs and writing programs for these solutions.
4. To solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, branch and bound and writing programs for these solutions.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Apply knowledge of basic programming constructs to design algorithms and programs.
CO2: Analyze clearly different possible solutions to a program and select the most efficient one. (for eg, Sorting and Searching)
CO3: Demonstrate functioning of linear data structures (stack, queue and linked list) and apply them in required applications
CO4: Write neat code by selecting appropriate data structure and demonstrate a working solution for a given problem
CO5: Distinguish between linear and non-linear data structures and determine their applicability
CO6: Analyze performance of algorithms by determining time and space complexity.

Contents –

Unit	Content	Teaching hours
1	**Introduction to Data Structures** Introduction to Data & Data Structures, types of Data Structure -Primitive & Non-Primitive, Arrays, Types of Arrays, Memory representation of Array, traversing linear array, inserting and deleting elements in array. Structures, Functions, Recursive Functions, Introduction to time & space complexity.	**8**
2	**Searching, Sorting and Hashing** Searching – Linear Search and Binary Search.Sorting – Bubble sort, Selection sort, Insertion sort, Hashing: Hashing (Hash Function), Hash Collision,Collision resolution : Chaining Open Addressing, Linear Probing, Quadratic Probing, Double Hashing.	**6**

3	**Stacks and Queues** Stacks: Definition, Array Representation of stacks. Evaluation of a Postfix Expression, Transforming Infix into Postfix expression. Queues: Definition, Array Representation of Queues, Types of Queues: Circular Queues, Double ended Queue, Priority Queue, Stack using two queues, Queue using two stack.	**8**
4	**Linked List** Linked List: Representation of Linked Lists in memory, Insertion & deletion from Linked List, Traversing & Searching a Linked List, Types of Linked List: Circular Linked List and Doubly Linked Lists, Stack and Queue Implementation using Linked List, Addition of two Polynomials using Linked List.	**8**
5	**Tree Data Structures** Introduction to Tree and Terminologies , Binary Trees, Properties of Binary Trees, Array and liked representation of Binary Trees, Traversals-In order, Post order, Preorder, Binary Search trees- Definition, Insertion, Deletion, Traversal, and Searching. Evaluation of Expression, AVL Trees.	**8**
6	**Graph** Definitions, Terminologies, Matrix and adjacency list representation of graphs. Traversal Methods: Breadth First Search and Depth First Search, Minimum Spanning Tree, Prim's & Krushal Algorithm.	**7**

Text Books: 1. "Data Structures with C", S. Lipschutz, Mc Graw Hill.2010

2. " Schaum's Outline of Data Structures", Lipschutz, S., McGraw-Hill

3. "Fundamentals of Data Structures in C++", E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.1995

4. "Data Structures using C", E. Balaguruswamy, McGraw Hill 2013

Reference Books:

1. "Data Structures using C" Aarron M. Tenenbaum, Yedidyah Langsam,Moshe J. Augensten , PHP

Online Resources: 1. https://archive.nptel.ac.in/courses/106/106/106106127/

2. https://nptel.ac.in/courses/106102064

Course code: 22IUCD0502L **Course name: Java Programming Lab Course category:** PCC
Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** TW–60, PR–40
Pre-requisites: Computer Fundamentals, Programming Fundamentals, Problem solving and algorithms, OOP, Data Structures, Web Technolgies.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Understand and develop program using Object Oriented methodology in Java.
LO2: Implement inheritance relationships in Java classes to promote code reuse.
LO3: Apply multi threading concepts in Java programs to achieve concurrent execution.
LO4: Implement Exception Handling.
LO5: Create Javaprogramsthatinvolvegraphicsand applets.
LO6: Create Java programs for handling I/O and filestreams.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Set up a Java Programming development environment and test by implementing a small program.	2
2	Develop programs to demonstrate use of if statements and its different forms.	2
3	Develop programs to demonstrate use of - a) Switch—Case statement Conditional if(?:)	2
4	Develop programs to demonstrate use of Looping Statement 'for'	2
5	Develop programs to demonstrate use of 'while'. - do- while	2
6	Develop a program for implementation of explicit type conversion in Java.	2
7	Develop a program for implementation of Constructor.	2
8	Develop a program for implementation of different functions of String Class.	2
9	Develop a program for implementation of Arrays in Java.	2
10	Develop a program for implementation of Vectors in Java.	2
11	Develop a program for implementation of Wrapper Class to a) convert primitive into object. convert object into primitive	2
12	Develop programs which implement the concept of overriding.	2
13	Develop a program for implementation of Single and Multilevel inheritance.	2
14	Develop a program for implementation of interface	2
15	Develop a program to import different classes in package.	2
16	Develop a program for implementation of try, catch and finally block.	2
17	Develop programs for implementation of throw, throws clause.	2
18	Develop a program for implementation of multithreading operation.	2

19	Write a program to create animated shape using graphics and applets. You may use the following shapes: a) Lines and Rectangles. b) Circles and Ellipses. c) Arcs Polygons with fillPolygon method.	2
20	Develop a program for implementation of I/O stream classes.	2

Text Books: 1.PatrickNaughton-HerbertScheldt,TheCompleteReference-Java2(ThirdEdition)TMH 2.Deitel&Deitel:"HowtoProgramJAVA", PearsonEducation 3.E.Balagurusamy, ProgrammingwithJava, BPBPublication
Reference Books:
Online Resources: http://www.nptel.ac.in/courses/106105084/25https://docs.oracle.com/javase/8/docs

Course code: 22IUCD0503L **Course name: Software Engineering Lab Course** **category:** PCC	
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–00
Pre-requisites: Basic programming Concept	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Understand suitable Software Process model for software development.	
LO2: Create software requirement specifications.	
LO3: Apply Software modelling to create datad esigns	
LO4: Evaluate size and cost of software product.	
LO5: Apply Project management principl esin software development.	
LO6: Apply Quality Assurance and security principles in software development.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hourse
1	Write problem statement to define the project title with bounded scope of the project.	2
2	Select relevant process model to define activities and related tasks set for assigned project.	2
3	Gather application specific requirements for assimilate into RE (Requirements Engineering) model.	2
4	Prepare broad SRS (software requirement specification) for the above selected project.	2
5	Prepare use-cases and draw use-case diagram using Software Modeling Tool.	2
6	Develop data designs using DFDs (data flow diagram), Decision tables, and E-R (entity-relationship) diagram.	2
7	Draw class diagram, Sequence diagram, Collaboration diagram, State Transition Diagram for the assigned project.	2
8	Write test cases to validate requirements of assigned project from SRS document.	2
9	Identify risks involved in the project and prepare RMMM (Risk Management, Mitigation, and Monitoring) plan.	2
10	Estimate cost of the project using COCOMO (Constructive Cost Model) / COCOMO II approach for the assigned project.	2
11	Prepare SQA plan that facilitates various attributes of quality of process and product.	2
12	Write Test Cases to validate Requirement of Assigned project from SRS document.	2

Text Books:

1.Roger Pressman, *Software Engineering: A Practitioner's Approach* ,McGraw-Hill Higher Education, New Delhi, Seventh Edition,ISBN**:** 978-0-07-337597-7

2.Richard Fairley, *Software Engineering Concepts*,McGraw-Hill Education, New Delhi, 2001,ISBN-13**:** 9780074631218

Reference Books:

1.Deepak Jain**,** *Software Engineering: Principles and Practices*,Oxford University Press, New Delhi ,ISBN: 9780195694840

Online Resources:

1.http://www.rspa.com/spi/
2.www.tutorialspoint.com//software_engineering/
3.www.versionone.com/agile-101/
4.www.sei.cmu.edu
5.www.nptel.ac.in/courses/
6.https://techbeacon.com/secure-devops

Course code: 22IUCD0504L **Course category:** PCC		**Course name: Programming in Python Lab**
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Basic knowledge of computer**.**		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Display message on screen using Python Script or IDE		
LO2: Develop python program to demonstrate euse of operators.		
LO3: Perform operations on Data Structures in Python.		
LO4: Develop functions for given problem.		
LO5: Study about the various Package sin Python for data science		
LO6: Handle Exception		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Write Program to perform input/output operations Write a program to take input (integer, float, string) and print it.	2
2	**Program based on operators** 1. Write a program to simulate a simple calculator (+ - / * %) that takes two operands as input and displays the result. 2. Write a program to find area and perimeter of geometric objects. 3. The distance between two cities (in km.) is input through the keyboard. Write a program to convert and print this distance in meters, feet, inches, and centimeters. 4. Write a C program to interchange two numbers. **5.** Write a program to compute Fahrenheit from Centigrade.	2
	Programs based on Decision making. 1. Write a program to read marks from keyboard and your program should display equivalent grade according to conditions : (100-80) Distinction , (79-60) – First Class , (59-40) - Second Class, <40 - Fail 2. Write a program to input basic salary of an employee and calculate gross salary according to given conditions. a)Basic Salary <= 10000 : HRA = 20%,DA=80% b)Basic Salary is between 10001to20000 : HRA=25%, DA=90% c)Basic Salary>=20001:HRA=30%, DA = 95% 3. If the ages of three brothers are input through the keyboard, write a C Program to determine the youngest and oldest of the three. 4. Write a program to calculate overtime pay of employee. Overtime is paid at the	2

3	rate of Rs. 12.00 per hour for every hour worked above 40 hours. Assume that employee do not work for fractional part of an hour. 5. Write a program for checking the speed of drivers. If speed is less than 70, it should print "Ok". Otherwise, for every 5kmabove the speed limit (70), it should give the driver one demerit point and print the total number of demerit points. For example, if the speed is 80, it should print: "Points: 2". If the driver gets more than 12 points, the function should print: "License suspended"	
4	**Programs using while and for loops** 2. WAP to find factorial of given number 3. WAP to check whether given number is Palindrome or not 4. WAP to check whether given number is Armstrong or not 5. WAP to print Fibonacci series 6. Write a Python program which iterates the integersfrom1to 50.Formultiplesof three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz". 7. WAP to check whether given number is Perfect number or not 8. WAP to check whether given number is Prime number or not 9. Write C program to print given star and number patterns and reverse it. *** 1** **** 12** ***** 123** ****** 1234**	2
5	**Programs on string** 2. Write Python Program to find length of string without using len() function. 3. Count all letters, digits, and special symbols from a given string. 4. Python Program to Count the Number of Vowels in a String. 5. Python Program to Calculate the Number of Upper Case Letters and Lower Case Letters in a String. 6. Python Program to Check whether given string is palindrome or not	2
	Programs on List and Tuple 1. Write a Python program to sum all the items in a list. 2. Write a Python program to multiply all the items in a list 3. Write a Python program to get the largest number from a list. 4. Write a Python program to get the smallest number from a list 5. Write a Python program to count all element sin list and count Occurrences Of A List Item In Python 6. Write a Python program to create a tuple with different data types 7. Write a Python program to check whether an element exists within a tuple 8. Write a Python program to reverse atuple Write a Python program calculate the	2

6	product of all the numbers given intuple. Original Tuple:(2,4,8,8,3,2,9) Product - multiplying all the numbers of the said tuple: 27648	
7	**Programs on set and dictionary** 1. Write a Python program to concatenate following dictionaries to create a new one. Sample Dictionary : dic1={1:10, 2:20} dic2={3:30, 4:40} 2. dic3={5:50,6:60} Expected Result:{1:10,2:20,3:30,4: 40,5:50,6:60} 3. Write a Python program to check whether a given key already exists in a dictionary 4. Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (x, x*x) Sample Dictionary(n=5):Expected Output : {1: 1,2: 4,3: 9, 4: 16, 5:25} 5. Write a Python program to merge two Python dictionaries Write a Python program to get the maximum andminimum value in a dictionary 7. Write a Python program to create set difference, Union and intersection 8. Write a Python program to check if two given sets have no elements in common	2
8	**Programs using function** 1. Write Functions to calculate your trip's costs: Define a function called hotel _cost with one argument nights as input Define a function called plane _ride_ cost that takes a string, city, as input. Define a function called rental_ car _cost with an argument called days. Define a function called trip _cost that takes two arguments, city and days. Like the example above, have your function return the sum of calling the rental_car_cost(days), hotel_cost(days),and plane_ride_cost(city) functions. 2. Write a program into check a given number is even or odd using the function. 3. Write a function Exchange to interchange the values of two variables, say x and y. illustrate the use of this function in a calling function. 4. Write a program to find Sum of natural number using recursion. 5. Write a program to print Fibonacciseries Number using recursion	2
9	**Program using NumPy, Matplotli band Pandas library** 1. Write a program to perform matrix addition, substraction, multiplication. 2. Plot all types of graph using Matplotlib. 3. Write a program which performs basic operation of Pandas.	2
10	**Program on Binary Search** Write a program to find given element using Binary search	2
11	Write a program to find given element using Binary search	2

12	**Program on I/o Handling and Exception Handling** Write a program to handle user defined Exception for given problem.	2

Text Books:

1. K Nageshwara"PythonProgramming", ScitechPublicationsPvt.Ltd.
2. Kent D.Lee, "PythonProgrammingFundamentals", SecondEdition, Springer Publication.
3. Wes Mc Kinney, "PythonforDataAnalysis"O'ReillyPublication.

Reference Books:

1. Dr Nageshwar Rao ,"Core Python Programming" -Wiley

Online Resources:

1.ThePythonLanguageReference: http://docs.python.org/2/reference/index.html
2.ThePythonStandardLibrary: http://docs.python.org/2/library/
3.https://docs.scipy.org/doc/scipy/reference/tutorial/stats.html
4.http://matplotlib.org/api/mlab_api.html#module-matplotlib.mlab
5.http://conference.scipy.org/proceedings/scipy2010/pdfs/seabold.pdf
6.http://seaborn.pydata.org
7.https://www.datacamp.com/community/data-science-cheatsheets
8.PEP20--The ZenofPython :https://www.python.org/dev/peps/pep-0020/
9.https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html

Course Code:22IUCD0505L Course Name: Data Structure and Algorithm Lab Course Category: PCC
Credits: 01 **Teaching scheme:**P-2 **Evaluation Scheme:**TW–30, PR–20
Pre-requisites: Programming Language such as: C, C++
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Implement programs using arrays and structures to solve computational problems.
LO2: Design and demonstrate searching, sorting, and hashing algorithms using appropriate data structures.
LO3: Implement and test stack and queue operations using both arrays and linked list representations.
LO4: Develop programs to create, manipulate, and demonstrate various linked list structures and their operations.
LO5: Implement and evaluate different tree data structures such as binary trees, binary search trees.
LO6: Develop programs to represent and analyze graphs using adjacency matrix and adjacency list, and design efficient graph algorithms.

List of Experiments—

Sr. No.	Name of Experiment
1	Write a ‘C’ program to perform following Operations on Array: Create, Insert, Delete,Display.
2	Write a ‘C’ Program to Search a particular data from the given Array of Strings using Linear Search Method.
3	Write a ‘C’ program to Search a particular data from the given Array of numbers using Binary Search Method.
4	Write a ‘C’ Program to Sort an Array of numbers using Bubble Sort Method.
5	Write a ‘C’ Program to Sort an Array of numbers using Selection Sort Method
6	Write a ‘C’ Program to Sort an Array of numbers using Insertion Sort Method.
7	Write a 'C' Program to perform PUSH and POP Operations on Stack using an Array.
8	Write a 'C' Program to perform INSERT and DELETE Operations on Linear Queue using an Array.
9	Write a 'C' Program to perform INSERT and DELETE operations on Circular Queue using an Array.
10	Write a 'C' Program to Implement Singly Linked List with Operations: (i) Insert at beginning, (ii) Search, (iii) Display
11	Write a 'C' Program to Implement BST (Binary Search Tree) and Traverse in In-Order.
12	Write a program to implement Graph operations.

Text Books: 1. “Data Structures with C”, S. Lipschutz, Mc Graw Hill.2010

2. ” Schaum's Outline of Data Structures”, Lipschutz, S., McGraw-Hill

3. “Fundamentals of Data Structures in C++”, E. Horowitz, Sartaj Sahni, D.Mehta, Universities Press.1995

4. “Data Structures using C”, E. Balaguruswamy, McGraw Hill 2013
Reference Books: 1. “Data Structures using C” Aarron M. Tenenbaum, Yedidyah Langsam,Moshe J. Augensten , PHP
Online Resources: 1. https://archive.nptel.ac.in/courses/106/106/106106127/
2. https://nptel.ac.in/courses/106102064

SEM-VI

Course code: 22IUCD0601T Course name: Data Communication and Networking Course category: PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Understanding of computer hardware (CPU, memory, storage), Programming Fundamentals, Basic computer organization
Course Objectives: Understand Fundamentals of Data Communication, Learn Networking Models and Standards, Study Transmission Media and Devices
Course Outcomes: At the end of the course, the students will be able to -
CO1: Analyse the function of data communication and computerNetwork
CO2: Select relevant transmission media and switching Techniques.
CO3: Analyse the transmission errors with respect to IEEE standards
CO4: Configure various networking devices
CO5: Configure different TCP/IPservices

Contents –

Unit	Content	Teaching hours
1	**Introduction** Data Communications, Networks, Network Types, Internet History. Types of Network, the OSI model. Introduction to Physical Layer-1: Data and Signals, Digital Signals, Transmission Impairment, Data Rate limits, Performance, Digital Transmission: Digital to digital conversion.	**8**
2	**Fundamentals ofdata communication and Computer Network** Process of Data Communication and its Components. Protocols, standards, Standard organization, Bandwidth, Data Transmission Rate, Baud rate and bits per second. Modes of Communication Analog signal and Digital signal Fundamental of Computer network Classification of network Network Architecture	**6**
3	**Transmission media and switching** Communication media 3.2.Unguidedtransmission media Lines of Sight Transmission Point to Point Multiplexing Switching	**8**
4	**Error detection, correction and wireless communication** Types of error Error detection IEEE Standards 4.4.Wireless LANs BluetoothArhitecture Mobilegenerations	**8**

5	**Network Topologies And networkingDevices** Network Topologies Network Connecting Devices: Hub,Switch,Router,repeater,Bridge,Gateway, Modem, Wireless infrastructure Components	**8**
6	**Reference Models** OSI Reference Model TCP/IP Model: Layered Architecture, Data link Layer Introduction:Adddressingmechanismin Internet IP addressing IPv4 and IPv6 OSI and TCP/IP Network Model	**7**

Text Books: 1. Data communication and networking Author: Forouzan Behrouz
Reference Books: 1. Computer Networks Author: Tanenbaum Andrew S
Online Resources: 1**.** https://www.tutorialspoint.com 2. https://www.w3schools.com

Course code: 22IUCD0602T	**Course name: Advanced Java Programming**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Core Java, Object-Oriented Programming (OOP), Java AWT (GUI Programming Basics), Java Collections & Data Structures, Basic Networking Concepts, Basic Database/SQL Concepts	
Course Objectives:	
1. Understand and apply GUI programming concepts using AWT and Swing components, layout managers, and event-handling mechanisms to develop interactive Java applications.	
2. Implement advanced event. handling in Java using listener interfaces, adapter classes, and inner classes for robust user-interface functionality.	
3. Develop networked Java applications using sockets, TCP/IP, UDP, and internet addressing concepts.	
4. Perform database connectivity and operations using JDBC, including connections, statements, prepared statements, result sets, and transaction management.	
5. Utilize Java Collections Framework and generics to manage and manipulate data efficiently, including sets, lists, maps, and custom objects for real-world applications.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Create Javaprograms usingGUIframeworks suchasAWT or Swing.	
CO2: Apply knowledge of event handling mechanisms in AWT and Swingcomponents	
CO3: Write Java code to handle various events in aprogram.	
CO4:Apply networking concepts to communicate over networks. Also, handle databaseoperations	
CO5: Utilize Javacollections tomanageandmanipulate data efficiently.	
CO6: Create Java to handle HTTP requests and generate dynamic web content.	

Contents –

Unit	Content	Teaching hours
1	**Abstract Windowing Toolkit (AWT)** Component, container, window, frame, panel. Creating windowed programs and applets. AWT controls and layout managers: use of AWT controls: labels, buttons, checkbox, checkbox, group, scroll bars, text field text area. Use of layout managers: flow Layout, border Layout, grid Layout, card Layout, Grid bag Layout, menu bars, menus, dialogboxes, file dialog.	**7**
2	**Swings** Introduction to swing: Swing features, Difference between AWT and Swing. Swing Components JApplet, Icons and Labels, Text Fields, Combo Boxes .Buttons: The JButton, Check Boxes, RadioButtons. Advanced Swing Components :Tabbed Panes ,Scroll Panes, Trees, Tables, Progress bar, tooltips .MVC Architecture. Key Listener Interface, Mouse Listener interface, Mouse Motion Interface, Text Listener Interface, Windows Listener Interface.	**8**

3	**Event Handling** The delegation Event Model: Event sources,Event listeners Event classes: The Action Event class, the Item Event class, the Key Event class, the Mouse Event class, the Text Event class, the Window Event class .Adapter classes. Inner classes .Event listenerInterfaces :Action Listener Interface, Item Listener Interface,	**8**
4	**Networking Basicsand Database interaction** Socket Overview: Client/Server, Reserved Sockets, Proxy Servers, Internet Addressing. Java and the Net: The Networking Classes and interfaces.InetAddress: Factory Methods, Instance Methods. TCP/IP Client Sockets: Who is URL: Format, The URI Class URL Connection: TCP/IP Server Sockets. Datagrams : Datagram Packet , Datagram Server and Client Introduction to JDBC DDBC JDBC Architecture: Two tier models and three tier models Types of JDBC Drivers Driver Interfaces andDriver Manager class: Connection Interfacen Statement lnterface, Prepared Statement Interface Result Set Interface The essential JDBC Program	**8**
5	**Java Collections andGenerics** The Collections Framework, The Set Interface, Set Interface Methods, Set Implementation Classes, The List Interface, List ImplementationClasses, TheMap Interfaces, Map Implementation Classes, Utility classes, Generics in Java, Primitive wrapper classes--{Collection Framework: Array List class, Linked List class, List Iterator interface, Hash Set class, Linked Hash Set class ,Tree Set class, Priority Queue class,ArrayDequeclass,Mapinterface,HashMapclass,LinkedHashMapclass,TreeMapclass,Hashtableclass, Comparable and Comparator, Properties class}	**8**
6	**Servlets** The Life Cycle of a Servlet Creating simple Servlet: The Servlet API, javax. servlet Package, Servlet Interface, Servlet ConfigInterface, Servlet Context Interface, Servlet Request Interface, Servlet Response Interface, Generic Servlet Class The javax. Servlet .http Package: Http Servlet Request Interface, Http Servlet Response Interface, Http Session Interface, Cookie Class, Http Servlet Class, Http Session Event Class, Http Session Binding Event Class Handling HTTP Requests and Responses Handling HTTPGET Requests Handling HTTPPO STRequests. Cookies and Session Tracking	**6**

Text Books: 1.Java:TheCompleteReferenceySchildt,Herbert,Publisher:McGrawHillEducationIndia

2.Java2ProgrammingBlackBook–byHolzner,Stevenetal:DreamtechPress,Newdelhi

3.Java Server Programming Tutorial JAVA EE6 Black Book- by KogentLearnig Solutions: Dreamtech Press, New delhi

Reference Books:
Online Resources: 1.https://nptel.ac.in/courses/106105084/30 2.https://www.tutorialspoint.com/java 3.https://www.javatpoint.com/servlet-tutorial

Course code: 22IUCD0603T	**Course name: Software Testing Course category:** PCC
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic Programming Knowledge, **Software Engineering**	
Course Objectives:	
1. To introduce students to the fundamental concepts and principles of software testing.	
2. To enable students to understand and apply various software testing methods and techniques.	
3. To develop the ability to design and prepare test cases for different types and levels of software testing.	
4. To train students to design and prepare a comprehensive test plan for a software application.	
5. To equip students with the skills required to analyze software applications, identify defects, and prepare defect reports.	
6. To provide hands-on experience in evaluating software performance using automated testing tools.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Apply fundamental concepts and principles of software testing in practical scenarios.	
CO2: Apply appropriate software testing methods and techniques for different types of software systems.	
CO3: Design and develop test cases for various levels and types of software testing.	
CO4: Design and prepare a comprehensive test plan for a given software application.	
CO5: Analyze software behavior to identify defects and prepare detailed defect reports.	
CO6: Evaluate software performance by executing automated testing tools and interpreting performance metrics.	

Contents –

Unit	Content	Teaching hours
1	**Basics of Software Testing** Software Testing, Objectives of Testing Failure, Error, Fault, Defect, Bug Terminology. Test Case, When to Start and Stop Testing of Software (Entry and Exit Criteria).Verificationand Validation(V Model),QualityAssurance, Quality Control.	**8**
2	**Software Testing Methods** Methods of Testing: Static and dynamic Testing The box approach: White Box Testing: Inspections, Walkthroughs, Technical Reviews, Functional Testing, Code Coverage Testing, Code Complexity Testing. Black Box Testing: Requirement Based Testing, Boundary Value Analysis, Equivalence Partitioning,	**9**
3	**Types and Levels of Testing** Levels of testing: Unit Testing: Driver, Stub Integration Testing: Top-Down Integration, Bottom-Up Integration, Bi-Directional Integration Testing on web Application: Performance Testing: Load Testing, Stress Testing, Security Testing. Client-Server Testing. Acceptance Testing: Alpha Testing and Beta Testing, Special Tests: Regression Testing, GUI Testing,	**10**

4	**Test Management** Test Planning : Preparing a Test Plan, Deciding Test Approach, Setting Up Criteria for Testing, Identifying Responsibilities, Staffing, Resource Requirements, Test Deliverables, Testing Tasks Test Management: Test Infrastructure Management, Test People Management. Test Process: Base Lining a Test Plan, Test Case Specification. Test Reporting: Executing Test Cases, Preparing Test Summary Report.	**8**
5	**Defect Management** Defect Classification, Defect Management Process. Defect Life Cycle, Defect Template Estimate Expected Impact of a Defect, Techniques for Finding Defects, Reporting a Defect.	**5**
6	**Testing Tools and Measurements** Manual Testing and Need for Automated Testing Tools Advantages and Disadvantages of Using Tools Selecting a Testing Tool When to Use Automated Test Tools, Testing Using Automated Tools. Metrics and Measurement: Types of Metrics. Product Metrics and Process Metrics, Object oriented metric sin testing.	**5**

Text Books:

1. Sriniwasan Desikan, Gopalswamy Ramesh, *Software Testing: Principles and Practices*, Pearson Education, Pearson India, 2005.
   ISBN: 978-81-7758-121-8.
2. M. G. Limaye, *Software Testing: Principles, Techniques and Tools*, Tata McGraw-Hill Education, New Delhi, 2007.
   ISBN: 978-0-07-013990-9.

Reference Books:

1. Naresh Chauhan, *Software Testing: Principles and Practices*, Oxford University Press, Noida.
2. Yogesh Singh, *Software Testing*, Cambridge University Press, Bengaluru.
   ISBN: 978-1-107-65278-1.

Online Resources:

1.http://www.selenium.com
2.http://en.wikipedia.org/wiki/Test_automation
3.http://en.wikipedia.org/wiki/Software_testing#Testing_tools
4.http://www.softwaretestingsoftware.com
5.www.nptel.ac.in/courses/
6.www.toolsqa.com

Course code: 22IUCD0604T	**Course name: Data Science with Python**
Course category: PCC	
Credits: 3 **Teaching scheme:** L-3	**Evaluation scheme:** CA–40, ESE–60
Pre-requisites: Basic knowledge of Python Programming	
Course Objectives:	
1. Introduce the foundations of Data Science	
2. Develop Python programming skills for data science.	
3. Use NumPy and Pandas to manage, clean, transform, and analyze structured and unstructured data effectively.	
4. Perform data preprocessing, aggregation, and statistical operations to prepare datasets for meaningful insights.	
5. Create effective visualizations using Matplotlib and Seaborn to identify patterns, trends, and outliers in datasets.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Apply set of preliminary tools and techniques to perform datascience	
CO2: Apply various Python data structures to effectively manage various types of data	
CO3: Learn the fundamentals of some of the widely used python packages and apply the min to data analytics.	
CO4: Design applications applying various operations for data cleansing and transformation.	
CO5: Able to visualize data and statistics, interpreting the displays to gain information	
CO6: Learn how to make it easier to identify patterns, trends and outliers in large data sets	

Contents –

Unit	Content	Teaching hours
1	**Introduction to DataScience** Introduction of data science What is data, What is data science, History of data science, Data science life cycle, Why data science is important, Applications of data science, Why Python is necessary for data science, Types of Data: Qualitative Data, Quantitative Data, Basic terminologies of DS Getting started with Python Installation of Anaconda, Jupyter/pycharm/ spyder or any other python tool set Up and installation	6
2	**Overview of Paython** Basics of Python including data types, operators, variables, expressions, control structures using sample dataset, objects and functions. Python sequence data structures including String, Array, List, Tuple, Set, and Dictionary. Introduction to Various python libraries for data scienc	7
3	**Basic data Processing** Introduction of data processing, Why numpy requires for data processing, properties of numpy, Types of arrays: One dimensional, Two dimensional, Three dimensional , Data Types of NumPy, Array Attributes of NumPy , Array creations from existing and non existing values, Array manipulations, Universal Functions, arithmetic operations, Indexing and slicing	8

4	**Working with Pandas Data Pre processing** Features of Pandas, data structure in pandas, Series creation, accessing elements of series, vectorizing operations, Data Frame creations, viewing Data Frame elements, Basic Functionality of data structres, Descriptive functions, function applications: pipe, apply, apply map, re indexing, sorting, working with text data, statistical functions, aggregation, missing data, group by operations, merge join and concantation, visualization,Data pre processing	8
5	**Data Visualization** Basic visualizations with Matplotlib, Matplotlib: Sin wave plot, sub plot, bar, histogram ,scatterplot, contour plot, Quiver plot, box plot, heat map, pie chart, line graph , Matplotlib object oriented interface ,matplotlib classes ,multiple plots,3Dimesional plotting, Working with text, working with images	8
6	**Data Visualization with seaborn** Advanced visualizations with Seaborn,Features of Seaborn, importing datasets, seaborn figure style, color palette, histogram, visualizing pairwaise relationship, plotting categorail data, Linear realationship,Facet grid, Pair grid	8

Text Books:

1. K Nageshwara — *Python Programming*, Scitech Publications Pvt. Ltd.
2. M T Savaliya, R K Maurya, G M Magar — *Programming through Python*, STAREDU Solutions

Reference Books:

1. Daniel Y. Chen — *Pandas for Everyone: Python Data Analysis*, Pearson
2. Davy Cielen, Arno D.B. Meysman, et al. — *Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools*

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0605T **Course name: Database Management System Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–40, ESE–60
Pre-requisites:
Course Outcomes: At the end of the course, the students will be able to -
CO1: Recallsand understandsfundamentalconceptsrelatedtodatabases.
CO2: Applynormalizationrulestoensuredata integrity
CO3: Apply SQLcommandstocreateand managedatabases.
CO4: Apply PL/SQLcodetoimplementbusiness logicandmanipulatedatawithinthe database.
CO5: Createstoredproceduresandfunctionstoencapsulatereusablelogic.
CO6: Implement security Measures (e.g., access controls, encryption) to protect the database and ensure confidentiality.

Contents –

Unit	Content	Teaching hours
1	**Database System Concept** Concept of Data, Database, DBMS, advantages of DBMS over file processing system, application of database. Three-level Architecture for Database system. Data abstraction. Overall Structure of DBMS. Data Modelling. Data Modelling Using the E-R Model.	8
2	**Relational Data Model** Fundamentals of RDBMS: Key concept, Normalization and its types. Introduction to Structured Query Language: DDL commands Data Integrity Constraint DML commands DCL commands DQL commands SQL operators	6
3	**Interactive SQL and Advance SQL** In-built Functions: String Arithmetic Date and time Aggregate functions Queries using: Group by, Having, and Order by clause Joins: Inner and Outer JoinSub queriesViews and Joins. Sequences, Indexes, Synonyms.	8
4	**PL/SQL Programming** Introduction of PL/SQL, execution environment, PL/SQL data Types, Variables, Constants. Control Structure, Exception handling Cursors Procedures, Functions. Database Triggers	8
5	**Database security and TransactionProcessing** DatabasesecurityProtectingthedatawithindatabase-Databaseprivileges Transaction Database backup Database recovery	8
6	**Emerging DatabaseTechnologies** Object oriented and object relational databases Web databases Distributed Databases Dataware housing and datamining.	7

Text Books: 1.Database System Concepts Mc Graw Hill in ANSIC Korth Mc Graw Hill Education,NewDelhi,2015 2.Introduction to Database Management Systems 1 SRD Group McGraw Hill Education,New Delhi, 2015
Reference Books: 1.Introduction to Relational databases & SQL programming Allen McGraw HillEducation, New Delhi, 2015 2.Complete Reference:Mysql Vikram Vaswani McGraw HillEducation,NewDelhi.2015
Online Resources: 1.https://nptel.ac.in/course.html

Course code: CSS	**Course name: Communication skills**	**Course category:** PCC
Credits: 3	**Teaching scheme:** L-3	**Evaluation scheme:** CA–20, ESE–30
Pre-requisites: Communciaotn Basic		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Student will be able to utilize communication skills in different formal settings.		
CO2: Student will be able to select and apply the appropriate methods of communication in various situations.		
CO3: Student will be able to develop presentation skills and body language.		
CO4: Student will be able to apply learned interview skills.		
CO5: Student will be able to draft correct format of emails and business letters.		

Contents –

Unit	Content	Teaching hours
1	**Introduction to communication** Nature and scope of communication Process of communication Function of Communication Cross cultural communication Corporate communication-Effectiveness in Managerial communication.	5
2	**Non verbal communication** Introduction to nonverbal communication Aspects of body language (Kinesics) (Facial expression, eye contact, vocalic, paralanguage, gesture, posture, dress & appearance, haptics) **Tips for effective use of Non-verbal communication** Body language and confidence Use of body language at workplace.	5
3	**Interview Skills.** Introducton,Principles of Interview,Types of Interview,Preparing for Interview,Success in Interview.	5
4	**Group Discussion and Team Presentation.** Introduction to Group Discusssion. Methodology of Group Discussion. Guidelines for Group Discussion. Elements of effective presentation. Tips for effective team Presentation	5

5	**Formal Written Communication** **Office Drafting-** Notice, memo & e-mail. Job application with resume. Writing a Case Study. Report Writing	**5**
6	**Introduction to communication** Nature and scope of communication Process of communication Function of Communication Cross cultural communication Corporate communication-Effectiveness in Managerial communication	**5**

Text Books:

1. *Sanjay Kumar, Pushpa Lata* (Oxford University Press) ,Communication Skills ,Core textbook covering communication fundamentals and practice.
2. *Urmila Rai* (Himalaya Publications), Communication Skills Standard student textbook with explanations and exercises.
3. *E. Sureshkumar, P. Sreehari, J. Savithri,* English for Success(Foundation Books) ,Student-oriented book blending communication and English skill practice.

Reference Books:

1. *M. Ashraf Rizvi* , Effective Communication Skills**,** Commonly used as supplementary/reference material for communication theory and soft skills.

Online Resources:

1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills
2. https://www.skillsyouneed.com/ips/communication-skills.html
3. https://corporatefinanceinstitute.com/resources/management/communication/
4. https://www.coursera.org/articles/communication-skills

Course code:CSS-L	**Course name: Communication skills Lab**
Course category: Program Core Course	
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–00
Pre-requisites: 1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Student will be able to apply basic theory and principles in day to day communication	
LO2: Student will be able to effectively apply non-verbal skills.	
LO3: Student will be able to competently use communication skills in interview and presentation.	
LO4: Student will be able to use appropriate words as per the context.	
LO5: Student will be able to apply written communication in day to day life.	

List of Experiments–

Sr.No.	Name of Experiment	Pratical Hours
1	Give self-Introduction.	2
2	State the importance of body language in communication.	2
3	Prepare effective techniques for successful interview.	2
4	Prepare guidelines for presentation.	2
5	Design a power point presentation on any topic.	2
6	Organize debate on given topic.	2
7	Prepare job application with resume.	2
8	Discuss some important personal attributes required for a team.	2
9	Write some tips for professional e-mail writing.	2
10	Write a case study on any given situation.	2

Text Books:

1. *Sanjay Kumar, Pushpa Lata* (Oxford University Press) ,Communication Skills ,Core textbook covering communication fundamentals and practice.
2. *Urmila Rai* (Himalaya Publications), Communication Skills Standard student textbook with explanations and exercises.
3. *E. Sureshkumar, P. Sreehari, J. Savithri,* English for Success(Foundation Books) ,Student-oriented book blending communication and English skill practice.

Reference Books:

1. *M. Ashraf Rizvi* , Effective Communication Skills**,** Commonly used as supplementary/reference material for communication theory and soft skills.

Online Resources:

1. https://in.indeed.com/career-advice/resumes-cover-letters/communication-skills
2. https://www.skillsyouneed.com/ips/communication-skills.html
3. https://corporatefinanceinstitute.com/resources/management/communication
4. https://www.coursera.org/articles/communication-skills

Course code: 22IUCD0601L	**Course name: Data Communication and Networking Lab**	
Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–00
Pre-requisites:1 Understanding of computer hardware (CPU, memory, storage), Programming Fundamentals, Basic computer organization		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Analyse the function of data communication and computer Network		
LO2: Select relevant transmission media and switching Techniques.		
LO3: Analyse the transmission errors with respect to IEEE standards		
LO4: Configure various networking devices		
LO5: Configure different TCP/IP services		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Configure Peer to Peer Network with at least three hosts	2
2	Study of network IP	2
3	Connect the computers in Local Area Network.	2
4	Create desired standard network cable including cross cable and test by using cable tester	2
5	Connect computer using given topology	2
6	Connect computers using wireless media	2
7	Write a C program for CRC Error Detection	2
8	Create a network using Bluetooth	2
9	Share printer and folder in a network and transfer a file from one computer to another	2
10	Install operating system	2
11	Configure file server	2
12	Configure DHCP server	2

Text Books:
1. Data communication and networkin Author: Forouzan Behrouz

Reference Books:
1.ComputerNetworksAuthor:TanenbaumAndrewS

Online Resources:
1.https://www.tutorialspoint.com
2. https://www.w3schools.com

Course code: 22IUCD0602L **Course name: Advanced Java Programming Lab Course category:** PCC
Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** TW–60, PR–40
Pre-requisites: Core Java, Object-Oriented Programming (OOP), Java AWT (GUI Programming Basics), Java Collections & Data Structures, Basic Networking Concepts, Basic Database/SQL Concepts
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Create Java programs using GUI frameworks such as AWT or Swing.
LO2: Apply knowledge of event handling mechanisms in AWT and Swing components
LO3: Write Java code to handle various events in a program.
LO4: Apply networking concepts to communicate over networks. Also, handle data base operations
LO5: Utilize Java collections to manage and manipulate data efficiently.
LO6: Create Java to handle HTTP requests and generate dynamic web content.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Write a program to demonstrate the use of AWT components like Label, Textfield, TextArea, Button, Checkbox, RadioButton, List and Choice, etc.	2
2	Write a program to show use of different Layout.	2
3	Write a program using AWT to create a menubar where menubar contains menu items such as File, Edit, View and create a submenu under the File menu: New and Open.	2
4	Write a program using swing to display a ScrollPane and JComboBox in an Applet with the items — English, Marathi, Hindi, Sanskrit.	2
5	Write a program to create a JTable.	2
6	Write a program to create a JTree.	2
7	Write a program to launch a JProgressBar.	2
8	Write a program to demonstrate status of key on Applet window such as Key_Pressed, KeyReleased, KeyUp, KeyDown.	2
9	Write a program to demonstrate various mouse events using MouseListener and MouseMotionListener interface.	2
10	Write a program to demonstrate the use of JTextField and JPasswordField using Listener Interface.	2
11	Write a program to demonstrate the use of InetAddress class and its factory methods.	2
12	Write a program to demonstrate the use of URL and URLConnection class and its methods.	2
13	Write a program to implement chat Server using ServerSocket and Socket class.	2
14	Write a program to insert and retrieve the data from database using JDBC.	2
15	Write a program to demonstrate Collections Framework classes and interfaces and use of Generics.	2
16	Write a program to demonstrate the use of HttpServlet as a parameterized Servlet.	2
17	Write a Servlet program to send username and password using HTML forms and	2

	authenticate the user.	
18	Write a program to create Session using HttpSession class.	2

Text Books:
1.PatrickNaughton-HerbertScheldt,TheCompleteReferenceJava2(ThirdEdition)TMH
2.Deitel&Deitel: “HowtoProgramJAVA”, PearsonEducation
Reference Books:
1.E.Balagurusamy, ProgrammingwithJava, BPBPublication
2.Java2 BlackbookbyStevenHolzner
3.KathySieraa&BertBates,HeadFirstJava(2ndEdition),O’reilly
4.DarrelInce&AdamFreeman,programmingtheInternet withJAVA,Addison-Wesley
5.Dr.G.T.Thampy,ObjectOrientedProgramminginJava,DreamtechPress
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0603L	**Course name: Software Testing Lab**
Course category:PCC	
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:**TW–50, PR–00
Pre-requisites: Basic Programming Knowledge, **Software Engineering**	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Apply basic concepts of software Testing.	
LO2: Apply various software Testing Methods.	
LO3: Prepare Test cases for different type sand levels of Testing.	
LO4: Prepare Test plan for an Application.	
LO5: Identify bugs to create defect report of given application.	
LO6: Test software for Performance measures using automated testing tools	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Designtestcases Forsimplecalculator application.(BB Testing)	2
2	Design test cases for railway reservation form	2
3	Design test cases fore-commerce(Flipkart ,Amazon)login Form	2
4	Design test cases for Web Pages Testing any Web Sites	2
5	Write program and design test cases for the following Control and decision making statement. 1)For...Loop 2) Switch...case 3)Do...While 4)lf...else	2
6	Prepare test plan for an identified Mobile application.	2
7	Design test plan and test cases for Notepad (MS Window based) Application.	2
8	Prepare defect report after executing test cases for library management system	2
9	Prepare defect report after executing test cases for Withdrawn of amount from ATM Machine.	2
10	Prepare defect report after executing test cases for any login form.	2
11	Design and run test cases for WordPad (MS Windows based). Using an Automated tool.	2
12	Design and run test cases for MSWord application using an Automation Tool.	2

Text Books:

1. Sriniwasan Desikan, Gopalswamy Ramesh, *Software Testing: Principles and Practices*, Pearson Education, Pearson India, 2005.
   ISBN: 978-81-7758-121-8.
2. M. G. Limaye, *Software Testing: Principles, Techniques and Tools*, Tata McGraw-Hill Education, New Delhi, 2007.
   ISBN: 978-0-07-013990-9.

Reference Books:

1. Naresh Chauhan, *Software Testing: Principles and Practices*, Oxford University Press, Noida.
2. Yogesh Singh, *Software Testing*, Cambridge University Press, Bengaluru.
   ISBN: 978-1-107-65278-1.

Online Resources:

1.http://www.selenium.com
2.http://en.wikipedia.org/wiki/Test_automation
3.http://en.wikipedia.org/wiki/Software_testing#Testing_tools
4.http://www.softwaretestingsoftware.com
5.www.nptel.ac.in/courses/
6.www.toolsqa.com

Course code:22IUCD0604L	**Course name:Data Science with Python Lab**

Course category: Program Core Course
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:**CA–0, MSE–0, ESE–0, TW–30, PR–20
Pre-requisites: Basic knowledge of python programming
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1:Applysetofpreliminarytoolsandtechniquestoperformdatascience
LO2:ApplyvariousPythondatastructurestoeffectivelymanagevarioustypesofdata
LO3:Learnthefundamentalsofsomeofthewidelyusedpythonpackagesandapplythemintodataanalytics.
LO4:Design applications applying various operations for data cleansing and transformation.
LO5:Abletovisualizedataandstatistics,interpretingthedisplaystogaininformation
LO6: Learn how to make it easier to identify patterns, trends and outliers in large data sets

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Installation of Anaconda with set up of jupyter lab	2
2	Write a program on operators and control structure sin python	2
3	Write a program for list and tuple operations	2
4	Write a program for set and dictionary operations	2
5	Write a NumPy program to convert a list of numeric value into a one dimensional two dimesniaonal and three dimensional NumPy array.And perform all operations on that array	2
6	Write a numpy program to create an array using non existing values. Also WAP to perform arithmetic operations and indexing slicing	2
7	Write a Pandas program to read a dataset and show Data Frame and modify the default columns values and print the first 6 rows. Also find the number of rows and columns and data type ofe ach column of Dataframe.Performminimum5operationsonDataframe. Perform data cleaning and handle missing values.	2
8	Write a program to apply pipe, apply, applymap, re indexing, sorting,working with text data,statisticalfunctions,aggregationfunctionswithpandasDataframe	2
9	Consider dataset with student name, gender, PRNno, 5 semester result with marks of each subject, his mobile number, and city. Implement following in Python .Plot various graph sand charttovisualizestudent'sdata	2
10	Write a program on given data set which shows pair plot ,historgram, heatmapetc	2
11	Write a program to display to loreffectonhistrogram	2
12	Write a program to show Linear relationship ,Facet grid, Pair grid using visualization technique	2

Text Books:

1. K Nageshwara — *Python Programming*, Scitech Publications Pvt. Ltd.
2. M T Savaliya, R K Maurya, G M Magar — *Programming through Python*, STAREDU Solutions

Reference Books:

1. Daniel Y. Chen — Pandas for Everyone: Python Data Analysis, Pearson
2. Davy Cielen, Arno D.B. Meysman, et al. — Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: 22IUCD0605L **Course name: Database Management System Lab Course category:** PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites:1 Database Creation 2. Basics of Table Creation in DBMS 3.Basic SQL Queries
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Re calls and understands fundamental concepts related to databases.
LO2: Apply normalization rules to ensure data integrity
LO3: Apply SQL commands to create and manage databases.
LO4: Apply PL/SQLcode to implement business logic and manipulate data with in the database.
LO5: Create stored procedures and functions to en capsulatere usable logic.
LO6: Implement security measures (e.g., access controls, encryption) to protect the data base and ensure confidentiality.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	**Create a database** and perform: i) Assign primary key. ii) Modify the table structure - add column, change the data type of column. iii) Delete the column from table. iv) Insert, update, and delete the record from table. v) Retrieve data from the table according to condition given.	2
2	**Create and Execute DDL commands** using SQL. ii) Apply following Integrity constraints on table: iii) Primary key, Foreign key, Unique key constraint, Null, No Null, and Check constraint.	2
3	**Create and Execute DML commands using SQL.**	2
4	Write Queries using following operators: Arithmetic Operators Comparison Operators Logical Operators Set Operator Range Searching operators (Between) Pattern matching operators (Like)	2
5	Write Queries using following Functions: String Arithmetic Date and time Aggregate Functions	2
6	Execute Queries using the Select command with Where, Having Group by and order by clauses .Execute the queries for implementation of Inner And Outer Join	2
7	Execute Queries using the Select command with Where, Having Group by and order by clauses.	2
8	Execute the queries for implementation of Inner and Outer Join	2
9	Implement Views: i) Create different views.	2

	ii) Insert, modify, and delete records through views. iii) Delete the views.	
10	Create and Execute Indexes, Sequences, and Synonyms in SQL**.**	2
11	Write a PL/SQL program using if-then-else, for, while, and nested loop.	2
12	Creation of procedures, database triggers, and cursor**.**	2
13	Executing DCL commands using SQL**:** i) Create users. ii) Grant privileges to user. iii) Revoke privileges from users.	2

Text Books:

1. 1.Database System Concepts McGraw Hill in ANSIC Korth McGraw Hill Education,New Delhi,2015
2. Introduction to Database Management Systems1SRD Group McGraw Hill Education,New Delhi, 2015

Reference Books:

1. Introduction to Relational databases & SQL programming Allen McGraw HillEducation, New Delhi, 2015
2. Complete Reference:Mysql Vikram Vaswani McGraw Hill Education,NewDelhi.2015

Online Resources: 1.NPTEL / SWAYAM lectures.

SEM-VII

Course code: CSI24PCL401 **Course name: Artificial Intelligence** **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Linear Algebra, Calculus, Probability, Statistics, Python
Course Objectives:
1. Understand the basics of artificial intelligence and its subfields
2. To apply Problem Solving and algorithm application.
3. Applying Knowlwge and reasoning in Artificial Intelligence.
4. To Understand different forms of learning in AI.
Course Outcomes: At the end of the course, the students will be able to -
CO1: To understand the fundamentals of Artificial Intelligence.
CO2: To design smart system using different search strategies of Artificial Intelligence
CO3: Demonstrate knowledge of reasoning and knowledge representation for solving real world Problems.
CO4: Illustrate the construction of learning and expert system.

Contents –

Unit	Content	Teaching hours
1	**Introduction to AI**. Introduction: History & overview of Artificial Intelligence, Evolution of AI ,Applications of AI, Classification of AI Systems with respect to the environment, Artificial Intelligence vs Machine Learning. Types of AI Agent, Concept of Rationality, nature of environment, structure of agents. Turing Test in AI.	**7**
2	**Problem Solving**. Uninformed Search, depth first search, breadth first search Heuristic Search Techniques: Generate-and-Test; Hill Climbing; Properties of A* algorithm, Best first Search; Problem Reduction. Constraint Satisfaction problem: Interference in CSPs; Backtracking search for CSPs; Local Search for CSPs; structure of CSP Problem. Beyond Classical Search: Local search algorithms and optimization problem, local search in continuous spaces, searching with nondeterministic action and partial observation, online search agent and unknown environments	**7**
3	**Knowledge and Reasoning** Knowledge-Based Agent in Artificial intelligence: Architecture, Approaches to designing a knowledge-based agent. Knowledge representation: Techniques of knowledge representation, Propositional logic, Rules of Inference, First-Order Logic, Forward Chaining and backward chaining in AI. Reasoning in Artificial intelligence: Types of Reasoning and Probabilistic reasoning, Uncertainty.	**8**

4	**Learning & Game** Learning from examples: Overview of different forms of learning, Supervised learning, Unsupervised learning, Learning Decision Trees, regression and classification with linear model, SVM. Ensemble learning, Reinforcement learning. Artificial neural network Search under adversarial circumstances. Optimal decision in game, minimax algorithm, alpha-beta pruning, games with an element of chance imperfect real time decision, stochastic games, partially observable games, stat of art game program, alternative approaches	**8**

Text Books:
1. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition,Prentice Hall.
2. J. Gabriel, Artificial Intelligence: Artificial Intelligence for Humans (Artificial Intelligence, Machine Learning), Create Space Independent Publishing Platform, First edition, 2016

Reference Books: .
1. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI. 2010 2. S Kaushik, Artificial Intelligence, Cengage Learning, 1st ed.2011.
2. Ric, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata McGraw Hill.
3. Luger, G.F. 2008. Artificial Intelligence -Structures and Strategies for Complex Problem Solving, 6th edition, Pearson
4. Alpaydin, E. 2010. Introduction to Machine Learning.2nd edition, MIT.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCL402	**Course name:** Operating System
Course category: PCC	
Credits: 2 **Teaching scheme:** L -2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Computer Organization and Architecture & Fundamentals of Data Structures.	
Course Objectives: 1. Provide basic knowledge of computer operating system structures and functioning.	
2. Compare several different approaches to memory management, file management and process management.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Outline the basic concept of operating systems	
CO2: Examine the working of various scheduling/allocation approaches	
CO3: Understand deadlock, prevention and avoidance algorithms.	
CO4: Compare and contrast various memory management schemes.	

Contents –

Unit	Content	Teaching hours
1.	**Introduction –** Operating System overview objectives and functions, Types of Operating system, The Evolution of Operating System, Operating System Structure and Operations- System Services, Systems Calls, System Programs, System structure.	7
2.	**Process & CPU Scheduling -** Process Concept, Process Scheduling, and Operations on Processes, Inter process Communication, Threads, Interprocess Communication, Scheduling criteria	8
3.	**Process Synchronization -** Scheduling algorithms and performance evaluation.,Classical Problems of synchronization, Semaphores, Monitors, Deadlock- Principles of Deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.	7

4.	**Memory Management & File Management-** Basic concept, Logical and Physical address map, Memory Partitioning: Fixed Partitioning, Dynamic Partitioning, Paging, Segmentation, Virtual Memory - Basics of Virtual Memory – Page Replacement Algorithms: Optimal, First in First Out (FIFO), Not recently used (NRU) and Least Recently used (LRU). **File Management:** Overview, File Organization and Access, File Directories, File System structure, Allocation methods (contiguous, linked, indexed), directory implementation efficiency and performance	8

Text Books:

1. Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Operating System Concepts, Wiley Publication, 8th Edition, 2008.
2. William Stallings, Operating System: Internals and Design Principles, Prentice Hall,8th Edition.
3. Andrew S. Tanenbaum, Modern Operating System, PHI Publication, 4th Edition, 2015.

Reference Books:

1. D. M. Dhamdhere, Systems Programming and Operating Systems, McGraw-Hill,2nd Edition, 1996.
2. Garry Nutt, Operating Systems Concepts, Pearson Publication, 3rd Edition, 2003.
3. Harvey M. Deitel, An Introduction to Operating Systems, Addison-Wesley Publication, 2ndEdition,1990
4. Tom Adelstein and Bill Lubanovic, Linux System Administration, O'ReillyMedia

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCL403 **Course name: Data Science with R**

Course category: PCC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Knowledge of python programing
Course Objectives:
1. Gain proficiency in R programming for statistical analysis.
2. Use R to conduct statistical analyses, solve problems, and interpret data to inform decisions
3. Implement Monte Carlo simulations in R to model systems and estimate solutions.
4. Develop optimization algorithms in R to minimize and maximize functions in real-world problems.
Course Outcomes: At the end of the course, the students will be able to -
CO1: Understand to use and program in the programming language R
CO2: Understand to use R to solve statistical problems
CO3: Implement and describe Monte Carlo the technology
CO4: Implement minimize and maximize functions using R.

Contents –

Unit	Content	Teaching hours
1.	**Overview of R:** Introduction: Overview of R, R data types and objects, reading and writing data, sub setting R Objects, Essentials of the R Language, Installing R, Running R, Packages in R, Variable names and assignment, Operators, Integers, Factors, Logical operations, Control Statements: if else,for , while loops	**7**
2.	**Function and Classes:** functions, Introduction to Functions: built in and User Defined, preview of Some Important R Data Structures, Vectors, Character Strings, Matrices, Lists, Data Frames Classes Vectors: Generating sequences, Vectors and subscripts, Extracting elements of a vector using subscripts, Working with logical subscripts, Scalars, Vectors, Arrays, and Matrices, Adding and Deleting Vector Elements, Obtaining the Length of a Vector, Matrices and Arrays as Vectors Vector Arithmetic and Logical Operations, Vector Indexing, Common Vector Operations	**8**
3.	**Lists and Its Operations** Lists: Creating Lists, Use of List, Difference between Python and R list ,General List Operations, List Indexing Adding and Deleting List Elements, Conversion of List to Vector and matrix,Getting the Size of a List, DATA FRAMES, Creating Data Frames, Accessing Data Frames, show it graphically	**7**
4.	**Mathematical Operations** Factors And Tables: Factors and Levels, Common Functions Used with Factors, Working with Tables, Matrix/Array-Like Operations on Tables, Math Functions, Calculating a Probability, Cumulative Sums and Products, Minima and Maxima, Calculus, Functions for Statistical Distributions	**8**

Text Books:

1.R Programming for Data Science by Roger D. Peng

2.The Art of R Programming by Norman Matloff Cengage Learning India

Reference Books: 1.Hadley Wickham, Garrett Grolemund, R for Data Science: Import, Tidy, Transform, Visualize, and Model Data 1st Edition, O'Reilly 2.Tilman M. Davies, The book of R a first course in programming and statistics, no starch press
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCP401	**Course name: Artificial Intelligence Lab**
Course category: PCC	
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Linear Algebra, Calculus, Probability, Statistics, Python, Data Structures in python	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: To understand the fundamentals of Artificial Intelligence.	
LO2: To design smart system using different search strategies of Artificial Intelligence	
LO3: Demonstrate knowledge of reasoning and knowledge representation for solving real world Problems.	
LO4: Illustrate the construction of learning and expert system.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Study and understand simple reflex and Model Based Agent	2
2.	Program to implement Uninformed Search Technique: Breadth First Search	2
3.	Program to implement Uninformed Search Technique: Depth First Search	2
4.	Write a Program to solve 8 Queen Problem	2
5.	Program to implement Informed Search Technique: A* Algorithm	2
6.	Implement Forward Chaining Algorithm	2
7.	Implement backward Chaining Algorithm	2
8.	Implement Supervised & Unsupervised Learning using Python	2
9.	Implement Mini-Max algorithm for game playing (Alpha-Beta pruning)	2
10.	Chatbot in Python	2

Text Books:

1. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition, Prentice Hall.
2. 2. J. Gabriel, Artificial Intelligence: Artificial Intelligence for Humans (Artificial Intelligence, Machine Learning), Create Space Independent Publishing Platform, First edition, 2016

Reference Books:

1. Introduction to Artificial Intelligence & Expert Systems, Dan W Patterson, PHI. 2010 2. S Kaushik, Artificial Intelligence, Cengage Learning, 1st ed.2011.
2. Ric, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata Mc Graw Hill.
3. Luger, G.F. 2008. Artificial Intelligence -Structures and Strategies for Complex Problem
4. Solving, 6th edition, Pearson
5. Alpaydin, E. 2010. Introduction to Machine Learning. 2nd edition, MIT.

Online Resources:

1. www.tutorialspoint.com//ArtificialIntelligence/
2. https://swayam.gov.in/

Course code: CSI24PCP402	**Course name:Operating System Lab**
Course category:PCC	
Credits: 01 **Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: 1. Proficiency in the C programming language is essential for Linux programming.. 2. Gain a basic understanding of operating system concepts like process.	
Course Objectives: The course objective is to analyze fundamentals of the Linux operating system, including installation, configuration, administration, file management, and security.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Install and administer server services.	
LO2: Manage Users and Groups.	
LO3: Setup and manage policies.	
LO4: Implement File Services.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1	Download Install and configure Ubuntu LINUX Operating system on your machine.	2
2	Execute General purpose and process related command on your machine	2
3	Execute File and Directory manipulation related command on your machine	2
4	Write a shell script to find greatest among three numbers, Fibonacci series using conditional statement.	2
5	Write a shell script to list all files in a directory.	2
6	Write a shell script to find whether a give files present. Write a shell script to check read, write permission of given file.	2
7	Write a C Program to calculate total waiting and Turnaround time of n process with FCFS algorithm.	2
8	Simulate algorithm for deadlock prevention	2
9	Simulate a page replacement algorithm.	2
10	Write a C Program to implement FIFO page replacement algorithm	2

Text Books:

1. Abraham Silberschatz, Peter B. Galvin and Greg Gagne, Operating System Concepts, Wiley Publication, 8th Edition, 2008.
2. William Stallings, Operating System: Internals and Design Principles, Prentice Hall,8th Edition.
3. Andrew S. Tanenbaum, Modern Operating System, PHI Publication, 4th Edition, 2015.

Reference Books:

1. D. M. Dhamdhere, Systems Programming and Operating Systems, McGraw Hill, 2nd Edition, 1996.
2. Garry Nutt, Operating Systems Concepts, Pearson Publication, 3rd Edition, 2003.
3. Harvey M. Deitel, An Introduction to Operating Systems, Addison-Wesley Publication, 2 Nd Edition,1990
4. Tom Adelstein and Bill Lubanovic, Linux System Administration, O'ReillyMedia

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCP403	**Course name: Data Science with R Lab**	
Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1. Knowledge of python programing		
Course Objectives: 1.Gain proficiency in R programming for statistical analysis. 2. Use R to conduct statistical analyses, solve problems, and interpret data to inform decisions. 3. Implement Monte Carlo simulations in R to model systems and estimate solutions. 4. Develop optimization algorithms in R to minimize and maximize functions in real- world problem		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Understand to use and program in the programming language R		
LO2: Understand to use R to solve statistical problems		
LO3: Implement and describe Monte Carlo the technology		
LO4: Implement minimizes and maximizes functions using R		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Download and install R-Programming environment and install basic packages using install. packages() command in R.	2
2.	Learn all the basics of R-Programming (Data types, Variables, Operators etc,.)	2
3.	Implement R Loops with different example.	2
4.	Learn the basics of functions in R and implement with examples.	2
5.	Write R command to Illustrate summation, subtraction, multiplication, and division operations on vectors using vectors. Enumerate multiplication and division operations between matrices and vectors in R console	2
6.	Write R command to Illustrates the usage of Vector subsetting and Matrix subsetting Write a program to create an array of 3×3 matrixes with 3 rows and 3 columns	2
7.	Write an R program to draw i) Pie chart ii) 3D Pie Chart, iii) Bar Chart along with chart legend by considering suitable CSV file	2
8.	Create a CSV file having Speed and Distance attributes with 1000 records. Write R program to draw i) Box plots ii) Histogram iii) Line Graph iv) Multiple line graphs v) Scatter plotto demonstrate the relation between the cars speed and the distance.	2
9.	Create and manipulate factor variables and tables using R.	2
10.	Use R's built-in mathematical functions for data analysis.	2
11.	Mini Project	2

Text Books: 1.R Programming for Data Science by Roger D. Peng 2.The Art of R Programming by Norman Matloff Cengage Learning India
Reference Books: 1.Hadley Wickham, Garrett Grolemund, R for Data Science: Import, Tidy, Transform, Visualize, and Model Data 1st Edition, O'Reilly 2.Tilman M. Davies, The book of R a first course in programming and statistics, no starch press
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: MGM56VEL102	**Course name: Constitution of India**
Course category: PCC	
Credits: 2	**Teaching scheme:** L-2 **Evaluation scheme:** CA–30, ESE–20
Pre-requisites: Indian History	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand and Explain the Significance of Indian Constitution as the Fundamental Law of the Land.	
CO2: Exercise Fundamental Rights in Proper Sense at the Same Time Identify Responsibilities in National Building.	
CO3: Analyse the Indian Political System, Electoral Process, Emergency Provisions and Amendment Procedure.	
CO4: Understand the Powers and Functions of the Union, State and Local Governments in detail	

Contents –

Unit	Content	Teaching hours
1.	**Historical Background and Philosophy of Indian Constitution** Societies before and after the Constitution Adoption, Introduction to the Indian constitution, Making of the Constitution. Role of the Constituent Assembly, Preamble and Nature of Indian Constitution, Salient Features of Indian Constitution.	**8**
2.	**Fundamental Rights & Duties** Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion. Cultural and Educational Rights, Right to Constitutional Remedies, Fundamental Duties.	**8**
3.	**Directive Principles of State Policy and Other Important Constitutional Provisions** Meaning and Significance of Directive Principles, Economic Principles, Political Principles. Emergency Provisions under Indian Constitution, Election Commission.	**8**
4.	**Organs of Government** Centre and State Legislature, Centre and State Executive, Centre and State Judiciary, Local Self Government.	**6**

Text Books:

1. D. D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2013.
2. Pravin Kumar, Basic Mechanical Engineering, Pearson Education; Second Edition, 2018.
3. M. P. Jain, Outline of Indian Legal and Constitutional History, Lexis Nexis, 2014 .Constitutional Law of India - J.N.Pandey.

Reference Books:

1. Austin, the Constitution of India, Cornerstone of a Nation. Oxford University press 1966.

Online Resources: 1.https://onlinecourses.nptel.ac.in/

Course code: CSI24PCL404 Course name: Business Management & Financial Accounting **Course category: PCC**		
Credits: 2	Teaching scheme: L-2	Evaluation scheme: CA–60, ESE–40
Pre-requisites: Basic knowledge of business plan		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Analyse the Risk and Return of Alternative Sources of Financing.		
CO2: Estimate Cash Flows from a Project, including Operating, Net Working Capital and Capital Spending.		
CO3: Define Basic Terminology used in Finance and Accounts.		
CO4: Prepare Financial Statements		

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Business Management** Aims, Objective and Function of Business Management, Principles of Management. Concept of Business Finance, Goals & Objectives of Financial Management, Sources of Financing-Long Term Financing and Short Term Financing.	**7**
2.	**Business Capital Management** Concept of Business Working Capital, Significance, Types. Adequacy of Working Capital, Factors Affecting Working Capital Needs. Financing Approaches for Working Capital, Methods of Forecasting, Working Capital Requirements.	**8**
3.	**Basics of Financial Accounting** Meaning, Scope and Importance of Financial Accounting, Financial Accounting - Concepts and Conventions, Classification of Accounts. Rules and Principles Governing Double Entry Book-Keeping System, Meaning, Preparation of Journal, Ledger, Cash Book & Trial Balance.	**7**
4.	**Financial Statement Preparation and analysis** Preparation of Financial Statements. Profit & Loss Account, Balance Sheet. Ratio Analysis.	**8**

Text Books:

1. Financial Management by Khan & Jain, Text, Problem & Cases, Tata McGraw Hill Publication 5th Edition.
2. Tulsian Financial Management by Dr.P.C.Tulsian, S.Chand Publication 5th Edition.
3. Taxman's Financial Management by Ravi M. Kishore, Taxmann 2017 Edition.

Reference Books:

1. A Textbook of Financial, Cost & Management Accounting by Dr.P.Periasamy, Himalaya Publishing House.
2. 2. Fundamentals of Financial Management by Bhabhtosh Banerjee, PHI publication, 2nd Edition.

Online Resources:1. https://onlinecourses.nptel.ac.in/

Course code: CSI24FPJ405		**Course name:** Field Project
Course category: FP		
Credits: 2	**Teaching scheme: P-4**	**Evaluation scheme:** TW: 50, PR:–00
Pre-requisites: Basic Concepts of Web		
Course Objectives:		
1. To get familiar with client-server architecture.		
2. To explore recent web technologies**.**		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Design web pages using HTML , CSS and JavaScript.		
CO2: Build appropriate server-side scripts using PHP.		
CO3: Create dynamic web application using PHP and MySQL.		
CO4: Develop solution to real world problem using appropriate web technologies and content management system.		
Contents –		

Sr. No.	**List of Experiments**	**Practical Hours**
1	Develop informative web pages using HTML5 (Use all possible formatting for example font, colour etc.). Web page should include various images, links within the page, links to other pages for navigation, new tabs.	02
2	Design an online registration form for any application.	02
3	Use Cascaded Style Sheets(CSS) to style the web pages designed. Make use of border, margins, padding, navigation, dropdown list etc.	02
4	Design an XML document to store information about a student in an engineering college affiliated to MGM University. The information must include RegistrationNo, Name, Name of Brach, Year of Joining, and e- mail id. Make up sample data for 3 students. Create a CSS style sheet and use it to display the document.	02
5	Design, Develop and Deploy new web application using Bootstrap in consultation of course instructor.	02
6	Write a Java Script to design a simple calculator to perform the following operations: sum, product, difference and division.	02
7	Write a JavaScript that calculates the squares and cubes of the numbers from 0 to 10 and outputs HTML text that displays the resulting values in an HTML table format.	02
8	Develop and demonstrate a HTML5 file that includes JavaScripts that uses functions	02

	for the following problems: Parameter: A string Output: The position in the string of the left-most vowel. Parameter: A number Output: The number with its digits in the reverse order.	
9	Write a JavaScript code that displays text "TEXT- GROWING" with increasing font size in the interval of 100ms in RED COLOR, when the font size reaches 50pt it displays "TEXT-SHRINKING" in BLUE colour. Then the font size decreases to 5pt.	02
10	Write a PHP program to keep track of the number of visitors visiting the web page and to display this count of visitors, with proper headings.	02
11	Write a PHP program to display a digital clock which displays The current time of the server.	02
12	Write a PHP program to display the student records which are stored in the database.	02
13	Write the PHP programs to do the following: a. Implement simple calculator operations. b. Find the transpose of a matrix. c. Multiplication of two matrices. d. Addition of two matrices.	02
14	Develop web site using any CMS tool which falls into one of the categories blogs, social networking, News updates, Wikipedia, E commerce store. Website must include home page, and at least 3 pages.	02
15	Study Experiment/Project: Develop a web application project using the languages and concepts learnt in the exercises with a good looks. You can use any web technologies and Frameworks and databases.	02

Text Books:

1. Ralph Moseley & M.T. Savaliya,—Developing Web Applications ‖, Wiley publications, 2007
2. Kogent Learning Solutions Inc,Web Technologies, Black book, Dreamtech Press.2009

Reference Books:

1. IvanBayross, Web Enabled Commercial Application Development Using HTML, JavaScript, DHTML and PHP, BPB Publications.2005, Third Edition

2. StevenM. Schafer," HTML, XHTML and CSS", Wiley India Edition., 5th Edition 2011

Online Resources:

1. MicrosoftAcademyCoursesonHTML5&CSS.3
2. Udemy and Coursera Courses on Web Technologies.
3. Spoken Tutorial Course on PHP and MySql.
4. W3Schools, Tutorials Point tutorials on HTML,CSS, JavaScript,PHP,Bootstrap.

SEM-VIII

Course code: CSI24PCL451	**Course name: Machine Learning**
Course category: PCC	
Credits: 2 **Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Data Structures, Knowledge on statistical methods, python	
Course Objectives:	
1. To understand introduction to Machine Learning, Data Precprocessing, Performance Metrics.	
2. To apply Different Learning Algorithms and ML Techniques.	
3. To apply Classification and regression to ML Models.	
4. To understand advanced Machine Learning Techniques	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Recognize the characteristics of machine learning that makes it useful to real-world problems.	
CO2: Apply the different supervised learning methods of support vector machine and tree based models.	
CO3: Use different linear methods for regression and classification with their optimization through different regularization techniques.	
CO4: Implement and assess the results of advanced machine learning algorithms.	

Contents –

Unit	Content	Teaching hours
1	**Introduction to ML** Introduction to Machine Learning: History of ML, Examples of Machine Learning Applications Learning Types ML Life cycle, AI & ML, dataset for ML Data Pre-processing, Training versusTesting, Positive and Negative Class, Cross-validation.	7
2	**Learning** Types of Learning: Supervised, Unsupervised and Semi-Supervised Learning. Supervised: Learning a Class from Examples, Types of supervised Machine learning Algorithms, Unsupervised: Types of Unsupervised Learning Algorithm, 2.4 Dimensionality Reduction: Introductionto Dimensionality Reduction, Subset Selection Introduction to Principal Component Analysis.	7

3	**Classification & Regression** Classification: Binary and Multiclass Classification: Assessing Classification Performance, Handling more than two classes, Multiclass Classification-One vs One, One vs Rest. Regression: Assessing performance of Regression Error measures, Overfitting and Underfitting, Catalysts for Overfitting, VC Dimensions.	**8**
4	**Advanced Machine Learning Techniques** Decision Trees and Random Forests, Support Vector Machines (SVM), Clustering algorithms: K-Means, Hierarchical clustering, Neural Networks and Deep Learning basics, Introduction to TensorFlow and PyTorch.	**8**

Text Books:

1. Peter Flach: Machine Learning: The Art and Science of Algorithms that Make Sense of Data, Cambridge University Press, Edition 2012.

Reference Books:

1“Introduction to Machine Learning”,EthemAlpaydin, PHI 2nd Edition-2013.
2."Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido.
3. “Elements of Statistical Learning" by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
4."Python Machine Learning" by Sebastian Raschka and VahidMirjalili
5."Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto

Online Resources:

1. https://www.tutorialspoint.com/machine_learning/index.htm
2.https://www.geeksforgeeks.org/machine-learning/
3. https://onlinecourses.nptel.ac.in/

Course code:CSI24PCL452 Course name: Design and Analysis of Algorithm **Course category:** PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Basic programming knowledge, Data Structure
Course Objectives: 1. To learn fundamentals of algorithm design techniques. 2. To understand basic knowledge of computational complexity, approximation and randomized algorithms, selection of the best algorithm to solve a problem. 3. To design and analyse the performance of algorithms and to compare algorithms with respect to time and space complexity. 4. To develop proficiency in problem solving and programming
Course Outcomes: At the end of the course, the students will be able to -
CO1: Gain understanding of concepts of time and space complexity in terms of worst, average and best case using Asymptotic Notations.
CO2: Design standard algorithms such as sorting, searching, problems involving graphs etc. using various design techniques and compute complexity measures of these algorithms
CO3: Design algorithms for optimization problems.

Contents – CSI24PCL453

Unit	Content	Teaching hours
1	**Introduction to Algorithms:** Definition of Algorithms, Properties of Algorithms, Sorting techniques–Bubble Sort, Insertion sort, Selection sort, Asymptotic Notations, Performance Analysis of algorithms.	9
2	**Divide and Conquer:** Introduction to Divide and Conquer Technique, Binary Search, Min-Max, Merge Sort, Quick Sort, Heap sort	7
3	**Greedy Algorithm:** Introduction to Greedy Technique, optimal storage on tapes, Optimal Merge Patterns, Huffman Coding, Knapsack Problem, Activity Selection Problem, Job Sequencing with Deadline, Minimum Spanning Tree, Single- Source Shortest Path Algorithm	9
4	**Dynamic Programming:** Introduction to Dynamic Programming Longest common subsequence, Multistage graphs, All pair shortest path algorithm- Floyd-Warshall Algorithm, The Knapsack Problem, Traveling Salesperson, Application of Dynamic Programming	10

5	**Branch and Bound, Backtracking Strategy:** Backtracking: General strategy and characteristics, N-queen problem, Four– Queens Problem, Eight–Queen Problem, graph coloring, sum of subsets, Hamiltonian Circuit Problem Branch and Bound: Introduction, 15- Puzzle Problem, Comparisons between Backtracking and Branch and Bound	10

Text Books:

1. E. Horowitz, S. Sahni, and S. Rajasekaran, *Computer Algorithms*, 2nd ed. Hyderabad, India: Universities Press, 2008.
2. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed. Cambridge, MA, USA: MIT Press, 2009.

Reference Books:

1. B. K. Joshi, "Data Structures and Algorithms in C++", Tata McGraw Hill Education, 2010.
2. G. T. Heineman, Gary Pollice, Stanley Selkow, "Algorithms in a Nutshell", Shroff Publication, 1st Edition, 2008.

Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CSI24PCL453	**Course name: Data Analytics**
Course category: PCC	
Credits: 2 **Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Discrete Mathematics	
Course Objectives: 1. To provide students with a foundational understanding of Data Analytics, its importance, and the process of extracting meaningful insights from data. 2. To Gain familiarity with commonly used tools and technologies in Data Analytics. 3. To introduce machine learning concepts and techniques.	
1.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Define the concept of Data Analytics and its relevance in various domains.	
CO2: Apply various techniques for data cleaning and preprocessing.	
CO3: Gain familiarity with commonly used tools and technologies in Data Analytics.	
CO4: Apply hypothesis testing techniques to draw conclusions from data.	
CO5: Illustrate the fundamentals of machine learning and its applications.	

Contents –

Unit	Content	Teaching hours
1	**Introduction to Data Analytics with Excel** Data Analytics: An Overview, Importance of Data Analytics Types of Data Analytics: Descriptive Analysis, Diagnostic Analysis, Predictive Analysis, Prescriptive Analysis, Visual Analytics Life cycle of Data Analytics, Quality and Quantity of data, Measurement Data Types, Measure of central tendency, Measures of dispersion Finding mean, median, and mode, IQR, STDEV, VAR, for a given dataset using Excel functions Import dataset in Excel and perform data cleaning	7
2.	**Statistical Analysis** Graphical techniques, box plot, skewness and kurtosis, Descriptive Stats Outliers : python code and manual calculations Corelation and Regression Anova and Chi Square Scatter Diagram	7

3.	**Advanced Data Visualization and Chart Customization Techniques** Creating a Simple Chart, Charting Non-Adjacent Cells Creating a Chart Using the Chart Wizard, Modifying Charts, Moving an Embedded Chart, Sizing an Embedded Chart Changing the Chart Type, Changing the Way Data is Displayed, Moving the Legend Formatting Charts, Adding Chart Items, Formatting All Text, Formatting and Aligning Numbers, Formatting the Plot Area, Formatting Data Markers Pie Charts, Creating a Pie Chart Moving the Pie Chart to its Own Sheet Adding Data Labels, Exploding a Slice of a Pie Chart	8
4.	**Data Visualization using Python** Data Visualization, Data Analytics Tools for Decision Making, Overview of Matplotlib and its role in data visualization, Installing and setting up Matplotlib in Python Basic plotting with Matplotlib and legends to plots Changing figure size and aspect ratio Exporting and Saving Visualizations: Saving plots in different formats (PNG, PDF, SVG) Adjusting the resolution and quality of saved plots, creating interactive visualizations using Matplotlib widgets	8

Text Books:

1."Python for Data Analysis" by Wes McKinney
2. "An Introduction to Statistical Learning" by Gareth James et al.
3. "Statistics" by Robert S. Witte and John S. Witte
4. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron

Reference Books:

1."Hands-On Exploratory Data Analysis with R" by Rishi K. Narang
2. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
3."Practical Statistics for Data Scientists" by Peter Bruce and Andrew Bruce

Online Resources:

1. https://www.geeksforgeeks.org/
2. https://onlinecourses.nptel.ac.in/

Course code: CSI24PCP451	**Course name: Machine Learning Lab**	**Course category:** PCC
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Data Structures, Knowledge on statistical methods, python		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Recognize the characteristics of machine learning that makes it useful to real-world problems		
LO2: Apply the different supervised learning methods of support vector machine and tree based models.		
LO3: Use different linear methods for regression and classification with their optimization through different regularization techniques.		
LO4: Implement and assess the results of advanced machine learning algorithms.		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Write a python program to import and export data using Pandas library functions.	2
2.	Cleanse and preprocess a dataset, handling missing values and outliers.	2
3.	Use a dataset to implement and visualize a simple linear regression model.	2
4.	Evaluate machine learning models using accuracy, precision, recall, and F1 Score.	2
5.	Implement a decision tree and a random forest model on a dataset.	2
6.	Apply SVM to classify data points in a two-dimensional space.	2
7.	Implement K-Means clustering on a dataset.	2
8.	Build a basic neural network using a popular library like TensorFlow or PyTorch.	2
9.	Use a pre-trained model or train a basic model for image recognition.	2
10.	Case Study	2

Text Books:

1. Peter Flach: Machine Learning: The Art and Science of Algorithms that Make Sense of Data, Cambridge University Press, Edition 2012.

Reference Books:

1. “Introduction to Machine Learning”,EthemAlpaydin, PHI 2nd Edition-2013.
2."Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido.
3. “Elements of Statistical Learning" by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
4."Python Machine Learning" by Sebastian Raschka and VahidMirjalili
5."Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto

Online Resources:

1. https://www.tutorialspoint.com/machine_learning/index.htm
2. https://www.geeksforgeeks.org/machine-learning/
3. https://onlinecourses.nptel.ac.in/

Course code: CSI24PCP452 **Course Name: Design and Analysis of Algorithm Lab**
Course Category: PCC
Credits: 01 **Teaching scheme:** P-2 **Evaluation Scheme:** TW-30, PR-20
Pre-requisites: Basic programming knowledge, Data Structure**.**
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Gain understanding of concepts of time and space complexity in terms of worst, average and best case using Asymptotic Notations.
LO2: Design standard algorithms such as sorting, searching, problems involving graphs etc. using various design techniques and compute complexity measures of these algorithms.
LO3: Design algorithms for optimization problems.

List of Experiments–

Sr.No.	Name of Experiment
1.	Program to implement Insertion, bubble and selection sort.
2.	Program to implement Heap sort.
3.	Program to implement binary search algorithm using Divide & Conquer method.
4.	Program to implement max min algorithm using Divide & Conquer method
5.	Implementation of Merge Sort algorithm using Divide & Conquer method
6.	Implementation of Quick Sort algorithm using Divide & Conquer method.
7.	Program to implement knapsack problem using greedy method
8.	Program to implement job sequencing with deadlines using greedy method.
9.	Program to implement Prim's/Kruskal's Algorithm using Greedy method.
10.	Program to implement graph traversal technique: BFS and DFS
11	Program to implement 8-queens problem using backtrack method.

Text Books:

1. E. Horowitz, S. Sahni, and S. Rajasekaran, *Computer Algorithms*, 2nd ed. Hyderabad, India: Universities Press, 2008.
2. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed. Cambridge, MA, USA: MIT Press, 2009.

Reference Books:

1. B. K. Joshi, "Data Structures and Algorithms in C++", Tata McGraw Hill Education, 2010.
2. G. T. Heineman, Gary Pollice, Stanley Selkow, "Algorithms in a Nutshell", Shroff Publication, 1st Edition, 2008.

Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CSI24PCP453	**Course name: Data Analytics Lab**

Course category: PCC		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme:** TW–30, PR–20
Pre-requisites: Discrete Mathematics		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: Define the concept of Data Analytics and its relevance in various domains.		
LO2: Apply various techniques for data cleaning and pre-processing.		
LO3: Gain familiarity with commonly used tools and technologies in Data Analytics.		
LO4: Apply hypothesis testing techniques to draw conclusions from data.		
LO5: Illustrate the fundamentals of machine learning and its applications.		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	**Study and Perform Statistical Analysis in Excel:** a. Calculate mean, median, and mode for a given dataset using Excel functions (AVERAGE, MEDIAN, MODE). b. Calculate range, interquartile range (IQR), variance, and standard deviation using Excel functions (STDEV, VAR).	2
2.	**Create a table to execute the function using dashboard and perform operations:** **Create a Data Table** a. Import a sample dataset (e.g., sales data) into Excel. b. Converts the dataset into an Excel Table using the "Format as Table" feature and apply appropriate styles. c. Create a dashboard sheet that summarizes key metrics (e.g., total sales, average sales per region) using tables. d.Identify and remove duplicates from a dataset.	2
3.	Perform the different types of function using linear regression. a. Construct a box plot using the Insert Chart feature to identify the median, quartiles, and outliers of a dataset. b.Find outliers from given dataset c. Conduct a t-test to compare means between two groups	2
4.	**Calculate confidence** **Intervals & Apply Chi-square test for independence.** a. Calculate confidence intervals b. Conduct a Chi-square test	2

5.	**Create a simple chart to visualize the data sets.** a. A simple bar chart to visualize data sets b. A chart using non-adjacent cells to visualize data from different ranges. **Create a Chart Using the Chart Wizard** a Select the chart you created and experiment with the Chart Tools options b. Modifying Charts c. Moving an Embedded Chart Sizing an Embedded Chart	2
6.	**Change the chart type with adding data labels, axis format, and adjusting the gridlines.** **Change the Chart Type** a. Create a basic bar chart using a dataset and change its type to a different chart b. Experiment with different data display options, such as adding data labels, changing the axis format, and adjusting the gridlines Experiment with position and style of the legend	2
7.	**Design a pie chart:** a.Create a pie chart from a dataset b.Move the pie chart to a new worksheet for better visibility c. Emphasize a specific category by exploding a slice of the pie chart d. Customize the appearance of the pie chart for better presentation	2
8.	Generate and Save the plot in various formats: Create different types of plots.Write a Python script to save the plot in different formats: PNG, PDF, and SVG.	2
9.	Application of data analytics across various industries through case study	2
10.	Mini Project	2

Text Books:

1. "Python for Data Analysis" by Wes McKinney
2. "An Introduction to Statistical Learning" by Gareth James et al.
3. "Statistics" by Robert S. Witte and John S. Witte
4. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron

Reference Books:

1. "Hands-On Exploratory Data Analysis with R" by Rishi K. Narang
2. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville
3. "Practical Statistics for Data Scientists" by Peter Bruce and Andrew Bruce

Course code: CSI24VSP455 **Course name: UI UX Design Lab** **Course category:** VSEC
Credits: 2 **Teaching scheme:** P-4 **Evaluation scheme:** TW–60, PR–40
Pre-requisites:1. Knowledge of python programing, & Front End desing
Course Objectives: 1. Explain design thinking concept 2. Select appropriate visual design for given problem. 3. Create interactions using design tool 4. Create innovative design prototype for given applications.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Use design tool to collect user requirements.
LO2: Use any Design tool to create a 'blank project'.
LO3: Create library/repository of created assets in the design tool.
LO4: Use components and navigations to design quiz like page in design tool.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Use Design tool for user requirement collection and analysis	2
2.	Use Design tool for user requirement collection and analysis of various interfaces such as kiosks	2
3.	Recreate a given user interface using any open source design tool	2
4.	Create grid system for the given screen using any design tool	2
5.	Design given user interface using various components such as auto-layouts in the design tool	2
6.	Use horizontal scrolling to create pages for given website/ App	2
7.	Use vertical scrolling for a given website/ App	2
8.	Recreate given website for UI design, color, images, interactions, menu	2
9.	Create navigations for the given website/ App	2
10.	Design a quiz for given user interface	2
11.	Create any two gamification effects for given user interface in given scenario	2
12.	Create gamification for task completion in website such as LMS/ retail website/ banking website	2
13.	Create any five micro animations for the given user interface in given scenario	2
14.	Create prototyping with different interactions – tab, click, hover, delay. for the given user interface	2
15.	Convert created prototype in HTML page(s)	2
16.	Mini Porject	2

Text Books:

1. Kilian Langenfeld, Design Thinking for Beginners

Reference Books:

1. Jesse James Garrett , The Elements of User Experience: User-Centered Design for the Web and Beyond , New Riders Publishing, 201 West 103 Street, Indianapolis, IN 46290 800-545-5914,ISBN:978-0-321-68368-7
2. Fabio Staiano, Designing and Prototyping Interfaces with Figma, Packt Publishing Ltd, Grosvenor House, 11 St Paul's Square, Birmingham, B3 1RB ISBN-10: 180056418X

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24HSL456	**Course name: Entrepreneurship Development**
Course category: EEMC	
Credits: 2	**Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: To Develop Skills related to Various Functional Areas of Management (Marketing Management, Financial Management, Operations Management, Personnel Management etc.).	
CO2: To Develop Skills related to Project Planning And Business Plan Development.	
CO3: To Determine and Compare Various Skills of Successful Entrepreneurs, Business Organizations and Business Laws.	
CO4: To Examine Essentials to Avoid Failure in Entrepreneurship	

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Entrepreneurship Development** Entrepreneurship Development, Evolution of Entrepreneurship, Meaning, Definition and Concept of Enterprise. Concepts of Entrepreneurship, Role of Entrepreneurship in Economic Development, Factors Affecting Entrepreneurship. Entrepreneurial Development models and Theories, Entrepreneurial Trait Tests, Business Opportunity Identification.	**8**
2.	**Entrepreneur** Major Types of Entrepreneurship, Techno Entrepreneurship, Women Entrepreneurship, Social Entrepreneurship, Intrapreneurship (Corporate entrepreneurship), Rural Entrepreneurship, Family Business etc. The Entrepreneurial Decision Process, Entrepreneurial Environment – Political, Legal, Technological, Natural, Economic, Socio – Cultural etc.	**7**
3.	**Project Planning, Skills for Successful Entrepreneurs & Business Organizations and Business Laws:** Project Report, Project Appraisal, Setting Up an Industrial Unit – Procedure And Formalities in Setting Up an Industrial Unit; Business Plan Development. Communication Skills, Creativity and Problem-Solving, Innovation, Negotiation Skills, Risk Management. Types of Business Organizations – Sole, Proprietorship, Joint Hindu Family Business, Partnership, Limited Liability Partnership (LLP), Corporate Governance, Franchising, Business Laws in India to Start Business.	**8**
4.	**Concepts of Entrepreneurship Failure** Issues of Entrepreneurial failure, Reasons of Entrepreneurial Failure, Essentials to Avoid Failure in Entrepreneurship. Case Study of Failure in Entrepreneurship.	**7**

Text Books: 1. Dynamics of Entrepreneurship Development by Vasant Desai. 2. Entrepreneurship Development and small business management by Poornima M. Charantimath. 3. Dynamics of Entrepreneurship Development by Vasant Desai.
Reference Books: 1.Innovation and Entrepreneurship by Peter F. Drucker 2.Kathleen R Allen, Launching New Ventures, An Entrepreneurial Approach, Cengage Learning, 2016.Vijay Sathe, Corporate Entrepreneurship, Cambridge, 2009.
Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: MGM24VEL201 **Course name:** Energy Conservation & Management
Course category: VEC
Credits: 2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–30, ESE–20
Duration of ESE: 01 Hour
Pre-requisites: Industrial Automation
Course Objectives:
1. To introduce the fundamentals of energy conservation, Indian energy scenario, and national energy policies.
2. To develop understanding of energy auditing methodologies, instruments, and energy management systems.
3. To explain the relationship between energy use, environment, climate change, and sustainable development.
4.To identify and evaluate energy conservation opportunities in electrical and thermal systems.
Course Outcomes: At the completion of the course, students should be able to:
CO1: Explain energy conservation principles, Indian energy scenario, and provisions of the Energy Conservation Act and related policies.
CO2: Perform basic energy audits using standard procedures and instruments, and interpret energy audit reports.
CO3: Analyze the impact of energy usage on environment and climate change, and describe global and national mitigation mechanisms.
CO4: Identify and recommend suitable energy conservation measures for electrical and thermal systems

Contents –

Unit	Content	Teaching hours
1	**Fundamentals of Energy Conservation** **Energy Scenario:** Classification of Energy, Indian energy scenario, Sectorial energy consumption (domestic, industrial and other sectors), energy needs of growing economy, energy intensity, long term energy scenario, energy pricing, energy security, energy conservation and its importance, energy strategy for the future. **Energy Conservation Act 2001 and related policies:** Energy conservation Act 2001 and its features, notifications under the Act, Schemes of Bureau of Energy Efficiency (BEE) including Designated consumers, State Designated Agencies, Electricity Act 2003, Integrated energy policy, National action plan on climate change, ECSBC code for Building Construction 2024	08
2	**Energy Audit And Instruments** Definition, objectives, and types of energy audit (preliminary and detailed). Methodology of conducting energy audit. Energy audit report format. Instruments used in energy auditing: Lux meter, power analyser, data logger, flow meter, etc. Roll of BEE, Energy Auditors and managers. Management. Bureau of Energy Efficiency (BEE) and its role. Energy Conservation Act 2001 and its features. Star rating of	08

	appliances. ISO 50001: Energy Management System (basic overview). Government initiatives and programs (PAT, ECBC, UJALA, etc.) Case studies of energy audits in industries and commercial buildings.	
3	**Energy and environment, air pollution, climate change** United Nations Framework Convention on Climate Change (UNFCC), sustainable development, Kyoto Protocol, Conference of Parties (COP), Clean Development Mechanism (CDM), CDM Procedures case of CDM – Bachat Lamp Yojna and industry; Prototype Carbon Fund (PCF). Renewable Energy for Industrial development	06
4	**Energy Conservation Opportunities** Electrical: Buildings & Lighting Systems, Motors, Pumps, Transformers Power Transmission & Distribution Thermal: Boilers, Furnaces & Waste Heat Recovery Systems, Cogeneration Systems, HVAC, Cooling Towers & DG Systems	08

Text Books:

1. K. V. Sharma, P. Venkataseshaiah, Energy management and conservation I K International Publishing House Pvt. Ltd, ISBN-13:978-9381141298
2. . Chetan singh, solanki, Renewable energy technologies: a practical guide for beginners, PHI Learning Pvt. Ltd., 2008 ISBN 8120334345, 9788120334342.
3. Shukla, Shailendra K, Green building technologies, 2023, OM Publications New Delhi.
4. Sustainable Smart Homes and Buildings with Internet of Things, ramod Singh Rathore, Abhishek Kumar, Surbhi Bhatia, Arwa Mashat, Thippa Reddy Gadekallu, 2025 Scrivener Publishing LLC

Reference Books:

1. Sam Kubba, Handbook of Green Building Design and Construction, Butterworth-Heinemann; 1st edition (30 July 2012); CBSPD - NEW DELHI -110092 , ISBN13 :978-0123851284.
2. Bureau of Energy Efficiency (BEE), Energy Conservation & Sustainable Building Code (ECSBC) – India 2024.
3. J. Vijayalaxmi, Sustainable Building Performance Towards a Greener India, Springer.

SEM-IX

Course code: CSI24PCL501 Course name: Data story Telling and Visulization **Course Category:**PCC
Credits:2 **Teaching scheme:** L-2 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites: Fundamental Knowledge of data types and structures & statistics.
Course Objectives:
1. To introduce learners to the fundamental concepts of data storytelling.
2. To develop the ability to analyze and remove irrelevant information (clutter) and with storytelling.
3. To equip students with the skills to transform raw data into meaningful visual formats.
4. To enable learners to create advanced data visualizations using various statistical distributions**.**
Course Outcomes: At the end of the course, the students will be able to -
CO1: Identify the characters in data storytelling
CO2: Eliminate clutter to grab audience attention and Construct Storytelling for the given incident
CO3: Transform data to Visuals
CO4: Create data visualization using many distributions

.

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Data StoryTelling** Concept / Importance of Context Exploratory vs. explanatory analysis Who - Your audience, You, What – Action, Mechanism, Tone, How, Example What is Data Story, make a figure for the generals The 3-minute story, Big Idea, Storyboarding.Visual effects of Data Story Telling -Choosing an effective visual - Simple text, Tables, Graphs, Points, Bars - Vertical bar chart, Horizontal bar chart	**6**
2.	**Cluttering Decluttering & Storytelling** Clutter is our enemy - Cognitive load, Clutter Introduction to Gestalt principles of visual perception Decluttering: step-by-step Focus audience's Attention – Pre-attentive attributes signal where to look Pre-attentive attributes in text and graphs : Size, Color, Position on page Dissecting model visuals - line graph, 100% stacked bars Lessons in storytelling - The magic of story, Constructing the story, The narrative structure, The power of repetition, Tactics to help ensure that your story is clear Pulling it all together for data storytelling	**8**

3.	**Data Visualization** Introduction: Ugly, Bad, and Wrong Figures Visualizing Data: Mapping Data onto Aesthetics Coordinate Systems and Axes Directory of Visualizations Visualizing Amounts - Bar Plots Visualizing Distributions - Histograms and Density Plots. Empirical Cumulative Distribution Functions and Q-Q Plots	**7**
4.	**Visualizing Distributions and Proportions** Visualizing Many Distributions at Once Visualizing Distributions Along the Vertical Axis and Horizontal Axis Visualizing Proportions: A Case for Pie Charts and Side-by-Side Bars Titles, Captions, and Tables Choosing the Right Visualization Software and Image file formats Exploring free Data Visualization Tools for e.g. Tableau, Microsoft Power BI, Google Data Studio and Open heatmap	**9**

Text Books:

Reference Books:

1. Cole Nussbaumer Knaflic: “Storytelling with data – a data visualization guide for business professionals”: Cole Nussbaumer Knaflic Wiley ISBN: 978-1-119-00225-3
2. Claus O. Wilke: “Fundamentals of Data Visualization”: O’Reilly ISBN:978-1-492-03108-6

Online Resources:

1. NPTEL / SWAYAM lectures.
2. https://careerfoundry.com/en/tutorials/data- analytics-for-beginners/storytelling-with-data/
3. https://blog.hubspot.com/marketing/great-data- visualization-examples

Course code: CCI24PCL502	Course Name: Big Data
Course Category: PCC	
Credits: 2 **Teaching scheme: L-2**	**Evaluation scheme: CA–60, ESE–40**
Pre-requisites:Fundamentals of Programming ,Basic Knowledge of Databases and SQL	
Course Objectives:	
1. Understand the characteristics, challenges, and technologies of Big Data.	
2. Learn various Big Data storage and management techniques.	
3. Explore tools like Hive and Pig for data analysis and processing.	
4. Develop skills to implement Big Data processing solutions for real-world applications.	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand characteristics and technologies of Big Data..	
CO2: Apply relevant Big Data Storage techniques for a given application.	
CO3: Apply Big Data Technologies Hive and Pig to handle different nature of data.	
CO4: Analyze Big Data Processing Technique for a given application	

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Big Data Technologies** Introduction :Definition of Big Data, Characteristics of big data, types of big data Applications of big data Difference between big data and traditional data in terms of volume, velocity, variety, veracity, value, scalability Big data technology-definition, types : operational and analytical 1.5 Classification of big data technologies based on its purpose: Data Storage, Data Mining, Data Analytics, Data Visualization	**7**
2.	**Big Data Ecosystem and Architecture** 2.1 Data Storage Solutions: Hadoop architecture, HDFS features, HDFS architecture and operations 2.2 Map Reduce: The Map Task, The Reduce Task, grouping by key, Partitioner and combiners, Detailed Map-Reduce execution Workflow with basic word count, max-temperature finding example using python library 2.3 NoSQL databases: NoSQL data architecture , key value stores, family stores, graph stores and document stores, RDBMS vs NoSQL 2.4 MongoDB: Introduction to MongoDB, key features, Operation on database using MongoDB: CRUD	**8**

3.	**Handling data and Query with Hive and pig** 3.1 Apache Hive: Hive architecture, warehouse directory and megastore, Hive data types, loading data into tables, Hive file formats, Hive built in functions, Querying data, Sorting, Aggregation 3.2 Apache Pig: Architecture, execution modes, built-in functions, Pig operations: group and join, combine and split, filter	**7**
4.	**Big Data Processing Tools** 4.1 Apache Spark: Need for Apache Spark, Apache Spark architecture, Spark components, Resilient Distributed Datasets(RDD) transformations 4.2 Spark SQL, Spark Session, Data Frame operations 4.3 Apache Kafka: Need, benefits, Kafka Messaging System: Point to point and Public subscribe messaging system, Kafka cluster architecture	**8**

Text Books:

1. M. Vijayalakshmi, Radha Shankarmani – *Big Data Analytics*, Wiley, 2nd Edition, 2017, ISBN: 9788126565757
2. Tom White – *Hadoop: The Definitive Guide*, 4th Edition, O'Reilly Media, 2015, ISBN: 9781491901632

Reference Books:

1. Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia – *Learning Spark: Lightning-Fast Data Analytics*, O'Reilly Media, 2015, ISBN: 9781449358624
2. Pramod J. Sadalage, Martin Fowler – *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Addison-Wesley, 2012, ISBN: 9780321826626
3. Kieran Healy – *Data Visualization: A Practical Introduction*, Princeton University Press, 2018, ISBN: 9780691181622

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCL503 Course name: Advanced Database Management Course category: PCC
Credits: 2 Teaching scheme: L-2 Evaluation scheme: CA–60, ESE–40
Pre-requisites: Introduction to Database Management Systems (DBMS)
Course Objectives:
1.Enhance understanding of advanced database concepts beyond the basic relational model, including object-oriented, object-relational, and distributed databases.
Develop the ability to design, implement, and manage large-scale, complex, and distributed database systems.
Apply advanced querying and optimization techniques to improve the efficiency and performance of database operations.
Course Outcomes: At the end of the course, the students will be able to –
CO1: Execute various advance SQL queries related to Transaction Processing & Locking using concept of Concurrency control.
CO2: Demonstrate use of Database Object.
CO3: Perform PL/SQL programming using concept of Cursor Management, Error Handling, Package and Triggers.
CO4: Understand Functional Dependency and Functional Decomposition.

Contents –

Unit	Content	Teaching hours
1	**Advanced SQL** Transactional Control: Commit, Save point, Rollback DCL Commands : Grant and Revoke Types of locks: i. Row level locks ii. Table level locks iii. Shared lock iv. Exclusive lock v. Deadlock Synonym : Create synonym Sequences: Create and alter sequences Index : Unique and composite Views : Create/Replace, Update and alter views	**7**
2	**PL / SQL and Triggers** Basics of PL / SQL,Datatypes,Advantages Control Structures : Conditional, Iterative, Sequential Exceptions: Predefined Exceptions ,User defined exceptions Cursors: Static (Implicit & Explicit), Dynamic,Procedures & Functions Packages : Package specification, Package body, Advantages of package Fundamentals of Database Triggers Creating Triggers Types of Triggers : Before, after for each row, for each statement	**8**

3	**Functional Dependency and Decomposition** 3.1 Basics of Functional Dependency 3.2 Functional dependency diagram and examples 3.3 Full function dependency (FFD) 3.4 Armstrong's Axioms for functional dependencies 3.5 Redundant functional dependencies 3.6 Closures of a set of functional dependencies 3.7 Lossy Decomposition 3.8 Lossless join decomposition 3.9 Dependency-Preserving Decomposition	**8**
4	**Normalization &Transaction Processing** 4.1 Basics of Normalization 4.3 4.2 Normal Forms i. First Normal Form (1NF) ii. Second Normal Form (2NF) iii. Third Normal Form (3NF) 4.3 Introduction to transaction concepts 4.4 Concurrency 4.5 Methods for Concurrency control i. Locking Methods ii. Timestamp methods iii. Optimistic methods	**7**

Text Books: 1. Database Systems Concepts, design and Application, Singh, S. K. Pearson Education, New Delhi, 2012
Reference Books: 1.Sql/ Pl/SQLBayross, IvanBPB 2. An Introduction to Database SystemsDate, C. J. Pearson Education, New Delhi, 2012 3. Database System Concepts, Korth, HenryMGH
Online Resources: 1. NPTEL / SWAYAM lectures. 2.Software: Oracle 10e/11g express edition 3. DBMS:http://nptel.iitm.ac.in/video.php?subjectId=106106093 4. SQL Plus Tutorial: http://holowczak.com/oracle-sqlplus-tutorial/ 5. DatabaseTutorials:http://www.roseindia.net/programming-tutorial/Database- Tutorials

Course code:CSI24PCL504 Course Name: Data Privacy and Security
Course category: PCC
Credits: 2 Teaching scheme: L-2 Evaluation scheme: CA–60, ESE–40
Pre-requisites: Basic understanding of computer networks, algorithms, and introductory cryptography concepts
Course Objectives: 1. Introduce fundamental concepts of security, cryptography, and data protection.
2. Explain various symmetric and asymmetric encryption algorithms**.**
3. Discuss authentication, hashing, digital signatures, and data hiding techniques.
4. Introduce privacy frameworks, legal aspects, and ethical issues in data handling..
Course Outcomes: At the end of the course, the students will be able to **-**
CO1: Describe key concepts, goals, and mechanisms in data security and privacy.
CO2: Apply symmetric and asymmetric encryption algorithms for secure communication
CO3: Analyze the role of hashing, digital signatures, and data hiding in confidentiality and authentication.
CO4: Evaluate ethical, privacy, and legal aspects of data in digital systems.

Contents –

Unit	Content	Teaching hours
1.	**Introduction to Security and Cryptography** Security goals: confidentiality, integrity, availability; cryptographic attacks, services, and mechanisms; symmetric key cryptography: substitution and transposition ciphers, stream and block ciphers; overview of modern symmetric key ciphers.	**7**
2.	**Encryption Algorithms** Symmetric encryption: DES structure and limitations, AES transformations and key expansion; Asymmetric encryption: RSA, Elgamal, and overview of elliptic curve cryptography (ECC); key management and comparison of symmetric and asymmetric methods.	**8**
3.	**Hashing, Digital Signatures, and Data Hiding** Hash functions: SHA-512, WHIRLPOOL; digital signatures: process, services, and attacks; data hiding: watermarking, steganography, and simple LSB methods; applications of data hiding and watermarking.	**7**
4.	**Privacy, Legal, and Ethical Issues** Privacy concepts and principles; authentication and privacy; data mining, online tracking, and email security; introduction to legal frameworks: GDPR, IT Act (India); ethical issues and responsibilities in computer security	8

Text Books:

1. William Stallings, *Cryptography and Network Security: Principles and Practice*.
2. Behrouz Forouzan, *Cryptography and Network Security*.
3. Nina Godbole, *Information Systems Security: Security Management and Best Practices*.

Reference Books:

1. Bruce Schneier, *Applied Cryptography: Protocols, Algorithms, and Source Code in C*.
2. Charles Pfleeger & Shari Pfleeger, *Security in Computing*.
3. Daniel Solove, *Understanding Privacy*.

Online Resources:

1. NPTEL / SWAYAM courses on Network Security and Cryptography**.**

Course code: CSI24PCP501 Course name: Data story Telling and Visualization Lab
Course category: PCC
Credits: 1 Teaching scheme: P-2 Evaluation scheme: TW–30, PR–20
Pre-requisites:1. Fundamental Knowledge of data types and structures & statistics
Course Objectives:
1. To introduce learners to the fundamental concepts of data storytelling.
2. To develop the ability to analyze and remove irrelevant information (clutter) and with storytelling.
3. To equip students with the skills to transform raw data into meaningful visual formats.
4. To enable learners to create advanced data visualizations using various statistical distributions.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Practically identify and analyze the key elements and characters involved in data storytelling within real-time datasets.
LO2: Apply techniques to eliminate visual and narrative clutter, and construct compelling data-driven stories for given case scenarios.
LO3: Transform raw datasets into meaningful visual representations using appropriate data preprocessing and visualization tools.
LO4: Design and develop data visualizations using multiple distribution types (e.g., normal, binomial, Poisson, uniform, etc.) to interpret and communicate insights effectively.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Identify a project you are working on where you need to communicate in a data-driven way. Reflect upon and write the answers of following questions. a) What is a Stake? i) What are the benefits if your audience acts in the way that you want them to? ii) What are the risks if they do not? b) Form a Big Idea i) Articulate your point of view. ii) Convey what's at stake. Be a complete (and single) sentence.	2
2.	Determine audience, communication mechanism, desired tone and Select effective Visuals for any incident. (Below is an example for reference only). i) Who is your Audience? ii) List the primary groups or individuals to whom you'll be communicating. iii) If you had to narrow that to a single person, who would that be?	2

	iv) What does your audience care about? v) What action does your audience need to take? Example: Buses Bunching - Make a data story on bus bunching, (Bus Bunching means when a bus gets delayed and later causes multiple buses to arrive at a single stop at the same time.	
3.	Make a clutter free Data Story on any incident. a) Identify and eliminate clutter. b) Select suitable pre-attentive attributes. Explore affordances, accessibility, and aesthetics as per designers view.	2
4.	Create a data story for Vehicle (Bicycle/Bike/Car/Bus etc.) Rental System.	2
5.	Create a video (up to 5 minutes) telling a story on given Incidence. Record a video of yourself speaking, or narrate while showing visual props or sketches, or screencast a PowerPoint presentation, etc. Choose how you present the story. Produce a single video file (formatted as a .mov file)	2
6.	Construct a Case study on data storytelling for any Musical/Social App. Example: Spotify takes the data from our listening habits and spins it into an exciting audio and visual experience	2
7.	Implementation of a python program that performs data cleaning on any dataset.	2
8.	Create Bar chart for data visualization using Single distribution. Example: Histogram of the ages of the train/flight passengers.	2
9.	Create Bar chart for data visualization using Many distribution. Example: Histogram of the gender and ages of the train/flight passengers.	2
10.	Implementation of a python program that loads any dataset and plot a pie chart.	2

Text Books:
Reference Books: 1. Cole Nussbaumer Knaflic: "Storytelling with data – a data visualization guide for business professionals": Cole Nussbaumer Knaflic Wiley ISBN: 978-1-119-00225-3 2. Claus O. Wilke: "Fundamentals of Data Visualization": O'Reilly ISBN:978-1-492-03108-6
Online Resources: 1.NPTEL / SWAYAM lectures.
https://careerfoundry.com/en/tutorials/data- analytics-for-beginners/storytelling-with-data/
https://blog.hubspot.com/marketing/great-data- visualization-examples

Course code: CSI24PCP502	**Course Name: Big Data Lab**
Course category: PCC	
Credits: 1 **Teaching scheme: P-2** **Evaluation scheme: TW–30, PR–20**	
Pre-requisites: Fundamentals of Programming**,** Basic Knowledge of Databases and SQL	
Course Objectives:	
1. Understand the characteristics, challenges, and technologies of Big Data.	
2. Learn various Big Data storage and management techniques.	
3. Explore tools like Hive and Pig for data analysis and processing.	
4. Develop skills to implement Big Data processing solutions for real-world applications	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Understand Big Data characteristics and technologies through lab experiments.	
LO2: Apply Big Data storage techniques for given applications	
LO3: Use Hive and Pig to process different types of Big Data.	
LO4: Analyze Big Data processing techniques for specific applications.	

List of Experiments–

Sr.No.	**Name of Experiment**	**Practical Hours**
1.	Write a python program for analyzing the characteristics of any large dataset case study	2
2.	Install Single-node Hadoop cluster	2
3.	Write HDFS commands for file operations	2
4.	Write MapReduce program for word count operation	2
5.	Write MapReduce program to sort array elements	2
6.	Hive installation and Queries	2
7.	Apache Pig: a)Install Apache Pig b)Use Pig operators	2
8.	Develop spark program to perform basic operations on data frames using python	2
9.	Write Program to perform RDD transformations	2
10.	Develop program to produce and consume message and build simple pipeline	2

Text Books:

1. M. Vijayalakshmi, Radha Shankarmani – *Big Data Analytics*, Wiley, 2nd Edition, 2017, ISBN: 9788126565757
2. Tom White – *Hadoop: The Definitive Guide*, 4th Edition, O'Reilly Media, 2015, ISBN: 9781491901632

Reference Books:

1) Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia – *Learning Spark: Lightning-Fast Data Analytics*, O'Reilly Media, 2015, ISBN: 9781449358624
2) Pramod J. Sadalage, Martin Fowler – *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*, Addison-Wesley, 2012, ISBN: 9780321826626
3) Kieran Healy – *Data Visualization: A Practical Introduction*, Princeton University Press, 2018, ISBN: 9780691181622

.

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCP503 Course name: Advanced Database Management Lab
Course category: PCC
Credits: 1 Teaching scheme: P-2 Evaluation scheme: TW–30, PR–20
Pre-requisites:1. Introduction to Database Management Systems (DBMS)
Course Objectives:
1 Enhance understanding of advanced database concepts beyond the basic relational model, including object-oriented, object-relational, and distributed databases.
2. Develop the ability to design, implement, and manage large-scale, complex, and distributed database systems.
3. Apply advanced querying and optimization techniques to improve the efficiency and performance of database operations.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Execute various advance SQL queries related to Transaction Processing & Locking using concept of Concurrency control.
LO2: Demonstrate use of Database Object.
LO3: Perform PL/SQL programming using concept of Cursor Management, Error Handling, Package and Triggers.
LO4: Understand Functional Dependency and Functional Decomposition

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Perform queries for DCL Commands and Locks	2
2.	Implement authorization, authentication, privileges on database	2
3.	Perform queries to Create synonyms, sequence and index	2
4.	Perform queries to Create, alter and update views	2
5.	Implement PL/SQL programmes using control structures	2
6.	Implement PL/SQL programmes using Cursors	2
7.	Implement PL/SQL programmes using exception handling.	2
8.	Implement user defined procedures and functions using PL/SQL blocks	2
9.	Perform various operations on packages.	2
10.	Implement various triggers	2
11.	Practice on functional dependencies	2
12.	Practice on Normalization – using any database performs various normal forms.	2

Text Books: 1.Database Systems Concepts, design and Application, Singh, S. K. Pearson Education, New Delhi, 2012
Reference Books: 1.Sql/ Pl/SQLBayross, IvanBPB 2. An Introduction to Database SystemsDate, C. J. Pearson Education, New Delhi, 2012 3. Database System Concepts, Korth, HenryMGH
Online Resources: 1.NPTEL / SWAYAM lectures. 2.Software: Oracle 10e/11g express edition 3. DBMS:http://nptel.iitm.ac.in/video.php?subjectId=106106093 4. SQL Plus Tutorial: http://holowczak.com/oracle-sqlplus-tutorial/ 5. DatabaseTutorials:http://www.roseindia.net/programming-tutorial/Database- Tutorials

Course code: CSI24PCP504	Course Name: Data Privacy and Security Lab
Course category: PCC	
Credits: 1	**Teaching scheme: P-2** **Evaluation scheme: TW–30, PR–20**
Pre-requisites:1. Basic understanding of computer networks, algorithms, and introductory cryptography concepts.	
Course Objectives:	
1. Introduce fundamental concepts of security, cryptography, and data protection.	
Explain various symmetric and asymmetric encryption algorithms.	
3. Discuss authentication, hashing, digital signatures, and data hiding techniques.	
4. Introduce privacy frameworks, legal aspects, and ethical issues in data handling..	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Describe key concepts and goals of data security and privacy.	
LO2: Apply encryption algorithms for secure communication.CO4: Implement minimizes and maximizes functions using R	
LO3: Analyze hashing, digital signatures, and data hiding methods.	
LO4: Evaluate privacy laws and ethical issues.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Study of Security Goals and Attacks	2
2.	Design and implement simple substitution (Caesar, monoalphabetic) and transposition ciphers in Python or C.	2
3.	Demonstrate basic encryption and decryption using stream and block cipher logic. Compare performance and security differences	2
4.	Write a program to implement simplified DES encryption and decryption, showing key generation and rounds.	2
5.	Implement AES encryption using Python's cryptography library and analyze key expansion and transformation steps.	2
6.	Develop an RSA program for encryption and decryption. Demonstrate usage of public and private keys.	2
7.	Study and Application of Hash Functions -Use hash functions (MD5, SHA-256, SHA-512) to check message integrity and compare output lengths and collisions.	2
8.	Implement a digital signing and verification process using RSA or DSA. Demonstrate message authentication.	2
9.	Apply Least Significant Bit (LSB) steganography to hide text inside an image file. Demonstrate watermark detection and recovery.	2
10.	Analyze a small web application for common privacy risks (e.g., cookie tracking, data collection). Suggest mitigation techniques	2
11.	Prepare a short report comparing GDPR and the IT Act (India), focusing on data rights, consent, and penalties for violations	2

12	Case Study on Ethical and Privacy Issues- Research a real-world data breach. Discuss ethical, legal, and privacy implications.	2

Text Books:

1.William Stallings, *Cryptography and Network Security: Principles and Practice*.
2.Behrouz Forouzan, *Cryptography and Network Security*.
3.Nina Godbole, *Information Systems Security: Security Management and Best Practices*.

Reference Books:

1.Bruce Schneier, *Applied Cryptography: Protocols, Algorithms, and Source Code in C*.
2.Charles Pfleeger & Shari Pfleeger, *Security in Computing*.
3.Daniel Solove, *Understanding Privacy*.

Online Resources:

1.NPTEL / SWAYAM courses on Network Security and Cryptography.

Course code: CSI24PCL505	**Course Name:** Basics of CC
Course category: PEC	
Credits: 2 **Teaching scheme: L-2**	**Evaluation scheme:** CA–60, ESE–40
Pre-requisites:1. Computer Network.	
Course Objectives:	
1. Understand the fundamentals and history of cloud computing.	
2. Explore cloud services like IaaS, PaaS, SaaS, and others.	
3. Discuss methods to enhance cloud data and infrastructure security.	
4. Introduce virtualization and types of hypervisors.	
5. Compare cloud support plans and cost models.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
CO1: Explain mainframe, client-server, parallel, and distributed computing architectures.	
CO2: Illustrate cloud computing architectures and delivery models.	
CO3: Identify infrastructure security measures at network, host, and application levels.	
CO4: Explain virtualization concepts and types, including Type-I and Type-II hypervisors.	
CO 5: Apply knowledge of core services and pricing models of AWS, GCP, Azure and Databricks.	

Contents –

Unit	Content	Teaching hours
1	**Model computing** -Introduction to Mainframe architecture & Client-server architecture, Parallel & Distributed Computing, Cluster & Grid Computing, Definition and Evolution of Cloud Computing	4
2	**Cloud Computing and Cloud Services -** Cloud Computing architecture, cloud delivery models, cloud deployment models, Benefits and challenges, Service oriented architecture (SOA), Cloud Services: Infrastructure-as-a- Service (IaaS), Platform-as-a-Service (PaaS), Software-as-a-Service (SaaS), Identity-as-a-service (IDaaS), Storage-as-a-service.	8
3	**Cloud Security-** Cloud Security: Infrastructure Security – Network level security, Host level security, Application level Security. Data security and Storage - Data privacy and security Issues.	4
4	**Virtualization and SLA -** Virtualization: Introduction to virtualization, Hypervisor: Type-I & Type II, Types of Virtualization, Pros and cons of virtualization, Virtualization applications in enterprises: Server virtualization, Desktop and Application Virtualization, Storage and Network Virtualization SLA Management, SLA Management in Cloud.	8
5	**Cloud Platforms in Industry-** Amazon Web Services- Compute Services, Storage Services, Communication Services and Additional Services, GCP, Billing and pricing & Support plan, Pricing model of AWS & GCP, Different support levels offered by AWS and GCP,	6

	Google AppEngine-Architecture and Core Concepts, Application Life-Cycle, cost model. Microsoft Azure- Azure Core Concepts, SQL Azure	

Text Books:

1. Barrie Sosinsky, "Cloud computing Bible" Wiley India Pvt Ltd, 2011.
2. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi ,"Mastering Cloud Computing", McGraw Hill,2013

Reference Books:

1. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley.
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.

Online Resources: 1. NPTEL / SWAYAM lectures.

Course Code: CSI24PCL506 **Course Name:** Dev-Ops-I **Course Category:** PEC
Credits: 2 **Teaching scheme:** L-2 **Evaluation Scheme:** CA:60, ESE:40
Duration of External Exam: 2 Hrs
Pre-requisites: Software Development Life Cycle (SDLC), Version Control Systems, Basic Programming Knowledge.
Course Objectives: 1. Introduce software development concepts, SDLC models, and software quality practices. 2. Understand DevOps concepts and the role of version control in modern software development. 3. Develop practical skills in using Git for source code and version management. 4. Apply advanced Git features for effective branch, merge, and release management. 5. Familiarize students with current DevOps trends, tools, and industry practices.
Course Outcomes: At the end of the course, the students will be able to – **CO1**: Describe Study of various Software Development Methodologies and Cloud Computing. **CO2:** Understand the basic Linux commands with respect to Devops and basic of version control. **CO3:** Understand the concept of Devops commands and workflow in a systematic way. **CO4:** Apply the advanced concept of DevOps with real time applications. **CO5:** understand the concept of DevOps Methodology and tools in a systematic way.

Contents:

Unit No.	Content	Teaching Hours
1.	**Introduction to Software Development**: What is Software Development, Software Development Life Cycle, Traditional Models for SDLC, Elaborate software development life cycle: Waterfall model, iterative waterfall model, Prototype model, evolutionary model, spiral model, agile model, software testing and debugging, software verification and validation, Comparison of different life cycle models, User Interface Design, Coupling and Cohesion, Pham-Nordmann-Zhang Model (PNZ model), Schick-Wolverton software reliability model.	8
2.	**Introduction to DevOps and software version control:** What is DevOps, Industry Importance of DevOps, DevOps Lifecycle, Continuous Development, Continuous Testing, Configuration Management, Continuous Integration, Continuous Monitoring of software throughout its development life cycle, Understanding basics of version control, Control Concepts of different types of Version Control Systems	8

3.	**Basic Git:** Introduction to Git, Git lifecycle- Working Directory, Staging Area, Local Repository, Remotel Repository Common Git commands, Git Workflow, Working with Remote Repository, Version controlling using Git	4
4.	**Advanced Git:** Source code management with Git, Comparison commands, Branch management strategies, and merge strategies, Rebasing, Interactive rebase Stashing and Tagging.	4
5.	**DevOps Trends:** DevOps Market Trends, DevOps Engineer Skills, DevOps Delivery Pipeline, DevOps Ecosystem, Role of a DevOps Engineer, Devops Tools: Git, Docker, Selenium, Maven, Jenkins, Puppet, Ansible, Kubernetes, Nagios.	6

Text Books:

1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" Gene Kim, Kevin Behr, George Spafford.
2. Linux Pocket Guide: Essential Commands By Daniel J. Barrett, O'reilly.
3. "Learning Git" Anna Slater.

Reference Books:

1."The DevOps Toolkit: Kubernetes" by Viktor Farcic.

2. "Jenkins: The Definitive Guide: Continuous Integration for the Masses" by John Ferguson Smart.

3. Devops: A Software Architect's Perspective (Sei Series In Software Engineering) By Len Bass, Ingo Weber, Liming Zhu, Addison-Wesley.

Online Resources:

1. https://www.guvi.in/blog/best-websites-to-learn-devops/
2. https://www.geeksforgeeks.org/devops-tutorial/
3. GitHub Learning Lab.

Course code: CSI24PCP505	**Course Name:** Basics of CC Lab
Course category: PEC	
Credits: 1 **Teaching scheme: P-2**	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1. Computer Network.	
Course Objectives:	
1. Understand the fundamentals and history of cloud computing.	
2. Explore cloud services like IaaS, PaaS, SaaS, and others.	
3. Discuss methods to enhance cloud data and infrastructure security.	
4. Introduce virtualization and types of hypervisors.	
5. Compare cloud support plans and cost models.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Explain mainframe, client-server, parallel, and distributed computing architectures.	
LO2: Illustrate and compare cloud computing architectures and delivery models.	
LO3: Identify infrastructure security measures at network, host, and application levels.	
LO4: Explain virtualization concepts and types, including Type-I and Type-II hypervisors.	
LO5: Apply knowledge of core services and pricing models of AWS, GCP, Azure and Databricks.	

Contents –

Sr. No.	**List of Experiments**	**Practical Hours**
1.	Implementing Basic Concepts in Cloud Computing.	2
2.	Creating a Warehouse Application in SalesForce.com.	2
3.	Create an Application using Apex Programming in SalesForce.com.	2
4.	Implementation of SOAP Web services in C#/JAVA Applications.	2
5.	Implementation of Virtualization. Install Oracle Virtual box and create VMs.	2
6.	Install a C compiler in the virtual machine and execute a sample program.	2
7.	Setup account for AWS and GCP	2
8.	Create a Cloud Storage bucket using Amazon Simple Storage Service (Amazon S3).	2
9.	Working with AWS CLI	2
10.	Creating Azure Container Instances	2

11.	Case Study of Google App Engine	2

Text Books:

1. Barrie Sosinsky, “Cloud computing Bible” Wiley India Pvt Ltd ,2011.
2. Rajkumar Buyya, Christian Vecchiola, S. ThamaraiSelvi ,”Mastering Cloud Computing”, McGraw Hill,2013

Reference Books:

1. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley.
2. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, NewDelhi.

Online Resources:

1. NPTEL / SWAYAM lectures.

Course code: CSI24PCP506	**Course Name:** DevOps-I Lab
Course category: PEC	
Credits: 1 **Teaching scheme: P-2**	**Evaluation scheme:** TW–30, PR–20
Pre-requisites:1. Software Development Life Cycle (SDLC), Version Control Systems, Basic Programming Knowledge.	
Course Objectives:	
1. Understand the architecture and workflow of DevOps systems.	
2. Develop hands-on skills in version control using Git and GitHub.	
3. Install and configure virtualized and Linux-based development environments.	
4. Apply build, testing, and automation tools used in DevOps pipelines.	
5. Design and implement an end-to-end DevOps pipeline using industry tools.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1. Create and manage Git repositories using common Git commands.	
LO2. Create and manage Git repositories using common Git commands.	
LO3. Execute basic Linux commands and manage software packages using RPM and YUM.	
LO4. Build and test applications using Maven and Selenium.	
LO5. Implement a complete DevOps pipeline using Jenkins, Git, Maven, Docker, and Selenium.	

Contents –

Sr. No.	**List of Experiments**	**Practical Hours**
1.	Explore DevOps architecture	2
2.	Create a Git repository for a simple project, and then practice common commands such as git init, git clone, git commit, git push, and git pull. Tools: Git, GitHub (or any Git hosting service).	2
3.	Installation of Oracle VM Virtual Box and create Virtual Machine Practical.	2
4.	Installation of Linux, Implementation of basic Linux commands: chmod, grep, wget, chown, find, cat, echo, ifconfig, cp,ping,kill, tail, rm, rmdir, cd, mkdir, vi, mv.	2
5.	Installation of packages using RPM and YUM	2
6.	Installation of Git, Implementing common Git Command	2

7.	Repository creation in Git, Git Branch, source code management with Git	2
8.	Installation of Maven and Work	2
9.	Continuous Testing with Selenium	2
10.	Build a simple DevOps pipeline using Jenkins that performs the following tasks: 1. Pulls code from a Git repository. 2. Builds the code (using Maven or similar). 3. Runs automated tests (using Selenium). 4. Builds a Docker image of the application. 5. Deploys the Docker container to a test environment. Tools: Jenkins, Git, Maven, Docker, Selenium.	2
11.	Mini project	2

Text Books:

1."The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" Gene Kim, Kevin Behr, George Spafford.
2. Linux Pocket Guide: Essential Commands By Daniel J. Barrett, O'reilly.
3."Learning Git" Anna Slater.

Reference Books:

1."The DevOps Toolkit: Kubernetes" by Viktor Farcic.
2. "Jenkins: The Definitive Guide: Continuous Integration for the Masses" by John Ferguson Smart.
3. Devops: A Software Architect's Perspective (Sei Series In Software Engineering) By Len Bass, Ingo Weber, Liming Zhu, Addison-Wesley.

SEM-X

Course code: CSI24PCL551	**Course name: Business Intelligence**
Course category: PCC	
Credits: 1 **Teaching scheme:** L-1	**Evaluation scheme:** CA–30, ESE–20
Pre-requisites:1. Knowledge of python programming and Database	
Course Objectives:	
1. To provide students with a comprehensive understanding of Business Intelligence (BI) concepts, components, and framework, enabling them to interpret how BI supports strategic and operational decision-making.	
2. To develop the ability to design and analyze BI-oriented databases, including data warehousing structures that support efficient querying and reporting..	
3. To equip students with the skills required to perform Extraction, Transformation, and Loading (ETL) **processes**, ensuring proper data integration from multiple sources.	
4. To enable students to build business applications and generate meaningful BI reports, helping organizations derive insights for improved business performance.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
CO1: Describe Business Intelligence and its framework..	
CO2: Analyze Business intellifence Database Design.	
CO3: Implement Extraction, transformation and loading process.	
CO4: Develop Business Application and generate its report.	

Contents –

Unit	Content	Teaching hours
1.	**Introduction To Business Intelligence** BI Definitions & Concepts, Applications of Business intelligence ,Benefits of business intelligence Categories of BI analysis, Advantages and disadvantages of BI Business intelligence platforms, Types of BI Tools and Software BI Framework, Business intelligence platforms, Role of Data Warehousing in BI ,BI and big data Project Planning Process - Managing the BI Project ,Activities and Tasks,Estimating Techniques , Resource Assignment ,Task Dependencies, Resource Dependencies, Critical Path Method ,Project Schedules	**4**

2.	**Business Intelligence Database Designing** Database Design - Differences in Database Design Philosophies, Operational Databases, Operational Databases versus BI Target Databases, BI Target Databases Logical Database Design - The Star Schema, The Snowflake Schema Physical Database Design- Physical Dataset Placement, Partitioning ,Clustering ,Indexing ,Reorganizations ,Backup and Recovery ,Parallel Query Execution	**3**
3.	**Extraction, Transformation and Loading** ETL Development Source -Heterogeneous Data Sources ,Integrated BI Implementation Strategy Implementation Strategies for BI - Preparing for the ETL Process, Designing the Extract Programs, Designing the Transformation Programs ,Designing the Load,Programs Designing the ETL Process Flow- The Source-to- Target Mapping Document , The ETL Process Flow Diagram , The Staging Area ,Evaluating ETL Tools	**4**
4.	**Application Development and Enterprise Reporting** OLAP Tools ,OLAP Tool Features ,Multidimensional Analysis Factors , Multivariate Analysis OLAP Architecture-Presentation Services, OLAP Services , Database Services Development Environments -The Web Environment , Application Development Activities Introduction to enterprise reporting, concepts of Dashboards	**4**

Reference Books:

1. David Loshin, Business Intelligence : The Savvy Manager's Guide, Morgan Kaufmann ISBN-13:978-1-55860-916-7
2. Larissa T. Moss, Shaku Atre, Business intelligence roadmap, Addison Wesley ISBN: 0-201-78420-3
3. Mike Biere, Business intelligence for the enterprise, Pearson Publication ISBN 10: 0131413031 /ISBN 13: 9780131413030

Online Resources:

1.NPTEL / SWAYAM lectures.
2. https://learn.microsoft.com/en-us/training/modules/get-start ed-with-power-bi/2-using-power-bi
3. https://onlinecourses.nptel.ac.in/noc24_cs65
4. https://www.geeksforgeeks.org/power-bi-tutorial/

Course code: CSI24PCL552 **Course name: Natural Language Processing**
Course category: PCC
Credits: 3 **Teaching scheme:** L-3 **Evaluation scheme:** CA–60, ESE–40
Pre-requisites:1. Knowledge of programing language, Data Structures
Course Objectives:
1. To understand natural language processing concept
2. To get acquainted with word level, syntax analysis & language model.
3. To understand semantic analysis of NLP & model.
4. To design systems that uses NLP techniques
Laboratory Outcomes: At the end of the session, the students will be able to -
CO1: Have a broad understanding of the field of natural language processing.
CO2: Get knowledge of morphology, language model & POS
CO3: Understand concept of Lexical Semantics
CO4 : Understand NLP techniques to design real world NLP applications

Contents –

Unit	Content	Teaching hours
1.	**Introduction :** History of NLP, Study of Human Language, NLP pipeline, levels of NLP , Knowledge in language processing , Ambiguity in Natural language , stages in NLP, challenges of NLP , Applications of NLP, Linguistic Resources. **Text Normalization** : Tokenization, sentence segmentation, Stopwords, stemming, lemmatization. Regular Expressions & Automata :Regex fundamentals, Regular sets, grammars, Finite automata & relationships among RE, RG, FA	**12**
2.	**Word Level Analysis :** Morphology analysis –survey of English Morphology, Inflectional morphology & Derivational morphology, Lemmatization, finite state transducers (FST) , Morphological parsing , N gram language model. **Syntax Analysis :** Part-Of-Speech tagging(POS)- Tag set for English (Penn Treebank) , Rule based POS tagging, Stochastic POS tagging, Issues –Multiple tags & words, Unknown words. Introduction to CFG, Sequence labelling: Hidden Markov Model (HMM)	**12**

3.	**Semantic Analysis :** Lexical Semantics, Attachment for fragment of English- sentences, noun phrases, Verb phrases, prepositional phrases, Relations among lexemes & their senses –Homonymy, Polysemy, Synonymy, Hyponymy, WordNet, Robust Word Sense Disambiguation (WSD) ,Dictionary based approach	**10**
4.	**Pragmatic analysis :** Discourse –model & structure, reference resolution, reference phenomenon , syntactic & semantic constraints on co reference **Applications of NLP :** Sentiment Analysis, information Extraction ,Machine Translation , Text Summarization ,Grammar and Text Analysis , Modern NLP Applications : Virtual Assistants ,Chatbots and Customer Service , Speech Recognition ,Spam Filters , Predictive Text , Ethics & fairness in NLP systems	**11**

Text Books:

1. Daniel Jurafsky, James H. Martin ―Speech and Language Processing , Prentice Hall,
2. Christopher D.Manning and Hinrich Schutze, ― Foundations of Statistical Natural Language Processing ―, MIT Press.

Reference Books:

1. Siddiqui and Tiwary U.S., Natural Language Processing and Information Retrieval, Oxford University Press
2. Daniel M Bikel and Imed Zitouni ― Multilingual natural language processing applications‖ Pearson
3. Alexander Clark (Editor), Chris Fox (Editor), Shalom Lappin (Editor) ― The Handbook of Computational Linguistics and Natural Language Processing
4. Steven Bird, Ewan Klein, Natural Language Processing with Python, O'Reilly

Online Resources: 1.NPTEL / SWAYAM lectures, Tutorials Point

Course code: CSI24PCL553	**Course Name: Deep Learning**
Course category: PCC	
Credits: 2 **Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Duration of External Exam: 2 Hrs	
Pre-requisites:1. Data Structure, Probability, Python Programming, ML	
Course Objectives:	
1.To learn how neural networks work.	
2.To understand common deep learning models like CNN and RNN.	
3.To apply deep learning for images, text, and simple applications.	
4.To use DL libraries like TensorFlow or PyTorch	
Laboratory Outcomes: At the end of the session, the students will be able to -	
CO1: Explain basic concepts of deep learning.	
CO2: Build neural networks.	
CO3: Use Convolutional Neural Networks for image tasks	
CO4: Use RNN for sequence/text tasks.	

Contents –

Unit	Content	Teaching hours
1.	**Basics of Deep Learning & Neural Networks :** What is Deep Learning? Neuron model & activation functions, Perceptron and simple neural networks, Training a network (loss, backpropagation, gradient descent), Problems: overfitting, regularization, dropout .	**8**
2.	**Building Deep Networks :** Multi-layer neural networks, Optimization (SGD, Adam – simple explanation), Hyper parameters (learning rate, batch size, epochs), Introduction to TensorFlow / PyTorch , Building a simple classification model	**7**
3.	**Convolutional Neural Networks (CNN) :** Convolution operation (simple idea), Layers: convolution, pooling , Basic CNN models (LeNet, simple custom CNN), Image classification examples (MNIST / CIFAR), Data augmentation	**8**
4.	**Sequence Models & Generative Models :** RNN basics , LSTM / GRU (simple understanding), Text tasks (sentiment analysis), Autoencoders (basic idea), GANs (simple idea of generator + discriminator)	**7**

Text Books:

1. *Deep Learning* by Goodfellow, Bengio, and Courville
2. Deep Learning using Python – Lovelyn Rose, Ashok Kumar, Karthik Renuka (Wiley)
3. *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow* – Aurélien Géron

Reference Books:

1. Neural Networks and Deep Learning – Charu Aggarwal
2. Deep Learning with Python – François Chollet
3. S. R. M. Prasanna — Speech, Audio, Image and Biomedical Signal Processing using Neural Networks (Springer)
4. Bayya Yegnanarayana — Artificial Neural Networks (Prentice-Hall India)

Online Resources: 1.NPTEL / SWAYAM lectures.

Course code: CSI24PCP551	**Course name: Business Inetlligence Lab**
Course category: PCC	
Credits: 02 **Teaching scheme:** P-4	**Evaluation scheme:** TW–60, PR–40
Pre-requisites:1. Knowledge of python programing and Database	
Course Objectives: 1. To provide students with a comprehensive understanding of Business Intelligence (BI) concepts**,** components, and framework, enabling them to interpret how BI supports strategic and operational decision-making.	
2. To develop the ability to design and analyze BI-oriented databases, including data warehousing structures that support efficient querying and reporting..	
3. To equip students with the skills required to perform Extraction, Transformation, and Loading **(**ETL**)** processes, ensuring proper data integration from multiple sources.	
4. To enable students to build business applications and generate meaningful BI reports, helping organizations derive insights for improved business performance.	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Describe Business Intelligence and its framework.	
LO2: Analyze Business intellifence Database Design.	
LO3: Implement Extraction, transformation and loading process.	
LO4: Develop Business Application and generate its report.	

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Explore any Business Intelligence tool (Power BI / Tableau / Qlik). Connect it to a sample dataset (e.g., Superstore or Sales). Identify and demonstrate essential BI features such as importing data, applying filters, creating basic charts, and exploring drill-down capabilities. The objective is to understand the BI interface and navigation structure.	2
2.	Using a real-world dataset (Retail/HR/Banking), perform Descriptive, Diagnostic, Predictive, and Prescriptive analytics tasks. Based on the visual insights generated, prepare a short analytical summary explaining how BI helps decision-making in that domain. Identify benefits and limitations of BI observed during analysis.	2
3.	Choose any business domain (Retail, Telecom, Healthcare, Banking) and design a complete BI framework. Include data sources, ETL processes, data warehouse layers, OLAP analysis, dashboards, and decision-making components. Represent the framework through diagrams and justify each component's role.	2
4.	Design a project plan for implementing a BI solution in an organization. Prepare a Work Breakdown Structure (WBS), list project tasks, define task dependencies, and estimate effort. Create a Critical Path Method (CPM) network diagram and Gantt chart using MS Project/Excel to show timelines and resource allocation.	2
5.	Analyze the differences between operational databases (OLTP) and BI target databases (OLAP). Using a given dataset, identify tables suitable for OLTP operations and design analytical queries typical for OLAP. Provide a comparison report showcasing structural, functional, and performance differences.	2

6.	Given a business scenario (e.g., online retail sales), design a Star Schema consisting of one fact table and multiple dimension tables. Identify measures, dimension keys, hierarchies, and grain of the fact table. Draw the star schema diagram and justify your design decisions.	2
7.	Convert the star schema designed in Practical 6 into a Snowflake Schema. Normalize dimension tables to eliminate redundancy. Compare both schemas in terms of storage, query performance, and maintenance. Submit diagrams and a comparison report.	2
8.	Implement the fact and dimension tables of the designed schema using SQL (MySQL/PostgreSQL). Apply physical design optimizations such as indexing, partitioning, clustering, and parallel query execution (if available). Evaluate and document performance improvements using EXPLAIN/ANALYZE statements.	2
9.	Load data from multiple heterogeneous sources (CSV, Excel, JSON, SQL, Web API). Identify schema mismatches, missing values, and inconsistent formats. Prepare a summary of challenges in integrating such diverse data and propose preprocessing steps required before ETL.	2
10.	Prepare a Source-to-Target Mapping document for a selected dataset. Include column mapping, data type conversions, transformation logic, validation rules, error-handling strategy, surrogate key handling, and business rules. This document will act as the blueprint for the ETL implementation.	2
11.	Using Talend / Pentaho / SSIS / Python (Pandas), develop a complete ETL pipeline to extract data from multiple sources, transform it (cleaning, joining, deriving new columns, handling missing data), and load it into the target data warehouse tables. Validate that the loaded data meets the mapping document specifications.	2
12.	Design an ETL workflow with sequential and parallel execution paths. Represent the workflow using a process-flow diagram. Implement the workflow in an ETL tool and schedule it to run automatically using a scheduler (cron/Windows Task Scheduler). Submit logs and screenshots of task execution.	2
13.	Perform OLAP operations (slice, dice, roll-up, drill-down, pivot) on a multidimensional dataset using a BI tool or SQL OLAP functions. Demonstrate how each operation changes the view of data and supports analytical insights. Present results with visualizations.	2
14.	Design a basic OLAP cube for a business dataset showing dimensions, hierarchies, and measures. Explain the OLAP architecture (Presentation Layer, OLAP Services Layer, and Database Layer). Create the cube using SSAS or any OLAP-supported tool and demonstrate simple cube queries.	2
15.	Develop an interactive BI dashboard for a business scenario (Sales, HR, Finance, Retail). Include KPIs, slicers/filters, drill-through pages, and charts. The dashboard should provide actionable insights supported by visual storytelling techniques.	2
16.	Generate enterprise-level reports (summary & detailed) using a reporting tool (Power BI / Tableau / Google Data Studio). Apply formatting, conditional highlighting, grouping, and parameters. Export reports to PDF/Excel and demonstrate scheduled refresh if available.	2
17.	Develop a simple BI web application using Python Flask / Streamlit / Dash. The application must include data import, charts, filters, and an interactive dashboard. Demonstrate how such BI apps can be deployed internally within organizations.	2

18.	Mini Project Develop a complete BI solution for a any one domain as per students choice Tasks include: • Identify business problem • Collect/prepare data • Design ETL pipeline • Create Data Warehouse schema • Apply OLAP analysis • Build dashboard + enterprise report • Present final BI insights to faculty panel	2

Reference Books:

1. David Loshin, Business Intelligence : The Savvy Manager's Guide, Morgan Kaufmann ISBN-13:978-1-55860-916-7
2. Larissa T. Moss, Shaku Atre, Business intelligence roadmap, Addison Wesley ISBN: 0-201-78420-3
3. Mike Biere, Business intelligence for the enterprise, Pearson Publication ISBN 10: 0131413031 /ISBN 13: 9780131413030

Online Resources:

1. 1.NPTEL / SWAYAM lectures.
2. https://learn.microsoft.com/en-us/training/modules/get-start ed-with-power-bi/2-using-power-bi
3. https://onlinecourses.nptel.ac.in/noc24_cs65
4. https://www.geeksforgeeks.org/power-bi-tutorial/

Course code: CSI24PCP552 **Course name: Natural Language Processing Lab** **Course Category: Program** Core Course		
Credits: 01	**Teaching scheme:** P-2	**Evaluation scheme :** TW–30, PR–20
Pre-requisites:1. Knowledge of python programing, Data Structure		
Course Objectives: 1. To study available libraries & dataset related to NLP		
2. To get acquainted with designing skill of fundamental NLP operations.		
3. Design any application of NLP		
Laboratory Outcomes: At the end of the session, the students will be able to -		
LO1: know available libraries & dataset related to NLP		
LO2: design and implement fundamental NLP operations		
LO3: design an application of NLP		

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Installation of different Python libraries for NLP	2
2.	Loading and Exploration of NLP Datasets using NLP Libraries.	2
3.	Implementation of read, write, and append operations in files	2
4.	Implementation of tokenization concepts (word, sentence)	2
5.	Implementation to find synonyms and antonyms of words	2
6.	Implementation of stop-word removal from documents	2
7	Implementation of unigram, bigram, and trigram concepts	2
7.	Implementation of Part-of-Speech (POS) tagging	2
8.	Implementation of stemming and lemmatization	2
9.	Implementation of Named Entity Recognition (NER)	2
10.	Implementation of a complete NLP task using data cleaning and preprocessing	2
11.	Installation of different Python libraries for NLP	2

Text Books:

1. Daniel Jurafsky, James H. Martin ―Speech and Language Processing , Prentice Hall,
2. Christopher D.Manning and Hinrich Schutze, ― Foundations of Statistical Natural Language Processing ―, MIT Press.

Reference Books:

1. Siddiqui and Tiwary U.S., Natural Language Processing and Information Retrieval, Oxford University Press
2. Daniel M Bikel and Imed Zitouni ― Multilingual natural language processing applications‖ Pearson
3. Alexander Clark (Editor), Chris Fox (Editor), Shalom Lappin (Editor) ― The Handbook of Computational Linguistics and Natural Language Processing

4. Steven Bird, Ewan Klein, Natural Language Processing with Python, O'Reilly
Online Resources: 1.NPTEL / SWAYAM lectures, Tutorials Point

Course code: CSI24PCL553 **Course name: Deep Learning Lab** **Course category :PCC**
Credits: 1 Teaching scheme: P-2 Evaluation scheme : TW–30, PR–20
Pre-requisites: Programming Knowledge, Basic Math, basics of Machine Learning
Course Objectives:
1. To Build The Foundation Of Deep Learning 2. To Understand How To Build The Neural Network. 3. To enable students to develop successful deep learning concepts
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Learn the fundamental principles of deep learning.
LO2 : Identify the deep learning algorithms for various types of learning tasks in various domain
LO3: Implement deep learning algorithms and solve real-world problems.

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Setting up the environment for the practical execution of deep learning.	2
2.	Installation of various libraries for deep learning practicals.	2
3.	Implementation of the XOR problem using DNN.	2
4.	Face recognition using CNN.	2
5.	Language modelling using RNN.	2
6.	Image classification using MNIST / CIFAR or other datasets.	2
7	Sentiment analysis using LSTM.	2
8.	Part-of-speech tagging using a sequence-to-sequence architecture.	2
9.	Machine translation using an encoder–decoder model.	2
10.	Image augmentation using GANs.	2
11.	Mini project on real-world application.	2

Text Books:

1. *Deep Learning* by Goodfellow, Bengio, and Courville
2.Deep Learning using Python – Lovelyn Rose, Ashok Kumar, Karthik Renuka (Wiley)
3.Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow – Aurélien Géron

Reference Books:

1.Neural Networks and Deep Learning – Charu Aggarwal
2.Deep Learning with Python – François Chollet
3.S. R. M. Prasanna — Speech, Audio, Image and Biomedical Signal Processing using Neural Networks (Springer)
4.Bayya Yegnanarayana — Artificial Neural Networks (Prentice-Hall India)

Online Resources: 1.NPTEL / SWAYAM lectures

Course code: CSI24PCL554	**Course name:** CC Using AWS	**Course category:** PEC
Credits: 2	**Teaching scheme:** L-2	**Evaluation scheme:** CA–60, ESE–40
Duration of External Exam: 2 Hrs		
Pre-requisites: Basics of CC.		
Course Objectives:		
1. Host a basic application (e.g., Employee Directory) on AWS.		
2. Understand the Amazon EC2 instance lifecycle and compute services.		
3. Explain Docker and Kubernetes and their architectures.		
4. Understand S3 storage components, including buckets, objects, and keys.		
5. Configure IAM users, groups, and password policies.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Identify the components of AWS Global Infrastructure.		
CO2: Describe the Amazon EC2 instance lifecycle and container services.		
CO3: Define containerization concepts and tools like Docker and Kubernetes.		
CO4: Explain S3 components, storage classes, and versioning.		
CO5: Create and manage IAM policies, groups, and permissions**.**		

Contents –

Unit	Content	Teaching hours
1	**AWS Essentials -**AWS Global Infrastructure, Interacting with AWS, Security and the AWS Shared Responsibility Model, Protect the AWS Root User, Introduction to AWS Identity and Access Management, Role Based Access in AWS, Hosting the Employee Directory Application on AWS	6
2	**Introduction to Amazon Elastic Compute Cloud-**Compute as a Service on AWS, Amazon EC2 Instance Lifecycle, Container Services on AWS,What is Serverless?, Introduction to AWS Lambda, Networking on AWS, Introduction to Amazon VPC, Amazon VPC Routing, Secure Your Network with Amazon VPC Security.	6
3	**Introduction to Containerisation -**What is Docker & Kubernetes?, Architecture of Docker & Kubernetes,AWS Elastic container services, Amazon Elastic Kubernetes service, Google Kubernetes engine.	6
4	**Cloud Storage -**S3 Storage :Buckets, Objects, and Keys, S3 Data Consistency, S3 Storage Classes,S3 Management , Versioning, S3 Bucket Security, Amazon S3 Glacier Archive Storage, S3 Glacier Vaults and Archives , Shared File Systems at AWS	6
5	**Security Services -**Creating IAM Users and Groups, Defining a Password Policy, Enabling MFA Protection, Creating IAM Policies, Creating Permission Boundaries, Exploring AWS Security Tools, Exploring CloudTrail Events, Setting up AWS	6

	Organizations and RAM	

Text Books:

1. Danilo Poccia , “AWS Lambda in Action: Event-driven Serverless Applications”, Manning Publication, 2016
2. Mark Wilkins , “Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud”, Addison-Wesley Professional,2019 Computing”, McGraw Hill,2013
3. Andreas Wittig and Michael Wittig, “ Amazon Web Services in Action”, Manning Publications, 2015

Reference Books:

1. Raoul Alongi , “AWS: The Most Complete Guide to Amazon Web Services from Beginner to Advanced Level”, Kindle Edition, 2019
2. Ben Piper (Author), David Clinton , “AWS Certified Cloud Practitioner Study Guide:CLF-C01 Exam 1st Edition”, Sybex, 2019

Course Code: CSI24PCL555	**Course Name:** DevOps-II	**Course Category:** PEC
Credits: 2	**Teaching scheme:** L-2	**Evaluation Scheme:** CA:60, ESE:40
Duration of External Exam: 2 Hrs.		
Pre-requisites: Software Development Basics, Version Control (Git),Operating Systems & Networking		
Course Objectives: **1.** Use version control tools for collaborative development.. **2.** Implement Continuous Integration and Continuous Deployment.. **3**. Apply containerization and orchestration in DevOps. **4.** Practice monitoring, logging, and security in DevOps. 5. Apply cloud and IaC practices to achieve DevOps Zenith.		
Course Outcomes: At the end of the course, the students will be able to –		
CO1: Apply version control and collaboration using Git.		
CO2: Design and implement CI/CD pipelines.		
CO3: Deploy applications using Docker and Kubernetes.		
CO4: Implement monitoring, logging, and DevSecOps practices		
CO5: Build scalable and automated cloud infrastructure using IaC and CI/CD.		

Contents:

Unit No.	Content	Teaching Hours
1.	**Version Control and Collaboration: Version Control Systems (VCS):** Introduction to Git and, GitHub/GitLab/Bitbucket, **Git Basics:** Cloning, branching, committing, merging, resolving conflicts, **Collaboration with Git**: Forking, pull requests, and code reviews, **Branching Strategies**: Git Flow, feature branching, trunk-based development, **Team Collaboration**: Integrating with CI/CD tools, using version control for DevOps	6
2.	**Continuous Integration and Continuous Deployment (CI/CD): CI/CD Overview:** Introduction to Continuous Integration and Continuous Deployment/Delivery, **CI Tools**: Jenkins, GitLab CI, Travis CI, CircleCI, **Setting up a CI Pipeline**: Automating builds and tests with CI tools, **CD Pipelines:** Automating the deployment process to various environments (staging, production), **Test Automation:** Integrating unit, integration, and UI tests into the CI/CD pipeline	6
3.	**Containerization and Virtualization:** Introduction to Containers: Differences between containers and virtual machines, Docker Basics: Docker containers, images, Docker file, and Docker Compose, Containerization Workflow: Creating and managing containers, building Docker images, Kubernetes Basics: Overview of Kubernetes for container orchestration, Using Containers in DevOps: Benefits of containers in CI/CD, micro services, and scalability.	6
4.	**Monitoring, Logging, and Security in DevOps:** Monitoring in DevOps: Importance of application and infrastructure monitoring (Prometheus, Grafana), Log Management: Tools like ELK Stack (Elasticsearch,	6

	Logstash, Kibana), Fluentd, Alerting and Metrics: Setting up alerts, performance monitoring, and health checks, DevSecOps: Integrating security practices into the DevOps lifecycle, Security Tools: Vulnerability scanning, secrets management (HashiCorp Vault)	
5.	**Cloud Infrastructure & DevOps Zenith:** Cloud Platforms in DevOps: Role of AWS, Azure, and GCP in modern infrastructure, Infrastructure as Code (IaC): Concept, benefits, and automation using Terraform and Cloud Formation, Configuration Management: Basics of Ansible for consistent and repeatable deployments, **DevOps Zenith Concept:** Achieving the highest maturity level in DevOps through full automation, IaC-driven infrastructure, reliable CI/CD pipelines, monitoring, security integration, and scalability, Cloud Automation & Reliability: Autoscaling, high availability, and cost-optimized infrastructure as key indicators of DevOps Zenith.	6

Text Books:

1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" by Gene Kim, Kevin Behr, and George Safford.
2. "Accelerate: The Science of Lean Software and DevOps: Building and Scaling High Performing Technology Organizations" by Nicole Forsgren, Jez Humble, and Gene Kim.
3. "DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations" by Gene Kim, Patrick Debois, John Willis, and Jez Humble.

Reference Books:

1. Version Control with Git" by Jon Loeliger and Matthew McCullough.
2. "Pro Git" by Scott Chacon and Ben Straub
3. "Docker: Up & Running" by Karl Matthias and Sean P. Kane
4. Kubernetes Up & Running: Dive into the Future of Infrastructure" by Kelsey Hightower, Brendan Burns, and Joe Beda.

Online Resources:

1. Virtual Labs
2. GitHub Repositories

Course code: CSI24PCL556		**Course name:** Image Processing
Course Category: PEC		
Credits: 03	**Teaching scheme:** L-3	**Evaluation scheme:** CA–60, ESE–40

Pre-requisites:
1. Any one programming skill with

Course Objectives:

1. Demonstrate basic image operations for pre-processing
2. Introduce filtering techniques for image smoothing and sharpening
3. Introduce image segmentation techniques for object detection
4. Demonstrate restoration techniques for noisy images.
5. Explore advanced concepts like object recognition and feature extraction

Course Outcomes: At the end of the session, the students will be able to –

CO1: Explain image sampling, quantization, and basic operations.
CO2: Apply smoothing and sharpening filters to improve image quality
CO3: Explain segmentation techniques like thresholding and edge detection
CO4: Apply DCT and Wavelet Transform for image compression.
CO5: Analyse the use of machine learning in image recognition applications

Content

Unit	Content	Teaching Hours
1.	**Introduction to Image Processing -** Definition and Applications of Image Processing, Components of an Image Processing System, Types of Digital Images: Binary, Grey scale, and Color, Image Sampling and Quantization, Basic Operations: Image Addition, Subtraction, Multiplication, and Division	8
2.	**Image Enhancement in Spatial and Frequency Domains -** Basics of Image Enhancement , Spatial Domain Techniques: Point Processing, Histogram Equalization, and Filtering, Frequency Domain Techniques: Fourier Transform and Filtering, Smoothing and Sharpening Filters, Noise Removal Techniques	10
3.	**Image Segmentation and Morphological Processing** - Thresholding Techniques for Image Segmentation, Edge Detection: Sobel, Prewitt, and Canny Operators, Region-Based Segmentation: Region Growing and Splitting, Morphological Operations: Dilation, Erosion, Opening, and Closing, Morphological Applications in Image Analysis	9
4.	**Image Compression and Restoration** - Basics of Image Compression: Lossy vs. Lossless, Compression Techniques: JPEG, PNG, and GIF, Transform Coding: Discrete Cosine Transform (DCT) and Wavelet Transform, Image Restoration Techniques: Noise Modelling and Inverse Filtering, Applications of Image Compression in Multimedia	9
5.	**Advanced Topics in Image Processing**- Object Recognition and Pattern Matching, Introduction to Machine Learning in Image Processing, Feature Extraction: SIFT, SURF, and HOG, Image Processing in AI Applications: Facial Recognition, Medical Imaging, Introduction to OpenCV for Image Processing	9

Text Books: 1. Fundamentals of Digital Image Processing by Anil K. Jain, Prentice Hall 2. Image Processing, Analysis, and Machine Vision by Milan Sonka, Vaclav Hlavac, and Roger Boyle, 4th Edition,Cengage Learning
Reference Books: 1. Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods, 4th edition, Pearson Education

Course code: CSI24PCL557	**Course name:** Quantum Computing
Course category: PEC	
Credits: 03	**Teaching scheme:** L-3 **Evaluation scheme:** CA: 60, ESE: 40
Pre-requisites: **1.** Any one learn programming skill	
CourseObjectives:	
1. Relate the differences between the classical and quantum system.	
2. Apply the concepts of complex numbers to quantum computing.	
3. Examine the various algorithms using quantum programming	
4. Develop various quantum applications.	
Course Outcomes: At the end of the session, the students will be able to -	
CO1: Understand the basics of quantum computing.	
CO2: Understand the background of Quantum Mechanics.	
CO3: Analyze the computation models.	
CO4: Model the circuits using quantum computation.	
CO5: Understand the quantum operations such as noise and error–correction.	

Content

Unit	Content	Teaching Hours
1.	**Basic Quantum Theory**: Global Perspectives – Quantum Bits – Quantum Computation – Quantum Algorithms – Experimental Quantum Information Processing – Quantum Information.	9
2.	**Algorithms:** Deutsch's Algoirthm, Deutsch-Jozsa Algoirthm Simon's Periodicity Algorithm, Grover's Search Algorithm, Shor's Factoring Algorithm	9
3.	**Programming Languages:** Quantum Assembly Programming, Toward Higher-Level Quantum Programming, Quantum Computation Before Quantum Computers	9
4.	**Theoretical Computer Science:** Deterministic and Nondeterministic Computations, Probabilistic Computations, Quantum Computations **Cryptography:** Quantum Key Exchange Algorithms-BB84, B92 and EPR Protocols, Quantum Teleportation	9
5.	**Information Theory:** Classical Information and Shannon Entropy, Quantum Information and von Neumann Entropy, Classical and Quantum Data Compression, Error Handling Codes	9

Text Books:

1. Noson S Yanofsky and Mirco A. Mannucci, Quantum Computing for Computer Scientists, Cambridge University Press
2. Vishal Sahni, Quantum Computing, Tata McGraw-Hill Edition

Reference Books:

1. N. David Mermin, "Quantum Computer Science: An Introduction", Cambridge University Press, 2007.
2.Bernhardt, Quantum Computing for Everyone, The MIT Press

Course code: CSI24PCP554 **Course name:** CC Using AWS Lab **Course category:** PEC
Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Duration of External Exam: 2 Hrs
Pre-requisites: Basics of CC.
Course Objectives:
1. Host a basic application (e.g., Employee Directory) on AWS.
2. Understand the Amazon EC2 instance lifecycle and compute services.
3. Explain Docker and Kubernetes and their architectures.
4. Understand S3 storage components, including buckets, objects, and keys.
5. Configure IAM users, groups, and password policies.
Laboratory Outcomes:At the end of the course, the students will be able to -
LO1: Identify the components of AWS Global Infrastructure.
LO2: Describe the Amazon EC2 instance lifecycle and container services.
LO3: Define containerization concepts and tools like Docker and Kubernetes.
LO4: Explain S3 components, storage classes, and versioning.
LO5: Create and manage IAM policies, groups, and permissions.

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Create and configure a Cloud Storage bucket using Amazon S3 to store and manage object.	2
2.	Operate AWS services through the AWS CLI by executing basic commands for resource management.	2
3.	Design and deploy a Custom VPC with subnets, route tables, and internet gateways for secure networking.	2
4.	Configure Security Groups to control inbound and outbound traffic for EC2 instances.	2
5.	Provision Reserved Instances and demonstrate cost optimization strategies in AWS.	2
6.	Install and verify the CloudWatch Agent on EC2 instances to enable monitoring.	2
7.	Create and test a CloudWatch Alarm to trigger notifications based on performance metrics.	2

8.	Design and implement DynamoDB tables with appropriate keys and attributes for NoSQL storage.	2
9.	Build and query a data warehouse using Amazon Redshift and perform serverless querying with Amazon Athena on S3 data.	2
10.	Create and manage IAM Users and Groups with appropriate policies for access control.	2
11.	Deploy EC2 instances using AWS CloudFormation templates to automate infrastructure provisioning.	2
12	Provision and manage a Google Compute Engine instance using Terraform for Infrastructure-as-Code deployment.	2

Text Books:

1. Danilo Poccia , "AWS Lambda in Action: Event-driven Serverless Applications", Manning Publication, 2016
2. Mark Wilkins , "Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud", Addison-Wesley Professional,2019 Computing", McGraw Hill,2013
3. Andreas Wittig and Michael Wittig, " Amazon Web Services in Action", Manning Publications, 2015

Reference Books:

1. Raoul Alongi , "AWS: The Most Complete Guide to Amazon Web Services from Beginner to Advanced Level", Kindle Edition, 2019
2. Ben Piper (Author), David Clinton , "AWS Certified Cloud Practitioner Study Guide:CLF-C01 Exam 1st Edition", Sybex, 2019

Course Code: CSI24PCP555	**Course Name:** Dev-Ops-II Lab	**Course Category:** PEC
Credits: 1 **Teaching Scheme:** P-2	**Evaluation Scheme:** TW–30, PR–20	
Duration of External Exam: 3 Hrs.		
Pre-requisites: Programming Skills, Cryptographic Fundamentals, Tools & Software Knowledge		
Course Objectives:		
1. Understand DevOps principles, culture, and lifecycle		
2. Apply Git for version control and collaborative development.		
3. Design and implement Continuous Integration and Continuous Deployment pipelines.		
4. Use containerization and orchestration tools such as Docker and Kubernetes.		
5. Implement monitoring, logging, and real-world DevOps case studies		
Laboratory Outcomes: At the end of the course, the students will be able to –		
LO1. Use Git for version control, collaboration, and code review.		
LO2. Build and manage CI/CD pipelines for automated delivery.		
LO3. Deploy applications using Docker, Docker Compose, and Kubernetes.		
LO4. Apply monitoring and logging using Prometheus, Grafana, and ELK Stack.		
LO5. Analyse and implement DevOps solutions through case studies.		

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Collaborating Using Git - Fork, Pull Requests, and Code Reviews	2
2.	Setting up a Continuous Integration (CI) Pipeline	2
3.	Explore Continuous Deployment (CD) Pipeline	2
4.	Apply Docker Basics - Containerization	2
5.	Understand Docker Compose - Multi-Container Application	2
6.	Introduction to Kubernetes - Basic Cluster Setup	2
7.	Monitoring with Prometheus and Grafana	2
8.	Apply Log Management with ELK Stack	2
9.	Create cloud infrastructure using Terraform, enable autoscaling and verify resource creation.	2

10.	Case Study/Implementation	2
11	Mini project	2

Text Books:

1. "The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" by Gene Kim, Kevin Behr, and George Safford.
2. "Accelerate: The Science of Lean Software and DevOps: Building and Scaling High Performing Technology Organizations" by Nicole Forsgren, Jez Humble, and Gene Kim.
3. "DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations" by Gene Kim, Patrick Debois, John Willis, and Jez Humble..

Reference Books:

1. Version Control with Git" by Jon Loeliger and Matthew McCullough.
2. "Pro Git" by Scott Chacon and Ben Straub
3. "Docker: Up & Running" by Karl Matthias and Sean P. Kane
4. Kubernetes Up & Running: Dive into the Future of Infrastructure" by Kelsey Hightower, Brendan Burns, and Joe Beda.

Online Resources:

1. https://git-scm.com/book/en/v2
2. https://docs.gitlab.com/ee/ci/

.**Course code:** CSI24PCP556 **Course name:** Image Processing Lab **Course category:** PEC
Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites: Mathematical fundamentals like matrices, basic linear algebra.
Course Objectives: 1**.** Demonstrate basic image operations for pre-processing. 2. Introduce filtering techniques for image smoothing and sharpening. 3. Introduce image segmentation techniques for object detection 4. Demonstrate restoration techniques for noisy images 5. Explore advanced concepts like object recognition and feature extraction
Laboratory Outcomes: At the end of the course, the students will be able to - **LO1:** Explain image sampling, quantization, and basic operations. **LO2:** Apply smoothing and sharpening filters to improve image quality. **LO3:** Explain segmentation techniques like thresholding and edge detection. **LO4:** Apply DCT and Wavelet Transform for image compression. **LO5:** Analyse the use of machine learning in image recognition applications**.**

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Load and display images in Python/Matlab, convert between color spaces (RGB to grayscale).	2
2.	Implement histogram equalization to enhance image contrast.	2
3.	Apply smoothing (e.g., Gaussian filter) and sharpening filters (e.g., Laplacian filter).	2
4.	Perform image enhancement in the frequency domain using Fourier Transform.	2
5.	Implement Sobel, Prewitt, and Canny operators for edge detection.	2
6.	Apply global and adaptive thresholding for image segmentation.	2
7.	Demonstrate dilation, erosion, opening, and closing using binary images.	2
8.	Implement JPEG compression using DCT and analyze compression ratios.	2
9.	Extract features using SIFT and HOG for object recognition.	2
10.	Build a simple object recognition system using OpenCV and a machine learning model.	2

Text Books: 1. Fundamentals of Digital Image Processing by Anil K. Jain, Prentice Hall 2. Image Processing, Analysis, and Machine Vision by Milan Sonka, Vaclav Hlavac, and Roger Boyle, 4th Edition,Cengage Learning
Reference Books: 1. Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods, 4th edition, Pearson Education 2. Computer Vision: Algorithms and Applications by Richard Szeliski, Springer
Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CSI24PCP557 **Course name:** Quantum Computing Lab **Course category:** PEC
Credits: 1 **Teaching scheme:** P-2 **Evaluation scheme:** TW–30, PR–20
Pre-requisites: Basics of linear algebra and quantum mechanics is required
Course Objectives:
1. Relate the Differences between the classical and quantum System.
2. Apply the concept of complex numbers to quantum computing
3. Examine the Various algorithms using quantum programming
4. Develop various quantum applications.
Laboratory Outcomes: At the end of the course, the students will be able to -
LO1: Understand the basics of quantum computing.
LO2: Understand the background of Quantum Mechanics.
LO3: Analyze the computation models.
LO4: Model the circuits using quantum computation.
LO5: Understand the quantum operations such as noise and error–correction.

Contents –

Sr. No.	List of Experiments	Practical Hours
1.	Write a program to convert a complex number from Cartesian to polar and vice versa.	2
2.	Write a program to check whether a given square matrix is Hermitian or not, Unitary or not.	2
3.	Write a program that accepts two matrices and constructs their tensor product.	2
4.	Write a program to perform the little marble experiment	2
5.	Write a program that simulates the first quantum system. The user should be able to specify how many points the particle can occupy	2
6.	Design and implement a quantum computer emulator in the language of your choice.	2
7.	Study Algorithms-BB84, B92 and EPR Protocols	2
8.	Study Quantum Teleportation	2
9.	Write a program that lets the user choose how many 02 letters the source alphabet has, and then enter the probability distribution. The program should visualize it, and compute its Shannon entropy	2

10.	Write a program that lets the user enter two qubits and their corresponding probabilities	2

Text Books: 1. Noson S Yanofsky and Mirco A. Mannucci, Quantum Computing for Computer Scientists, Cambridge University Press 2. Vishal Sahni, Quantum Computing, Tata McGraw-Hill Edition
Reference Books: 1. N. David Mermin, "Quantum Computer Science: An Introduction", Cambridge University Press, 2007. 2. Chris Bernhardt, Quantum Computing for Everyone, The MIT Press
Online Resources: 1. NPTEL / SWAYAM lectures.

Course code: CSI24VSP559 **Course Name:** Mobile Application Development Lab **Course category:** VSEC
Credits: 02
Pre-requisites: Basic Programming Knowledge, Basic Knowledge of Databases and SQL
Course Objectives:
1. To understand and interpret the architecture, features, and core components of the Android operating system.
2. To configure and utilize the Android development environment effectively for building and testing applications.
3. To design and develop Android applications using various user interface components and layout techniques.
4. To create database-driven Android applications by integrating data storage, retrieval, and management mechanisms.
5. To develop advanced Android applications that implement security features and manage permissions for safe and controlled access to system resources.
Laboratory Outcomes: At the end of the session, the students will be able to -
LO1: Interpret the features of android operating system
LO2: Use after configuring Android development environment.
LO3: Develop android applications using UI components and layouts
LO4: Create database driven Android applications.
LO5: Develop advanced Android applications that require relevant permissions for security

List of Experiments–

Sr.No.	Name of Experiment	Practical Hours
1.	Install Android IDE and create Android virtual device	2
2.	Develop a program to display “Hello World” onscreen	2
3.	Explore the directory structure in Android IDE	2
4.	Develop android application using View Text and Edit Text.	2
5.	Develop a program to implement Button, Image Button and Toggle Button	2
6.	Develop a program to design Checkbox and Radio button.	2
7.	Develop a program to implement Progress Bar	2
8.	Develop a program to create a login form using the above UI controls	2
9.	Write program to implement Linear layout and Constraint layout to create any registration form with Custom Toast Alert	2
10.	Develop a program to implement Grid View, Image View, Scroll View, List View for any management system like library management/hotel management	2
11	Develop a simple calculator which uses grid layout and GUI concepts	2
12	Develop a splash screen in android	2
13	Design and develop any convertor application like temperature convertor /currency convertor/ volume convertor	2

14	Design and develop a simple countdown timer	2
15	Develop a program to implement Date Picker in application	2
16	Develop a program to implement Time Picker in application	2
17	Develop a program to create two simple activities for Login application	2
18	Develop a program to implement new Activity using explicit intent and implicit intent to open any other website	2
19	Develop a program to implement services like bluetooth/wifi	2
20	Develop a program to implement a broadcast receiver to switch between different modes like Airplane mode/Silent Mode/Loud Mode	2

Text Books:

1.Dixit, Prasanna Kumar – *Android*, Vikas Publication, New Delhi 2014, ISBN: 9789325977884
2.Hortan, John– *Android Programming for Beginners*, Packet Publications, 2015, ISBN : 978-1-78588-326-2

Reference Books:

1. Pradeep Kothari**–** *NAndroid Application Development*,Kogent Learning Solutions ISBN :9789351194095.
2.Maclean David , KomatineniSatya, Allen Grant**–** *Apress Publications, 2015, ISBN :978-1-4302-4680-0*

Online Resources:

1) https://onlinecourses.swayam2.ac.in/nou21_ge41/preview.
2) **https://developer.android.com/training/data-storage/sqlite**